Which Side Is Wrong? What Contradiction Flags Do to Retrieval-Augmented Generation Under Misinformation

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Abstract

Retrieval-augmented generation systems increasingly flag contradicting retrieved passages to the generator. We ask what such a flag must say, and how reliable its detector must be, to counter misinformation. We vary detector and flag wording on fixed passages for seven generators. For four, a neutral flag ("these passages may conflict") moves answers along the prompt's answer policy: towards abstaining when it asks for one answer, towards listing both sides when it asks for every valid answer. The other three barely move or move the other way. A flag naming the false side helps when right, but generators also follow wrong ones: with direction randomised, three of four generators followed wrong flags about as often as right ones, also on new questions, and gpt-6.1-terra followed 95%; Claude Sonnet, the most accurate there, followed 23%. Such flags pay only above a break-even direction precision that differs by generator (exploratory). An account frozen in advance, assuming follow rates transfer, predicted effects on unseen questions with mean absolute error 1.4 points, but a baseline ignoring which questions are flagged scored 1.8, and the account overstated harm on conflict-free questions. Directional results come from ConflictBank, whose synthetic misinformation concerns facts the generators rarely recall, yet whose true answer detectors pick from the two candidates without passages and whose false side a bag-of-words classifier reads from style; all are scored by string matching. We derive a rule, not tested end to end, for choosing a detector or none, and release code and runs.

1 Introduction

Retrieval-augmented generation (RAG) (Lewis et al., 2020) hands a language model passages that may disagree. Some disagreements are genuine; others are planted or stale misinformation (Zou et al., 2025; Wang et al., 2025b). Prompting a model about conflicts can help (Cattan et al., 2025), and recent systems pair contradiction detectors with prompts that flag the conflict to the generator (Wang et al., 2026). They report detection F1 and end-to-end accuracy, but not how one leads to the other. We ask what a flag must say, and what its detector must get right, for the flag to help against misinformation.

We fix the passages and vary two things independently: the *detector* (an LLM, an oracle built from benchmark labels, or a synthetic detector of chosen quality) and the *format* that passes its output to the generator, from a neutral "these passages may conflict" flag to a *directional* flag naming the likely misinformation. Answer keys that state a wrong answer without marking any passage control for what a flag conveys. We run seven generators on six benchmarks, chiefly RAMDocs (Wang et al., 2025b) and ConflictBank (Su et al., 2024), and registered hypotheses in stages, most after earlier results were known (§4).

The paper makes three claims, each scored by string matching:

- **A neutral flag mainly shifts the answer policy** (§5.1). Asked for one answer, four of seven generators abstain more (gpt-4o-mini on misinformed and correct answers alike); asked for every valid answer, they add the misinformation. The other three barely move or move the other way, and where misinformation passages outnumber the correct one the flag can raise selective accuracy. H2 and H8 were registered before any confirmatory data, H15 and H21 after earlier results; risk–coverage counts are exploratory.

- **Generators defer to directional flags, so a wrong flag can cost more than a right one gains** (§5.2). With direction randomised on true contradictions (H31, registered before its runs but after earlier results), three of four generators follow wrong flags about as often as right ones, also on new questions; the most accurate

of the four follows nearly all, Claude Sonnet few. The implied break-even direction precision is exploratory.

- **A frozen account of these follow rates predicts within a benchmark and shows where it fails** (§5.3). Predictions committed before any flagged answer existed came close on unseen ConflictBank questions (H32, prospective), though a baseline that ignores which questions are flagged was nearly as close. They overstated harm on clean questions and on HealthContradict, whose labels are unverified.

The numbers behind claims 2 and 3 come from ConflictBank, whose misinformation is LLM-written about facts the generators rarely know, and whose false side a bag-of-words classifier picks from passage style alone (§4), and detectors pick its true answer from the two candidates without any passage (§5.2). Detectors and generators may therefore find the false side without reasoning about the contradiction. HealthContradict and RAGuard use real documents but serve only as label-transfer stress tests, because their labels, not verified facts, decide which side counts as false. Whether a detector's verdict passed as an answer key conveys as much as its flag depends on the scorer and is unresolved (§5.4). We make no claims about model families. We release the harness, the pre-registration and every run.[1]

2 Related Work

Knowledge conflicts and benchmarks. Models can prefer memorised knowledge over contradicting context (Longpre et al., 2021), yet coherent contradicting evidence readily persuades them (Xie et al., 2024), depending on the strength of their prior (Wu et al., 2024). Among the conflict types surveyed by Xu et al. (2024), we study inter-context conflicts, where models tend to rely on one passage (Chen et al., 2022) and weigh relevance over credibility cues (Wan et al., 2024). RAMDocs (Wang et al., 2025b) mixes genuine ambiguity, misinformation and noise; ConflictBank (Su et al., 2024) generates passages for true and substituted facts; WikiContradict (Hou et al., 2024) pairs genuinely contradicting Wikipedia passages; CONFLICTS (Cattan et al., 2025) annotates search results with conflict types. FaithEval (Ming et al., 2025), ConfRAG (Yuan et al., 2026), QACC (Liu et al., 2025) and WhoQA (Pham et al., 2024) cover related settings.

Telling the model about conflicts. Prompting a model to reason about the conflict type helps, and supplying the gold type helps more (Cattan et al., 2025). Told that passages may conflict, models acknowledge both sides more often (Hou et al., 2024); the hint helps on contradictory contexts and hurts on consistent ones (Gross et al., 2026). Warned that passages may contain errors, models rarely find them (Chen et al., 2024), and vigilant prompting gives limited protection against generated misinformation (Pan et al., 2023). Our neutral flag is that warning, made pair-specific and true.

Telling the model which passage is wrong. Hong et al. (2024) pass a discriminator's marks on perturbed passages into the prompt; credibility-aware generation (Pan et al., 2024) and CrAM (Deng et al., 2025) weight passages by credibility, and TACS (Yu et al., 2024) masks untruthful context, close to our removal. Some of these vary the signal's quality (Hong et al., 2024; Pan et al., 2024). Models follow injected hints and asserted views, wrong ones included (Turpin et al., 2023; Sharma et al., 2024), and reasoning models often use a hint without saying so (Chen et al., 2025). Advice improves a decision-maker only when its errors fall where the decision-maker is weaker (Bansal et al., 2021; Mozannar and Sontag, 2020), and whether abstaining is a failure depends on a wrong answer's cost (Wen et al., 2025; Song et al., 2025). We add a randomised manipulation of direction on fixed pairs, answer-key controls, and a per-generator break-even.

Detection, resolution and poisoning. ConflictRAG (Wang et al., 2026) trains a pairwise conflict detector with an LLM fallback; we did not test it as a detector. LLMs detect conflicts but struggle to pinpoint them (Gokul et al., 2025; Wang et al., 2024; Jiayang et al., 2024). They favour the more frequent side, the side matching their prior, and LLM-generated passages (Jin et al., 2024; Kortukov et al., 2024; Tan et al., 2024), which matters for generated benchmarks such as ConflictBank. MADAM-RAG (Wang et al., 2025b), Astute RAG (Wang et al., 2025a), Self-RAG (Asai et al., 2024), Corrective RAG (Yan et al., 2024) and InstructRAG (Wei et al., 2025) handle conflicts without an explicit detector; RobustRAG (Xiang et al.,

[1] Code, the pre-registration and all run outputs are in the anonymised supplementary material and will be released publicly on acceptance.

2024) and TrustRAG (Zhou et al., 2025) defend against poisoned passages (Zou et al., 2025). We ask instead how to pass a detector's output to the generator.

3 Framework

A question comes with retrieved passages. A *detector* returns flagged pairs of passages and, if it is directional, a *suspect* side for each pair: the passage it judges false. A *format* turns the passages and the detector's output into the generator's prompt. Every prompt ends with an instruction to give a `Final answer(s):` line; all but the minimal skeleton state an answer policy. The *multi-answer* skeleton asks for every valid answer when the passages support several; the *one-answer* skeleton says the question has exactly one; the *minimal* skeleton states no policy. All three allow "unknown"; the *forced* skeleton is one-answer without it (Appendix A).

Formats. *Plain* gives passages and question only. The *neutral flag* heads each flagged passage with `[POTENTIAL CONFLICT with passage(s): j]` and lists the conflicting pairs. The *directional flag* instead heads suspect passages with `[LIKELY MISINFORMATION; conflicts with passage(s): j]`. *Remove* deletes the suspect passages. Controls separate what a flag conveys: a one-wrong flag (one side is false, not which), an *answer key* and a *verdict key* (Table 1); pick and hint controls are in Appendix A. A *single call* names suspect passages and answers in one prompt; *closed book* gives no passages.

Detectors. The *LLM detector* is one call that lists contradicting pairs as JSON, separating them from ambiguity, complementary and temporal relations, and names the likelier misinformation side (or "both" or "unclear"). It runs on gpt-4o-mini (temperature 0) unless stated, and on other models in the crossed and prospective designs. The *oracle* flags exactly the gold contradiction pairs, as the benchmark's labels define them, and names the labelled misinformation side. A *synthetic detector* is built from gold labels with a chosen pair precision, recall and direction precision d: each flag on a true pair names the false side with probability d, independently per pair. The randomised test (H31) uses $d = 0.5$ on every true pair (one per ConflictBank question). The heuristic CAAG detector, NLI, a random placebo and a 36-cell synthetic grid are in Appendices B and I.

Term	Meaning
Cell	one detector paired with one generator; *crossed* designs pair several of each
Direction precision	share of the detector's one-side flags on true pairs that name the misinformation passage (d if synthetic)
Pair precision, recall	share of flagged pairs that are true contradictions; share of true contradictions flagged
Follow rates f_R, f_W	f_R: of questions answered wrongly with no flag, the share a right flag makes correct; f_W: of those answered correctly, the share whose answer under a wrong flag contains the misinformation
Break-even p^*	direction precision at which randomised flags on true pairs have zero net effect, $-\Delta_W/(\Delta_R - \Delta_W)$
Answer key	plain prompt plus "The answer X is false" for each wrong answer; no passage marked
Verdict key	an answer key whose X is the answer the detector's suspect passage supports, by the benchmark's labels

Table 1: Terms. A directional flag on a true pair is *right* if it names the false side and *wrong* if it names the correct one; Δ_R and Δ_W are their effects on accuracy in the randomised test (H31).

The follow-rate account. A right flag is *followed* when the flagged answer is correct where the no-flag answer was wrong; a wrong flag is followed when the flagged answer contains the misinformation answer (matched in either direction) where the no-flag answer was correct. Abstaining is neither (PR§16.3). Generator G's follow rates f_R^G and f_W^G are these shares in the randomised test. For detector D and question i, let $r_i = 1$ if D flags the true pair and names its false side, $w_i = 1$ if D marks a correct passage as suspect, and $a_i = 1$ if G's no-flag answer is correct. Over n questions, the account predicts the directional flag's effect as

$$\hat{\Delta}(G, D) = \tfrac{1}{n} \textstyle\sum_i \left[r_i(1 - a_i) f_R^G - w_i a_i f_W^G \right]. \quad (1)$$

Unflagged questions keep their no-flag answer, so they contribute zero. The equation applies f_W, measured as adoption, as the probability of losing a correct answer. Every term except the two rates is observed before any flagged answer exists. Equation (1) is therefore bookkeeping plus one empirical claim: that a generator's follow rates transfer across detectors and questions. It omits right flags on correct answers, wrong flags on wrong answers, and abstention.

4 Experimental Setup

Data. Table 2 gives each benchmark's role. **RAMDocs** (Wang et al., 2025b) gives each question 1–12 documents typed correct, misinformation or noise. A pair is a *contradiction* if one side is misinformation and the answers differ. A *conflict* question has a contradiction pair; a *clean* ques-

Benchmark	Role here	What it cannot support
RAMDocs	claims 1–2: neutral and directional flags (H1–H7, H9–H11); grid; 300-question hold-out	entity-aligned pairs; natural misinformation
ConflictBank	claims 1–3: answer policy, randomised direction (H31), crossed and prospective cells (H30, H32–H36)	real documents; generator priors; a side not readable from style (R1)
RAGuard	label-transfer stress test (H26)	a factual oracle
Health-Contradict	label-transfer stress test (H37)	verified labels
Wiki-Contradict	genuine conflicts (H8, H28)	which answer is false
CONFLICTS	underpowered null	any positive claim

Table 2: What each benchmark supports. The quantitative core of claims 2 and 3 rests on ConflictBank alone; the other benchmarks add breadth to claims 1 and 2 and stress tests, not independent confirmation.

tion has none. On the 300-question hold-out split (139 conflict, 126 clean, 35 excluded), 104 clean questions have several valid answers, and the pairing rule does not align same-name entities, so at least 37% of oracle pairs join passages about different entities (Appendix D).

ConflictBank (Su et al., 2024): for a Wikidata fact, an LLM writes one passage for the true object and one for a substituted object. We sample 300 conflict questions (correct passage, misinformation passage, two distractors about other subjects) and 100 clean ones (correct passage, three distractors); each has one gold answer. A *majority* variant gives each conflict question three misinformation passages. ConflictBank-fresh, 400 more questions with no shared subject, serves the prospective test. Three properties limit what ConflictBank can show. The generators rarely know the facts: closed-book accuracy is 4.3–20.0% for every generator run without passages. With one passage per side, agreement between passages cannot reveal the false side. And the false side is legible from style. A bag-of-words logistic regression, trained on 124,769 ConflictBank facts that share no subject with ours, gives the misinformation passage the higher score in every conflict pair of both samples (pairwise accuracy 1.000), also with each passage's answer string masked (1.000). Its heaviest features are genre words, such as "studies", "study", "exclusive interview" for the false side (R1, exploratory). Its clean questions test only false flags on off-subject passages.

Stress tests. RAGuard (Zeng et al., 2025) pairs PolitiFact claims with Reddit posts that an LLM labelled misleading or supporting according to whether GPT-4 judged the claim correctly with the post in context; an LLM judge finds only 6% of sampled pairs to contradict. HealthContradict (Zhang et al., 2026) pairs yes/no health questions with one real web page per side (247 items, pages cut to 4,000 characters). Neither verifies that the labelled-misleading side is false; on HealthContradict an LLM judge confirms both pages' labels for only 37% of items. We therefore treat both as label-transfer stress tests. **WikiContradict** (Hou et al., 2024) pairs genuinely contradicting Wikipedia passages; a correct answer names both sides. CONFLICTS (Cattan et al., 2025) (237 questions) has too few misinformation conflicts to detect effects. Appendix H details every benchmark.

Generators. The main generator is gpt-4o-mini (OpenAI et al., 2024) (temperature 0.1, at most 256 output tokens, seeds 0–2). We also run gpt-4.1, Qwen2.5-7B-Instruct (Qwen Team et al., 2024) (4-bit, local), two reasoning models (gpt-6-luna, three seeds; gpt-6.1-terra, one seed), Claude Haiku 5.5 (three seeds) and Claude Sonnet 5.5 (one seed). The reasoning and Claude models run at their default temperature with up to 4,096 output tokens; Claude runs without JSON mode (Appendix H). On ConflictBank the main prompt is the one-answer skeleton.

Scorers. The primary scorer, fixed before the confirmatory run, is a one-way string matcher (bidirectional for WikiContradict's both-sides score): an answer is correct if, after normalisation, every gold answer and no wrong answer is a whole-word span of its final-answer line (Wang et al., 2025b); the *bidirectional* variant, for robustness, also accepts an answer inside the gold string. *Misinformation share* is the share of answers containing a wrong answer; *selective accuracy* is accuracy among answers that do not abstain. An outside-family LLM judge (gemini-3.5-flash-lite) re-scored all 23,204 responses of the pooled key comparisons and agrees with the matcher on 94.3%. The matcher's misses are one-sided and depend on the condition: the judge alone calls correct 8.7% of oracle directional-flag responses, against 6.7% of answer-key responses (Appendix D). The matcher thus misses more correct answers under the flags than under the keys, so a contrast near zero must hold under both scorers. Responses without a final-answer line are scored on their whole text, as registered (PR§5),

so some count as correct; they are 0.15% of all responses, but 51 of 979 run–condition cells exceed 1%, up to 20.3% in one gpt-4.1 condition (R9, Table 29). No human has yet labelled outputs; blind audit sheets are released unlabelled, and they and a within-family judge check are described in Appendix D.

Registration. "PR§N" is section N of our preregistration. We registered in three tiers. (i) H1–H9 were frozen before any confirmatory data, after pilots on 200 RAMDocs questions. Thirteen addenda followed (PR§9–PR§21); ten carry hypotheses, and PR§20 registers only re-analyses (R1–R19). (ii) H10–H31, H37–H38 and H40–H41 were registered before their runs but after earlier results were known; three items were committed after their runs had begun (PR§20.1), and parts of H24, H27 and H28 reuse runs that predate them (Appendix H). (iii) H32–H36 are prospective: their predictions were committed before any flagged answer existed. We call only tiers (i) and (iii) confirmatory (commit times: Table 34); tier (ii) is registered but not confirmatory. Everything else is exploratory, including R1–R19 and the descriptive PR§21 runs C1–C11 (Appendices F, G).

Statistics. We average each question's score over seeds and report paired differences with question-bootstrap 95% CIs and sign-flip permutation tests (Efron and Tibshirani, 1993; Dror et al., 2018). A registered test takes the Holm-adjusted p (Holm, 1979) within its hypothesis's family; the exploratory condition comparisons form one Holm family per metric (636 for accuracy). "Significant" means under Holm; CIs are unadjusted. A null difference is not read as equivalence without a TOST (Lakens, 2017) at ± 3 points, a margin set after the confirmatory run. By the APIs' usage reports, all runs cost $218.39 (Appendix H).

5 Results

Sections 5.1–5.3 each state one claim with its status and scope; numbers use the one-way matcher unless stated. Appendix B gives every condition, Table 4 each claim's status and Appendix J every registered test.

5.1 Claim 1: a neutral flag shifts policy

A neutral flag on the gold contradictions says that two passages disagree, not which one is wrong. Under RAMDocs' multi-answer prompt it changes gpt-4o-mini's conflict-question accuracy by −6.7 points [−10.8, −3.1] from 22.1% (H2, registered before any confirmatory data). Where answers go depends on the prompt's policy (Figure 1).

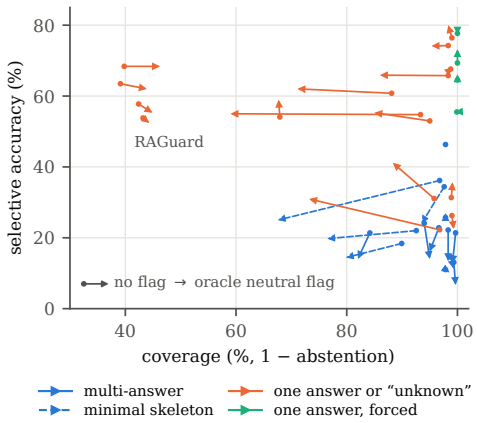


Figure 1: Risk–coverage operating points on conflict questions in 34 settings (benchmark, generator and prompt). Each arrow runs from no flag to the oracle neutral flag; colour and line style give the prompt's answer policy; the left-hand cluster is RAGuard. Coverage is 1 − abstention. One-way matcher; exploratory (R16; values in Table 30). Over 1% of responses lack a final-answer line for ConflictBank Qwen2.5-7B (multi-answer) and gpt-4.1 (one answer) and RAMDocs gpt-4.1 and Qwen2.5-7B (R9, Table 29).

One answer asked for: abstention. On ConflictBank's one-answer prompt the flag raises abstention by 34.3 points for gpt-4o-mini and 17.0 for gpt-6-luna (H21), and also for gpt-4.1 and Qwen2.5-7B (exploratory). gpt-4o-mini loses misinformed and correct answers alike: 45% of its misinformed and 37% of its correct answers become abstentions. Its selective accuracy shows no detectable change (54.8% to 55.0%, +0.2 [−4.8, +5.2]) under the ratio estimand of PR§15.4.5, which replaced the registered per-question mean after the data were seen (H21: −1.0 [−4.9, +2.9]). Strict accuracy, which counts abstention as wrong, falls by 18.7 points (H15): the flag lowers coverage without lowering risk. Across the 16 one-answer settings (CIs excluding zero, unadjusted), abstention rises in 7 and falls in 2, both on RAGuard, where few flagged pairs contradict; selective accuracy rises in 5 and falls in 0, and strict accuracy rises in 4. These gains come mostly from the majority variant, where misinformation passages outnumber the correct one and the added abstentions fall mostly on misinformed answers, and

from RAGuard. Three generators barely move: abstention goes from 1.2% to 0.9% for Claude Haiku, 1.0% to 1.7% for Claude Sonnet and 1.7% to 4.7% for gpt-6.1-terra (H21 tests Haiku and terra; terra's CI excludes zero, but it is not significant under Holm). Under forced choice its only significant accuracy effect is Claude Haiku's +3.9 (H20).

Every valid answer asked for: misinformation. Under the multi-answer prompt coverage barely moves and the model lists both sides: across the 10 settings the misinformation share rises in 7 and selective accuracy falls in 7 (CIs excluding zero, unadjusted). Claude Haiku on RAMDocs moves the other way: its misinformation share falls by 7.4 points (significant under the bidirectional matcher only, R12). The minimal skeleton, which states no policy, raises abstention in 4 of its 4 settings (same criterion). On WikiContradict, where both sides are valid and a correct answer names both, the same compliance is rewarded: the flag raises accuracy by 6.5 points (H8).

Cost of a wrong answer. If an answer containing the misinformation (matched in either direction) costs c and a correct one gains 1, gpt-4o-mini's one-answer utility change turns positive between $c = 1$ and $c = 2$ (-5.3 and $+8.0$) and is significantly positive only at $c = 4$ ($+34.7$); under the multi-answer prompt it is negative at every c (Table 22). *Status:* H2 and H8 are registered in tier (i), H15, H20 and H21 in tier (ii); the risk–coverage counts and utility are exploratory, and pooled summaries describe only these 34 settings (Table 21). The outside-family judge moves sampled contrasts by -1.7 to $+3.3$ points (Appendix D), so claim 1 is about direction, not size.

5.2 Claim 2: deference to directional flags

An oracle's direction helps. Naming the labelled false side on the gold pairs beats the neutral and one-wrong flags in every registered comparison on RAMDocs and ConflictBank (Table 4). Over no flag it gains 16.3 to 36.4 points on ConflictBank's one-answer prompt for every generator, and on RAMDocs 13.4 for gpt-4o-mini and up to 39.6 for gpt-6.1-terra, though not significantly for Qwen2.5-7B or minimal-prompt gpt-4o-mini (H10).

Randomised direction (H31). Each of the 300 ConflictBank conflict questions was flagged on its true pair, naming the false side for a random 159 and the correct side for the other 141. Because a coin flip assigns the direction, the effects measure deference to the flag, whatever cues the passages carry. A right flag raises accuracy by +17.6 to +33.5 points (Δ_R) and a wrong flag lowers it by 31.2 to 64.5 (Δ_W; Table 3). Their sum $\Delta_R + \Delta_W$, twice the net effect at an even mix, is significantly negative for 3 of 4 generators; for gpt-4o-mini it is -1.9 [-12.2, $+8.6$] (H31c). Three of the four follow wrong flags about as often as right ones; gpt-6.1-terra, the most accurate of the four, follows 95% of wrong flags against 67% of right ones. Claude Sonnet, more accurate still (one seed), follows only 23% (Table 33), so deference does not simply grow with accuracy. On 300 new questions the other three again lose more from a wrong flag than they gain from a right one (H40b), and stating 55% rather than 95% confidence lowers wrong-flag following for 3 of 4 generators (H41).

	Accuracy		Follow rate		Flag effect		
Generator	Closed	Plain	f_R	f_W	Right	Wrong	p^*
gpt-4o-mini	10.0	51.1	57 [46, 67]	51 [40, 62]	+29.4	−31.2	0.52 [0.43, 0.60]
gpt-6-luna	16.9‡	53.8	72 [60, 83]	77 [67, 86]	+33.5	−44.9	0.57 [0.51, 0.64]
Claude Haiku	11.8	66.8	63 [49, 76]	64 [55, 73]	+24.1	−49.2	0.67 [0.61, 0.74]
gpt-6.1-terra	20.0	69.7	67 [53, 80]	95 [89, 99]	+17.6	−64.5	0.79 [0.72, 0.85]

Table 3: Deference to a randomised directional flag on ConflictBank (H31; 300 conflict questions, one-answer prompt, one-way matcher). *Closed*: closed-book accuracy (%). *Plain*: accuracy with the passages and no flag in the H31 run (%). For gpt-6-luna and gpt-6.1-terra the H31 run drew its own no-flag answers, so Plain differs from their one-answer no-flag accuracy elsewhere in the paper (PR§20.1). *Right*, *Wrong*: the registered H31 effects Δ_R and Δ_W (points) of a flag that names the false passage (right) or the true one (wrong). f_R, f_W: follow rates (%, registered descriptives without a test, PR§16.3; question-bootstrap 95% CIs below): among questions answered wrongly without a flag, the share of right-flag responses that are correct; among questions answered correctly, the share of wrong-flag responses that contain the misinformation answer (bidirectional match); abstentions count as neither. p^*: break-even direction precision $-\Delta_W/(\Delta_R - \Delta_W)$ (exploratory; bootstrap 95% CI below). ‡: more than 1% of responses lack a final answer (R9).

Break-even direction precision. Randomised flags on true pairs with direction precision p change accuracy by $p\Delta_R + (1-p)\Delta_W$, positive only above p^* (Table 1). Across the four H31 gen-

erators p^* rises with accuracy, from 0.52 for gpt-4o-mini to 0.79 for gpt-6.1-terra (exploratory; 0.81 with terra's other no-flag draw, R7): an accurate generator has more correct answers for a wrong flag to overturn, and terra also follows wrong flags more often; Claude Sonnet's is 0.56. A RAMDocs grid registered before any confirmatory data agrees: synthetic detectors with perfect pair precision and recall help gpt-4o-mini significantly only at direction precision 1 (Appendix I).

Scope. On ConflictBank the real detectors' direction precision is 68.8–97.0% (R3), higher on facts their model knows closed-book, by +3.2 to +18.7 points (R2; Table 28), and in the same direction on new questions (C9). Yet they barely need the passages: given only the question and the two candidate answers, they pick the true one 80–90% of the time, and when the answers are swapped between the passages, 3 of 4 still follow the answer more than the passage's style (C1; Table 32). With three misinformation passages per question, the gpt-4o-mini detector's direction precision falls from 68.8% to 49.4% (PR§11.3). *Status:* the H31 effects are registered (tier ii); follow rates are registered descriptives without a test (PR§16.3); p^* is exploratory.

5.3 Claim 3: a frozen account predicts

Equation (1) was written after the crossed cells on our 300 ConflictBank questions had been seen (Table 19). Frozen with each generator's H31 follow rates, it was applied to ConflictBank-fresh: five detectors crossed with the four H31 generators under the same one-answer prompt (20 cells); Qwen2.5-7B is new as a ConflictBank detector, and most other pairs had been crossed before, on other questions. Predictions were committed after the detections and no-flag answers existed and before any flagged answer did (H32–H36, prospective; Table 26).

Conflict questions. H32 required a mean absolute error of at most 3 points, a calibration slope whose CI excludes 0 and includes 1, and a lower error than each generator's mean crossed-cell effect (4.1) and follow rates of 1 (5.3). All hold: the error is 1.4 points [1.2, 2.4] (predicting zero: 8.7), the slope 0.88 [0.77, 1.00]. The criteria were set knowing the retrospective fit, and two baselines without a criterion come close (R6, exploratory): replacing each per-question term by its cell mean scores 1.8, and one pair of rates for all four generators scores 1.6 (paired CIs in Appendix F). Neither is detectably worse, so most of the predictive power comes from cell-level bookkeeping and follow rates of the right size, not from which questions are flagged or from each generator's own rates. Within generators, predicted and observed effects correlate at 0.97 (H34). The sign is right in all 18 cells predicted at least 3 points from zero, but only 2 of them are predicted harms, both with gpt-6.1-terra generating: with the Qwen2.5-7B detector it was predicted to lose 10.1 points and lost 8.3; with the gpt-4o-mini detector, 3.1 and 4.0. H33 also required each predicted harm to be significant; only the first is, so H33 fails. A fresh randomised run gives gpt-4o-mini follow rates of 50% and 50%, inside the frozen intervals (H36); for the other three (H40) they stay inside except gpt-6.1-terra's f_W (86%), and the formula predicts their net effects within 2.2 points, though most of those flagged answers existed before the freeze (Appendix G).

Where it fails. On clean questions (H35) the account predicts −4.9 points on average against −1.4 observed. Cells whose detector flags no clean question are zero by construction; over the 8 that flag some, the mean error is 8.9 points [6.3, 11.9] (R6). f_W was measured where following a wrong flag means adopting the misinformation; on a clean question it can only mean dropping a correct answer, and it does not transfer. On HealthContradict the account overstates harm for all three generators (H37d; Table 27): gpt-4o-mini −8.2 predicted against −2.7 observed, gpt-6-luna −19.4 against −8.6, Claude Haiku −16.6 against +1.1. Label noise (a reversed label makes a "wrong" flag right) or the generators' higher prior there (closed-book above plain accuracy) could explain this; the first awaits human labels (PR§20.3).

5.4 What does not hold, or is not settled

Answer keys. If a flag conveys only the answer it implies, a key should do as well. Under the outside-family judge the flag leads by +2.5 [+0.9, +4.2] points over the answer key and +2.4 [+1.1, +3.7] over the verdict key (H38), so neither equivalence holds within ±3. Under the string matcher no registered comparison finds the oracle flag significantly better than an answer key (H23) or the LLM flag better than its verdict key (H30), but only 2 of the 9 H23 TOSTs show equivalence; pooled (unregistered), the flag's advantage is −0.1

[−1.5, +1.2] and +1.2 [−0.1, +2.5], equivalent within ±3 under the matcher only (PR§19.2). Whether a key conveys what the flag conveys stays unresolved until humans decide between the scorers.

Stress tests. On HealthContradict each item has one page per yes/no side, so an oracle flag on the misinformation page effectively states the answer; its gain (+5.5 to +13.8 points, H37a) measures deference to the released labels. On RAGuard the answer key beats the oracle flag for 3 of 4 generators, by +7.1 to +33.2 points, and falls below it for gpt-4o-mini (−8.5; R14; CIs exclude zero, unadjusted): there the flag conveys less than the answer it implies, and all four generators run closed-book (64.1–74.1%) do better without the posts. CONFLICTS shows no significant change.

Genuine conflicts. Where both sides are valid, a directional flag and removal discard valid answers (H28; Appendix J). On WikiContradict the LLM directional flag lowers the share of answers giving both sides by 6.5 to 29.0 points; on RAMDocs' same-name questions removal costs 21.2 to 29.3.

Prevalence. A real detector also flags clean questions, so its net effect depends on how common conflicts are. For the gpt-4o-mini detector's directional flag, 17 of 31 settings have a break-even conflict share above which it helps (2% to 74%, median 22%); in 7 it helps and in 7 it hurts at every share (R18; Table 31), so no single conflict share decides.

6 Discussion and a Decision Rule

A neutral flag is not a weak directional flag: it says only that passages disagree and leaves the response to the answer policy. Whether abstaining or listing both sides helps depends on what a wrong answer costs (Wen et al., 2025). A directional flag is advice, and generators take it whether it is right or wrong, much as models defer to asserted views (Sharma et al., 2024). As in human–AI teams, advice pays only when its errors fall where the receiver is weaker (Bansal et al., 2021; Mozannar and Sontag, 2020): an accurate generator has more correct answers for a wrong flag to overturn. Why the most accurate of the four H31 generators defers most to wrong flags while Claude Sonnet rarely does, we cannot tell from their outputs, and a model's stated reasons need not reveal it (Chen et al., 2025).

A rule for choosing a detector, or none. We derived it from these runs; it has not been tested end to end (see Limitations), and its numbers come from the ConflictBank regime of §4.

1. *Scope.* Consider direction only against planted misinformation; where genuine conflicts are common (same-name entities, contested facts), use neither direction nor removal (H28), since no detector we tested tells them apart.

2. *Measure the generator.* Estimate f_R, f_W and p^* with a small randomised-direction run. Direction precision below p^* predicts a net loss on flagged true pairs; whether p^* should veto a detector is untested, and exceeding it is not enough. With Claude Haiku generating, the gpt-4o-mini detector (direction precision 68.8%, near $p^* = 0.67$) loses 4.1 points on conflict questions, because its right flags fall mostly where Haiku is already right (Table 25); p^* assumes the two err independently.

3. *Choose the detector with the account.* Apply Equation (1) to a labelled sample with clean questions, reweighted to the expected conflict share; flag only if the prediction is positive, and measure the clean-question effect directly, since the account overstates it. On the fresh cells this picked the Claude Haiku detector for every generator, which gave gpt-6.1-terra +5.3 points over all fresh questions, clean included, against −3.5 with the gpt-4o-mini detector (PR§19.1); on the RAMDocs hold-out it flags 2% of clean questions (C7). We wrote the rule after seeing this, so it is an illustration, not a test.

4. *Check prevalence.* If the flag helps on conflict questions ($K > 0$) but hurts on clean ones ($C < 0$), flag only above a conflict share of $-C/(K - C)$.

5. *Neutral flag.* Use it only under a one-answer prompt where a misinformed answer costs well more than a correct one gains (§5.1); under a multi-answer prompt it mostly adds misinformation (Table 22) unless both sides are valid (H8).

6. *Single call.* Naming suspects and answering in one call halves the calls (no detectable difference for gpt-4o-mini, H18), but costs the reasoning models and Claude Haiku 37.7 to 61.0 points on clean questions (Appendix C).

Limitations

ConflictBank's false side is legible without the contradiction. The randomised-direction test, the follow rates, p^*, the crossed cells and the prospective test all use ConflictBank, whose two on-topic passages are both LLM-written around a substituted object. A bag-of-words classifier trained on other ConflictBank facts picks the misinformation passage on every conflict question of our eval set and of ConflictBank-fresh, also with the answer strings masked, and its heaviest words are genre words (R1, Appendix F). Detectors also name the false side more often on facts their own model knows closed-book (R2), and the generators know few of these facts (closed-book accuracy 4.3–20.0% for the 6 generators run closed-book). A detector's direction and a generator's no-flag accuracy can therefore come from style rather than from knowledge or from reasoning about the contradiction. H31 assigns direction by coin flip, so its follow rates still measure deference to the flag; but the no-flag accuracies, the real detectors' direction precision, p^* and the account were all measured where the truth is legible from style and from the answers themselves: given only the two candidate answers, the detectors pick the true one about as often as they name the false passage, and when the answers are swapped between the passages the stronger detectors follow the answer more than the style (C1, Appendix G). Whether these numbers hold for plausible real misinformation is untested. ConflictBank's clean questions add only off-subject passages, so they test off-topic false flags only.

Scorer and human audit. Every claim in the body is scored by string matching. The matcher's misses are one-sided and depend on the condition: it misses more correct answers under directional flags than under answer keys (Appendix E). Under an outside-family LLM judge neither pooled key equivalence holds (H38), so H23 and H30 are unresolved. The follow rates, p^* and the prospective test have no full judge re-score, and gpt-6.1-terra's follow rates have none at all. No human has labelled any output: the blind disagreement sheet that could settle the scorer question and the paired audit sheets are released unlabelled (Appendix D). On a 120-question sample the outside-family judge makes Claude Haiku's ConflictBank neutral-flag effect $+6.7$ [$+0.8$, $+13.3$] (matcher: $+3.3$ [-3.3, $+10.0$]). Responses with no final-answer line are scored on their whole text, as registered; they exceed 1% in 51 of 979 run–condition cells (R9). In the body these are the four settings named under Figure 1 and gpt-6-luna's closed-book cell in Table 3, both marked as PR§20.3 asks.

Real documents and other benchmarks. HealthContradict's labels are as released, and an LLM judge confirms both pages of an item for only 37% of items (R13). Label noise could therefore explain why the account overstated harm there, and with one page per side its oracle flag nearly states the answer, though an explicit key still adds up to $+16.3$ points (C3). RAGuard's labels are behavioural: a judge finds only 6% of sampled supporting–misleading pairs to contradict, and all four generators run closed-book are more accurate without the posts. We treat both as label-transfer stress tests. CONFLICTS has few misinformation conflicts and no passage labels, so it can neither confirm nor refute our findings. RAMDocs' document labels are imperfect and its pair rule does not align entities (§4). Our conflict-question comparisons detect effects of about 5.8 points with 80% power; smaller effects, and small harms on clean questions, may be missed.

Registration. H1–H9 were registered before any confirmatory data. H10–H31 and H37–H38 were registered in addenda committed before their runs but after earlier results were known; three items were committed after runs had started (PR§20.1). H32–H36 froze numeric predictions before their outcomes existed. Commit times are local; push times to the public remote are the only third-party record (Table 34), and we used no registry. H40–H41 were registered in PR§21 before their runs, but about two thirds of H40's flagged answers had been sampled in earlier cells (PR§21.11). Each addendum followed a review of earlier results, so registration fixes how an analysis is run more than which analyses are chosen. The ± 3 TOST margin was set after the confirmatory run, and the pooled TOSTs were added after the per-comparison ones. The account's form was chosen after the crossed cells were seen; only its frozen predictions (H32–H36) are confirmatory. p^*, prevalence and every PR§20 re-analysis are exploratory. H24 is mis-specified: generator accuracy enters both sides of its tests, so its support is mechanical. Holm controls error within each family, not across the programme: of 549 registered

tests, 288 are significant under Holm and 316 under a Benjamini–Hochberg FDR over all tests.

Models and runs. Before PR§21 we recorded the model names we called, not the served snapshots. gpt-6.1-terra, Claude Sonnet and Qwen2.5-7B ran one seed, as did gpt-4.1 on ConflictBank and in its second RAMDocs run (H13, H14). gpt-6.1-terra's follow rates and p^* rest on one default-temperature draw; its other no-flag draw moves p^* from 0.79 to 0.81 (R7). The reasoning and Claude models ran only at default temperature, Claude through an OpenAI-compatible endpoint without JSON mode. We draw no conclusion about model families: two Claude models and one 4-bit 7B open model cannot support one, and family is confounded with model age and capability.

Detectors and scope. Detector claims cover the detectors we tested: one LLM-detector prompt run on several models, CAAG, an NLI cross-encoder and synthetic detectors. We test no trained detector such as ConflictRAG's (Wang et al., 2026). Under its registered window NLI flags no ConflictBank pair; the 300-character window that does was chosen after seeing detection output (PR§17.6), and sentence-pair NLI flags every pair (C11). The published baselines are reimplementations (Appendix B). We study flags passed to the generator as text; learned interfaces such as soft prompts or credibility fine-tuning (Ye et al., 2026; Pan et al., 2024), and retrieval-side filtering, may trade off differently. A stated confidence is tested only with synthetic flags, where 55% lowers f_W for 3 of 4 generators (H41); real detectors' confidences are untested. We test one adversarial case, a detector that always names the wrong side, not an attacker who chooses the flagged pairs.

What would test these claims. Claims 2 and 3 need a benchmark whose false side is not legible from style or from the candidate answers and whose facts the generators know; ConflictBank offers neither, so we do not know how far the follow rates carry beyond it. The answer-key question needs human labels on the outputs where the two scorers disagree. The account has met only 2 predicted harms; more cells where a real detector is expected to hurt would show whether it predicts the case that matters most. The decision rule (§6), derived from these runs, has not been tested end to end; that needs a benchmark with on-topic clean questions at a realistic conflict share.

Ethics Statement

This work uses public benchmarks and commercial model APIs, with no human subjects. RAGuard's passages are public Reddit posts that can contain usernames; we use them as released, and our released RAGuard audit sheet replaces private users' handles with placeholders. The main risk is dual use: a false "likely misinformation" label, or removing a passage, can change a RAG system's answers. An inverted oracle that names the correct side as misinformation costs 7.0 points on RAMDocs and 8.6 on ConflictBank (multi-answer prompt) as a flag, and 21.8 on RAMDocs when the passage it names is removed. An adversary who can trigger a deployed detector could exploit this (Zou et al., 2025). We therefore report harm on clean questions alongside every gain, and argue that systems should measure a detector's document-level behaviour before acting on its flags.

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A Prompts

The prompts are verbatim from the code that produced the results. The examples use placeholder passages; in the runs every condition except removal shows the generator the same passages (checked against the run records).

Generation skeleton (default).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Minimal skeleton (no answer policy).

```
Answer the question using the passages below.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}
```

```
Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your answer; if there is more than
one, separate them with |>
```

Single-answer skeleton.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Closed book (single-answer version).

```
Answer the question from your own knowledge. The
question has exactly one correct answer.
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Instruction v1.

```
Some passages may contradict each other. Before
answering, check whether they do.
Ignore passages that appear to contain
misinformation, and if a disagreement reflects
genuinely different valid answers, report each one.
```

Instruction v2.

```
Some passages may contain false information. Before
answering, decide which passages are reliable by
checking them against each other.
Do not include an answer that is supported only by a
passage you judge to be unreliable.
```

Single call (inserted as the instruction).

```
Some passages may contain misinformation. First
decide which passages, if any, contain
misinformation, by checking them against each other
and against what you know, and write one line in
this format:
Suspect passages: <passage numbers, or none>
Then answer the question using only the other
passages.
```

Example: directional flag (passage 2 suspect).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 2]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [LIKELY MISINFORMATION; conflicts with
passage(s): 0]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction; passage
2 likely contains misinformation

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Example: one-wrong flag, single-answer skeleton.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [CONFLICT with passage(s): 2; one side
contains misinformation]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [CONFLICT with passage(s): 0; one side
contains misinformation]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction; exactly
one of the two contains misinformation (it is not
known which)

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Directional LLM detector.

```
You will see a question and {n} numbered passages
retrieved for it.
Some passages may contain misinformation: false
claims about the answer to the question.
Identify every pair of passages that CONTRADICT each
other about the answer to the question,
i.e. they cannot both be true about the same entity
at the same time. Passages that are about
different entities with the same name, give
complementary details, or describe different times
are NOT contradictions; note them as "ambiguity",
"complementary" or "temporal" instead.

For each contradiction, say which passage of the
pair is more likely the misinformation, judging
by agreement with the other passages and by
plausibility: its number, "both" if both are likely
false, or "unclear" if you cannot tell.

Question: {question}

{passages}

Return JSON only: {{"pairs": [{{"i": <int>, "j":
<int>, "relation":
"contradiction|ambiguity|complementary|temporal",
"misinformation": <int>|"both"|"unclear"}}]}}
Use the passage numbers shown. "misinformation" is
needed only for contradictions. Return {{"pairs":
```

```
[]}} if there are none.
```

LLM detector without direction.

```
You will see a question and {n} numbered passages
retrieved for it.
Identify every pair of passages that CONTRADICT each
other about the answer to the question,
i.e. they cannot both be true about the same entity
at the same time. Passages that are about
different entities with the same name, give
complementary details, or describe different times
are NOT contradictions; note them as "ambiguity",
"complementary" or "temporal" instead.

Question: {question}

{passages}

Return JSON only: {{"pairs": [{{"i": <int>, "j":
<int>, "relation": "contradiction|ambiguity|compleme
ntary|temporal"}}]}}
Use the passage numbers shown. Return {{"pairs":
[]}} if there are none.
```

Forced-choice skeleton (no "unknown").

```
Answer the question using the passages below. The
question has exactly one correct answer.
You must give your best answer; "unknown" is not
allowed.
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Pick instruction (with the neutral flag).

```
When flagged passages conflict, decide which side is
better supported by the other passages and by what
you know, and answer with that side only.
```

Example: neutral flag, single-answer skeleton.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 2]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [POTENTIAL CONFLICT with passage(s): 0]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: neutral flag without pairs.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [POTENTIAL CONFLICT]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
Some of the marked passages conflict with each
other.

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: one-wrong flag without pairs.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [IN CONFLICT; some marked passages contain
misinformation]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [IN CONFLICT; some marked passages contain
misinformation]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
Some of the marked passages contain misinformation
(it is not known which).

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: directional flag without pairs.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [LIKELY MISINFORMATION]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
Passages marked LIKELY MISINFORMATION probably
contain false information.
```

```
Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: answer key (also the verdict key, with the detector's answers).

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0:
<passage 0>

Passage 1:
<passage 1>

Passage 2:
<passage 2>

NOTE: The answer "<wrong answer>" is false.

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: hint (the detector model's own answer).

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0:
<passage 0>

Passage 1:
<passage 1>

NOTE: Another model that read these passages
answered: "<other model's answer>".

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: WikiContradict (default skeleton, neutral flag).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 1]:
<passage 0>

Passage 1 [POTENTIAL CONFLICT with passage(s): 0]:
<passage 1>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 1: possible contradiction

Question: <question>

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Paraphrase prompt (artefact probe, H29).

```
Paraphrase the passage below. Keep every fact, name,
number, date and place exactly as stated, including
any that seem wrong; do not add, remove or correct
information, and keep about the same length. Output
only the paraphrased passage.

Passage:
{passage}
```

LLM judge of the string matcher.

```
You are checking how a model's answer was scored.
Read the question, the correct answer(s), the wrong
(misinformation) answer(s), and the model's full
response. Judge only the model's final answer.

Question: {question}
Correct answer(s): {gold}
Wrong answer(s): {wrong}

Model response:
<<<
{response}
>>>

Reply with JSON only: {{"gives_all_correct":
true/false, "gives_wrong": true/false, "abstains":
true/false}}
- gives_all_correct: the final answer states every
correct answer (paraphrases and aliases count).
- gives_wrong: the final answer states any wrong
answer as an answer (mentioning it only to reject it
does not count).
- abstains: the final answer is "unknown" or
declines to answer.
```

B Additional Results

Tables 4 and 5 give the status of each claim the body makes. The hypothesis tables, which list every registered test, are in Appendix J.

Published baselines (H14). Astute RAG (Wang et al., 2025a) and TrustRAG (Zhou et al., 2025) lose same-name answers on RAMDocs (H14). On ConflictBank, the registered H14 compared our pipeline's multi-answer prompt with the baselines' own one-answer prompts. Under the matched one-answer prompt (our principal comparison), our pipeline beats them by +9.2 and +13.4 points (exploratory). Our TrustRAG port first omitted the repository's system message; sending it (PR§17.2) changes TrustRAG's accuracy by −1.0 [−3.4, +1.3] points on ConflictBank, −0.5 [−2.8, +1.8] under majority misinformation and −1.0 [−3.0, +0.9] on RAMDocs (−0.4 [−2.0, +1.4] for its multi-answer variant). Against the corrected TrustRAG, with policies matched (Holm over the four), the LLM directional flag changes accuracy by +14.4 [+9.3, +19.4] on ConflictBank (one-answer), +23.4 [+18.5, +28.7] on RAMDocs (multi-answer), −0.5 [−5.5, +4.6] under majority misinformation, and +1.8 [−2.8, +6.5] under the registered mismatched prompts. For gpt-4o-mini, a single call that names suspect passages before answering does not differ significantly from the pipeline (H18; Table 18), and its ±3 TOSTs do not show equivalence either (90% CIs [−4.2, +1.6], [−3.6, +3.1] and [−4.1, +0.6]). On ConflictBank's clean questions it changes gpt-4o-mini's accuracy by −1.0 [−4.0, +1.7] points. For gpt-4.1 (one seed, outside H18, exploratory) it gives +7.3 [+2.7, +12.0] on conflict and −5.0 [−10.0, 0.0] on clean questions.

NLI on ConflictBank (post hoc window). On ConflictBank, NLI first saw passage prefixes that usually omit the changed fact. Cutting each passage to the sentences closest to the question, up to the registered 1,500 characters, keeps both answers, but the pair still exceeds the cross-encoder's 512 tokens and NLI flags no pair (pair recall 0.0%). With 300 characters, about one sentence, chosen after seeing detection output (a disclosed deviation, PR§17.6), NLI finds 21.0% of the gold pairs at 24.7% pair precision and flags 36.0% of clean questions. Its flag changes conflict accuracy by −0.2 [−1.9, +1.3] points (removal: +7.6 [+4.1, +11.2]). Every NLI row on ConflictBank is exploratory.

Robustness to the metric and the labels. Table 11 repeats Table 8 with the bidirectional metric, and Table 12 repeats the pre-registered tests treating every suspect document as misinformation.

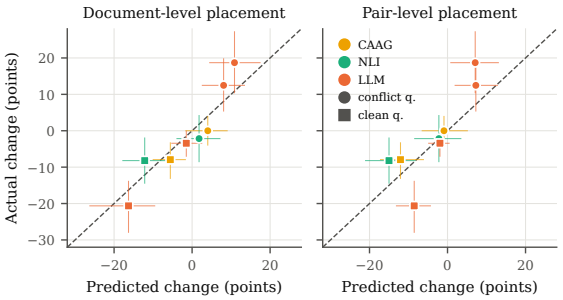


Figure 2: Effect of real detectors predicted from the synthetic grid (x, with the 95% bootstrap prediction interval) against the measured effect (y, with its 95% CI), for conflict (●) and clean (■) questions. Left: detectors located by document-level behaviour; right: by pair precision, recall and direction precision.

The heuristic detector. CAAG, the detector we set out to study, scores a pair only if its MiniLM (Reimers and Gurevych, 2019) cosine similarity reaches $\theta_s = 0.65$. Its score is the sum of a negation and contrast asymmetry term, a named-entity mismatch term (differing dates, numbers, money or percentages) and a temporal-marker term, multiplied by the similarity; a pair is flagged at $\theta_c = 0.35$. Its per-pair flag rate on the confirmatory split (6.22%) sets the random detector's rate. As a neutral flag, CAAG did not beat random flags of the same volume, and as a removal rule it only deleted correct passages (−7.9 points on clean questions, 0.0 on conflict ones). Its false flags mostly hit pairs that should both be reported: 134 of 221 on the hold-out are same-name ambiguity pairs, flagged by its entity-mismatch signal for differing dates or numbers. We read this as a warning about lexical contradiction cues generally (Gururangan et al., 2018), not a particular threshold. Retuning shows why F1 misleads: over a 121-setting sweep, CAAG's best pair F1 is 29.2, above NLI's, at $\theta_s = 0.40$ and $\theta_c = 0.10$, but it flags 43% of all pairs and 87% of clean questions, which our grid predicts is harmful under removal (Appendix I).

Original CAAG format. CAAG's flags in their original format, with typed lines and confidence scores ("factual conflict (confidence: 0.41)"), change accuracy by −0.9 points overall (−1.2 on

Claim	Evidence	Status
Claim 1: a neutral flag shifts the answer policy (§5.1)		
One-answer prompt allowing "unknown": four of seven generators abstain more; gpt-4o-mini's selective accuracy shows no detectable change; with more false than correct passages selective accuracy can rise	H15, H19, H21; Fig. 1; Tables 17, 30	(ii); H21 covers gpt-4o-mini and gpt-6-luna (gpt-4.1, Qwen2.5-7B: expl.); the selective ratio replaced the registered mean after the data were seen (PR§15.4.5); R16 expl.
Prompt asking for every valid answer: it adds the misinformation and lowers accuracy	H2, H9, H11, H12; Table 17	H2, H9: (i); H11, H12: (ii); misinformation marks expl.
Minimal skeleton: gpt-4o-mini and Qwen2.5-7B abstain more, gpt-4.1 adds the misinformation	H10; R17	(ii); R17 expl.
Exceptions: Claude Haiku, Claude Sonnet and gpt-6.1-terra barely move on the one-answer prompt; Haiku's misinformation share falls on RAMDocs	H19–H21, H25; Table 17	(ii); the RAMDocs Haiku mark is expl. and lost one-way (R12)
Where both sides are valid (WikiContradict), the same compliance is rewarded	H8	(i)
Utility at a misinformation cost c	Table 22	expl. (PR§15.4, PR§16.4)
Pooled summaries over the 34 settings	Table 21	expl.; these settings only; no claim rests on them
Claim 2: generators defer to directional flags (§5.2)		
An oracle directional flag beats the neutral and one-wrong flags on RAMDocs and ConflictBank, and no flag except for Qwen2.5-7B and minimal-prompt gpt-4o-mini	H3, H9–H12, H15, H16, H19, H20, H25	H3, H9: (i); the others: (ii)
Randomised direction: for three of four generators a wrong flag costs more than a right one gains	H31a–c; Table 3	(ii) (PR§16)
Three generators follow wrong flags about as often as right ones; gpt-6.1-terra follows them more often	Table 3; R4, R7, R8	registered descriptives without a test (PR§16.3); R4, R7, R8 expl.
On new questions randomised flags again cost more than they gain; follow rates stay in their eval intervals except gpt-6.1-terra's for wrong flags	H40a, H40b; Table 33	(ii) (PR§21); most flagged answers shared with earlier cells (PR§21.11)
Claude Sonnet, the most accurate generator on these questions, rarely follows wrong flags	C6; Table 33	expl. (PR§21, one seed)
A stated 55% confidence lowers wrong-flag following for some generators; right-flag effects do not change	H41; Table 33	(ii) (PR§21); synthetic confidences
The break-even direction precision p^* rises with accuracy across the four H31 generators (not for Claude Sonnet)	Tables 3, 33	expl.; not registered
Scope: ConflictBank's false side is readable from style and from the two candidate answers alone; detectors do better on facts their model knows, also on new questions; with more false than correct passages, direction falls	R1–R3; C1, C9; Tables 28, 32; PR§11.3	R1–R3 expl. (R3 first tabulated on 2026-11-02); C1, C9 expl. (PR§21); PR§11.3: registered as exploratory

Table 4: Status of each claim the body makes: claims 1 and 2 (continued in Table 5). *(i)*: registered before any confirmatory data (H1–H9). *(ii)*: registered in an addendum committed before the runs it covers but after earlier results were known (H10–H31, H37–H38, H40–H41; three items were committed after their runs had started, PR§20.1). *(iii)*: a numeric prediction frozen before the outcome existed (H32–H36). *Expl.*: exploratory: a re-analysis of data already seen, registered or not, including every R-number (PR§20; Appendix F), or a descriptive PR§21 run (C-numbers; Appendix G). Commit, push and first-run times of every registration are in Table 34.

Claim	Evidence	Status
Claim 3: a frozen account predicts within a benchmark (§5.3)		
Frozen, the account (Equation (1)) predicts effects on unseen ConflictBank questions, within generators too; gpt-4o-mini's follow rates recur on fresh questions	H32, H34, H36; Table 26	(iii) (PR§19); the account was defined after the crossed cells were seen
It does not detectably beat a cell-level marginal or a common-rate baseline: most of its power is cell-level bookkeeping	R6	expl.
Only one of the two predicted harms is significant	H33	(iii); not supported
It overstates harm on clean questions and on HealthContradict	H35, H37; Tables 26, 27	H35: (iii); H37: (ii), labels as released
Frozen predictions for three more randomised runs on fresh questions are within 3 points on average	H40c; Table 33	(ii); not prospective: most answers shared with earlier cells (PR§21.11)
Conditioning follow rates on the generator's prior barely changes the account on ConflictBank	R4, R5; Table 28	expl.
What does not hold, or is not settled (§5.4)		
Key equivalence is unresolved: under the judge the flag is ahead (H38); under the matcher only 2 of 9 registered H23 TOSTs and the unregistered pooled TOSTs pass	H23, H30, H38; R11	(ii); pooled TOSTs added after the per-comparison ones; H38 re-scores outputs already seen
Stress tests (label transfer): HealthContradict's oracle flag nearly states the answer, yet an explicit key adds to it for two generators; on RAGuard the key beats the flag except for gpt-4o-mini; CONFLICTS is underpowered	H26, H37; R13, R14; C3; Table 14	H26, H37: (ii); R13 (first tabulated on 2026-11-02), R14, C3, CONFLICTS: expl.
On genuine conflicts, direction and removal discard valid answers	H28	(ii)
In 17 of 31 settings a real detector's flag pays only above a break-even conflict share	Table 31; R18	expl. (PR§17.1, PR§20)
Decision rule (§6)		
Choosing a detector by the frozen account, checked over all fresh questions (clean included); the detector it picks flags few clean RAMDocs questions	PR§19.1; C7; Table 26	registered, descriptive; in sample; C7 expl. (PR§21)
Single call: no significant difference from the pipeline for gpt-4o-mini; it costs the reasoning models and Claude Haiku on clean questions	H18; Table 18	(ii); its TOSTs show no equivalence; clean-question losses expl.
The rule itself	–	derived from these runs; not tested end to end
Results the body does not use		
The account fits the crossed cells retrospectively	Table 19	expl., retrospective
Detector hit rate is equivalent for one of four reorder and paraphrase probes	H29	(ii); tests position and style, not content
The detector–generator reliability gap correlates with the gain	H24; Table 20	(ii) (PR§13; 5 of 30 cells existed); supported (mechanical)

Table 5: Status of each claim the body makes, continued from Table 4: claim 3, what does not hold, the decision rule, and three results the body does not use. Status codes as in Table 4.

Inverted oracle (names the correct side)	Conflict	Clean
RAMDocs, directional flag	−7.0 [−11.5, −2.9]	0.0
RAMDocs, removal	−21.8 [−28.8, −15.3]	0.0
ConflictBank (multi-answer), directional flag	−8.6	0.0

Table 6: Adversarial control, gpt-4o-mini: change from no flags, in points (exploratory; PR§9.5). On clean questions the inverted oracle flags nothing (prompt unchanged).

conflict questions, −1.6 on clean ones), no different from the uniform format.

Pilot study (exploratory). Before any confirmatory run, the 200-question pilot split, scored with the bidirectional metric, showed the same pattern: oracle neutral flags changed conflict-question accuracy by −10.5 points, directional flags by +11.8, and removing the suspect by +28.4. The pilots also showed that the original REMOVE strategy, which deletes an arbitrary side of each pair, removes correct documents, and led us to add directional formats, synthetic direction precision and document-level placement. All pilot runs are released.

Phase-0 audit of CAAG on NarrativeQA. CAAG was designed for long-document QA. On 100 NarrativeQA (Kočiský et al., 2018) questions from 50 stories (BM25 top-10 chunks from a single coherent story), it flags at least one pair in 54% of questions, 166 pairs in all. A single story rarely contradicts itself, so most flags are likely false. The flagged pairs are released for annotation.

			Accuracy		Oracle pairs				LLM detector		Single	LLM dir.
Generator	Prompt	Seeds	Closed	Plain	Flag	One wrong	Dir. flag	Remove	Flag	Dir. flag	call	clean q.
gpt-4o-mini	single	3	10.0	51.1	**−18.7**	**−35.7**	**+30.1**	**+35.0**	**−18.2**	+8.4	+7.6	−4.7
gpt-4o-mini	multi	3	–	21.6	**−8.8**	–	**+27.9**	**+66.4**	**−8.4**	**+20.0**	–	−1.0
gpt-4o-mini	minimal	3	–	35.0	**−18.1**	–	**+38.4**	**+51.2**	–	–	–	–
gpt-4.1	single	1	20.0	64.7	**−8.0**	**−35.3**	**+26.7**	**+26.0**	–	−2.3	+7.3	−3.0
gpt-4.1	multi	1	–	13.0	**−6.3**	–	**+23.0**	**+78.3**	**−6.7**	**+17.0**	–	0.0
Qwen2.5-7B	single	1	4.3	50.3	−3.3	−5.0	**+25.3**	**+35.7**	−4.0	+7.3	–	−1.0
Qwen2.5-7B	multi	1	–	23.7	**−8.3**	–	**+17.7**	**+63.7**	−8.3	+6.7	–	−1.0
gpt-6-luna	single	3	16.9	53.6	**−9.4**	**−29.1**	**+36.4**	**+37.0**	**−9.3**	+8.6	−6.3	−1.3
gpt-6.1-terra	single	1	20.0	73.0	−2.3	−5.3	**+17.0**	**+18.3**	−2.7	−11.7	**−21.0**	0.0
Claude Haiku 5.5	single	3	11.8	66.8	+0.7	+6.6	**+23.3**	**+22.6**	+0.7	−4.1	+1.7	−5.0
Claude Sonnet 5.5	single	1	–	75.7	+3.0	+4.0	**+16.3**	–	–	−2.7	–	−1.0
gpt-4o-mini	forced	3	–	55.4	−0.2	**+7.6**	**+24.2**	–	–	+4.0	–	−2.7
gpt-6-luna	forced	3	–	64.6	+1.6	**+10.3**	**+26.4**	–	–	−2.4	–	−1.3
Claude Haiku 5.5	forced	3	–	69.3	**+3.9**	**+5.1**	**+21.1**	–	–	−4.7	–	−1.3
gpt-6.1-terra	forced	1	–	77.7	−0.3	**+6.7**	**+14.7**	–	–	**−14.3**	–	0.0
Majority misinformation: three wrong-side passages, one correct												
gpt-4o-mini	single	3	–	21.6	+1.1	−5.0	**+32.7**	**+65.1**	−2.2	**+5.7**	+6.8	−4.7
gpt-6-luna	single	3	–	29.8	**+6.6**	+0.3	**+60.4**	–	–	**+7.1**	–	−1.3
gpt-6.1-terra	single	1	–	26.0	−3.7	**+22.0**	**+66.3**	–	–	**+9.0**	–	0.0
Claude Haiku 5.5	single	3	–	31.0	+4.3	–	**+57.3**	–	–	**+6.8**	–	−5.0
RAGuard: real Reddit posts labelled supporting or misleading (294 conflict claims)												
gpt-4o-mini	single	3	64.9	36.7	+3.2	−5.7	**+9.9**	+5.9	–	+0.6	–	+1.7
gpt-6-luna	single	3	64.1	23.2	0.0	**−7.0**	**+8.8**	−1.0	–	+3.1	–	−1.3
Claude Haiku 5.5	single	3	65.1	24.5	+0.3	**+5.9**	**+12.7**	+2.0	–	**+5.0**	–	+1.3
gpt-6.1-terra	single	1	74.1	24.8	+2.4	+0.7	**+6.8**	−1.0	–	+2.0	–	+1.0
Claude Sonnet 5.5	single	1	–	27.2	+4.4	–	**+20.4**	–	–	**+7.1**	–	+10.0

Table 7: ConflictBank conflict questions (300; one correct answer each) and RAGuard conflict claims (294; answer true or false): accuracy (%, one-way strict) closed-book and with the passages but no flags, then the paired difference in points made by each format. *forced*: the one-answer prompt without the option to answer "unknown". *Prompt*: the answer policy: *single* says there is one answer, *multi* asks for every valid answer (our default skeleton), *minimal* states no policy. *One wrong*: the neutral flag plus "exactly one of the two contains misinformation". *Single call*: one call that names the suspect passages, then answers. Last column: the LLM directional flag's change on the 100 clean questions. The LLM detector is gpt-4o-mini for every generator. **Bold**: significant under the paper's single scheme.

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	29.9	31.3	22.1	41.5	911
Prompt only					
Instruction (v1)	+1.6	+0.8	−1.2	+2.9	948
Instruction (v2)	+3.1	+2.3	+5.3	−1.1	951
Oracle detector					
Oracle, flag	−3.1	−3.5	**−6.7**	0.0	980
Oracle, directional flag	+6.2	+7.0	**+13.4**	0.0	998
Oracle, remove suspect	**+12.8**	**+14.5**	**+27.6**	0.0	825
Real detectors (exploratory)					
Random, flag	+1.0	+1.0	+0.7	+1.3	949
CAAG, flag	−0.4	−0.8	−0.7	−0.8	942
CAAG, remove	−3.3	−3.8	0.0	−7.9	809
NLI, flag	+0.2	+0.3	−0.5	+1.1	1021
NLI, remove	−5.7	−5.0	−2.2	−8.2	592
LLM, flag	−1.8	−1.3	−4.1	+1.9	970
LLM, directional flag	+5.0	+4.9	+12.5	−3.4	983
LLM, remove suspect	+0.3	0.0	+18.7	**−20.6**	718
Reference					
MADAM-RAG	+5.7	+5.7	+3.6	+7.9	8196

Table 8: RAMDocs hold-out, gpt-4o-mini, 3 seeds, one-way strict accuracy (primary). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

		Plain	Oracle, conflict q.			LLM, directional flag		LLM, remove suspect	
Generator	Skeleton	conflict	flag	dir. flag	remove	conflict	clean	conflict	clean
gpt-4o-mini	default	22.1	**−6.7**	**+13.4**	**+27.6**	+12.5	−3.4	+18.7	**−20.6**
gpt-4o-mini	minimal	20.4	**−5.3**	+5.0	**+18.9**	+3.6	−10.3	+11.0	**−19.6**
gpt-4.1	default	21.8	**−8.6**	**+18.2**	**+30.2**	+8.4	0.0	**+20.6**	**−24.3**
gpt-4.1	minimal	33.6	**−11.0**	**+12.7**	**+14.9**	–	–	–	–
Qwen2.5-7B	default	19.4	−5.8	+3.6	**+18.0**	+2.9	−7.1	+7.9	**−18.3**
Qwen2.5-7B	minimal	17.3	−5.8	+2.9	**+10.8**	–	–	–	–
gpt-6-luna‡	default	14.4	**−3.1**	**+39.1**	**+38.8**	**+29.7**	**−11.6**	–	–
gpt-6.1-terra‡	default	10.8	−0.7	**+39.6**	**+39.6**	**+38.8**	−0.8	–	–

Table 9: The main effects across generators and prompt skeletons (RAMDocs hold-out, 3 seeds, one-way strict accuracy). *Plain*: accuracy (%) on conflict questions with no flags. Other columns: paired difference from plain in points. *Default* skeleton tells the model to give every valid answer; *minimal* drops that policy (Appendix A). **Bold**: significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). The LLM detector is gpt-4o-mini for every generator. Qwen2.5-7B-Instruct runs locally (4-bit), seed 0. ‡: current reasoning models at their default temperature (gpt-6-luna 3 seeds, gpt-6.1-terra seed 0).

	Pair level			Clean-q.	Direc-	Removal, per question		Latency	Cost
Detector	P	R	F1	flagged	tion	misinfo gone	correct lost	(s)	($/1k q.)
CAAG (heuristic)	21.7	7.9	11.6	34.1	–	9.4	31.7	0.23	0.000
NLI cross-encoder	20.1	29.3	23.8	61.9	–	44.6	52.4	1.18	0.000
LLM (gpt-4o-mini)	37.2	20.5	26.4	76.2	88.3	61.2	65.9	0.80	0.181

Table 10: Detectors on the RAMDocs hold-out split (300 questions; percentages). *Clean-q. flagged*: share of conflict-free questions with at least one flag. *Direction*: share of correctly flagged contradictions whose suspect side is the misinformation. *Misinfo gone*: conflict questions whose misinformation documents are all removed when removing the suspect (or, without direction, the second passage). *Correct lost*: clean questions that lose at least one correct document. Latency is per question; CAAG and NLI run locally.

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	35.2	37.2	25.9	49.7	911
Prompt only					
Instruction (v1)	+1.3	+0.5	−1.0	+2.1	948
Instruction (v2)	+2.2	+1.4	+3.6	−1.1	951
Oracle detector					
Oracle, flag	−3.6	−4.0	**−7.7**	0.0	980
Oracle, directional flag	+7.0	**+7.9**	**+15.1**	0.0	998
Oracle, remove suspect	**+15.4**	**+17.5**	**+33.3**	0.0	825
Real detectors (exploratory)					
Random, flag	+0.4	+0.4	−0.2	+1.1	949
CAAG, flag	−0.9	−1.4	−1.9	−0.8	942
CAAG, remove	−4.0	−4.5	−1.0	−8.5	809
NLI, flag	+0.2	0.0	−1.4	+1.6	1021
NLI, remove	−8.0	−7.9	−3.1	−13.2	592
LLM, flag	−3.4	−3.4	−7.2	+0.8	970
LLM, directional flag	+4.7	+4.5	+14.1	−6.1	983
LLM, remove suspect	−2.1	−2.6	+18.9	**−26.5**	718
Reference					
MADAM-RAG	+5.7	+5.2	+3.6	+6.9	8196

Table 11: RAMDocs hold-out, gpt-4o-mini, 3 seeds, bidirectional strict accuracy (robustness). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

	Test (labels: *contains*)	One-way [95% CI]	p_{Holm}	Bidir.	p_{Holm}
H2	oracle+flag < plain (conflict)	−4.6 [−8.1, −1.5]	0.006	−5.2	0.003
H3	oracle+flag_directional > plain (conflict)	+11.6 [+5.6, +17.3]	<0.001	+13.3	<0.001
H3	oracle+flag_directional > oracle+flag (conflict)	+16.2 [+10.8, +21.8]	<0.001	+18.5	<0.001
H3	oracle+remove_suspect > plain (conflict)	+25.8 [+19.5, +32.2]	<0.001	+30.3	<0.001
H4	harmful cells, clean q. (dir. flag; removal); ρ	0/9; 8/9; $\rho = -0.08$ [-0.56, -0.02]			

Table 12: Label robustness: pre-registered tests with every suspect document treated as misinformation (173 conflict and 127 clean hold-out questions; this rule assigns the 35 questions the main analysis excludes). The bootstrap distribution of the H4 correlation is skewed: the estimate lies near the top of its percentile interval, which excludes zero although a permutation test gives $p = 0.75$; the bias-corrected interval [-0.32, +0.01] includes zero, so the correlation is not read as support (Appendix F, PR§20, R19).

CAAG setting	θ_s	θ_c	P	R	F1	Clean FPR
Default	0.65	0.35	21.7	7.9	11.6	34.1
Best pair F1	0.40	0.10	20.7	49.6	29.2	87.3
Best question F1	0.55	0.15	22.2	28.8	25.1	60.3

Signals (defaults)	Pair F1	Flag rate	Clean FPR
all signals	11.6	6.2	34.1
negation only	0.6	0.4	4.0
entity only	8.6	4.3	23.0
temporal only	0.0	0.0	0.0

Table 13: CAAG on the RAMDocs hold-out (%). Top: thresholds from a sweep over 121 settings ($\theta_s \in [0.40, 0.90]$, $\theta_c \in [0.10, 0.60]$). Bottom: each signal alone.

Condition	All	Lab.	Conflict	Clean
No flags (plain)	32.7	34.5	21.8	48.4
Instruction (v2)	+5.8	+5.2	+12.2	−2.6
Oracle, flag	−4.0	−4.5	**−8.6**	0.0
Oracle, directional flag	**+8.4**	**+9.6**	**+18.2**	0.0
Oracle, remove suspect	**+14.0**	**+15.8**	**+30.2**	0.0
LLM, directional flag	+5.4	+4.4	+8.4	0.0
LLM, remove suspect	+1.0	−0.8	**+20.6**	**−24.3**
NLI, remove	−7.0	−7.3	+0.2	**−15.6**

Table 15: Replication with gpt-4.1 as generator (RAMDocs hold-out, 3 seeds, one-way strict accuracy; detector fixed to gpt-4o-mini). Same conventions as Table 8.

Condition	All	Conflict	No conflict
No flags (plain)	65.8	46.9	73.6
Instruction (v2)	+0.6	+3.9	−0.8
CAAG, flag	+1.1	+2.4	+0.6
NLI, flag	−0.3	−0.5	−0.2
LLM, flag	+2.7	+5.3	+1.6
LLM, directional flag	+0.3	+1.4	−0.2

Table 14: CONFLICTS, 237 automatically scorable questions (69 conflict), gpt-4o-mini, 3 seeds; exploratory. Contains-match accuracy (%) and differences in points. These comparisons use their own Holm family (this table), more lenient than the paper's single exploratory family; bold would mark Holm-adjusted $p < 0.05$ in it, and no difference reaches it, so none is significant under either. CAAG: flags 94.2% of conflict and 92.9% of clean questions; NLI: flags 52.2% of conflict and 43.5% of clean questions; LLM: flags 98.6% of conflict and 83.9% of clean questions.

Condition	gpt-4o-mini	gpt-4.1
No flags (plain)	14.0	29.0
Instruction (v1)	**+8.7**	–
Instruction (v2)	**−6.7**	–
Oracle, flag	**+6.5**	**+19.0**
CAAG, flag	+0.7	–
NLI, flag	+3.2	–
LLM, flag	**+6.2**	–

Table 16: WikiContradict (253 genuine conflicts between two Wikipedia passages; the reference answer is both sides), 3 seeds (gpt-4.1: the oracle flag only). First row: share of answers giving both sides (%); other rows: difference in points, **bold** = significant under the paper's single scheme (H8 is a registered family; the gpt-4.1 column is exploratory).

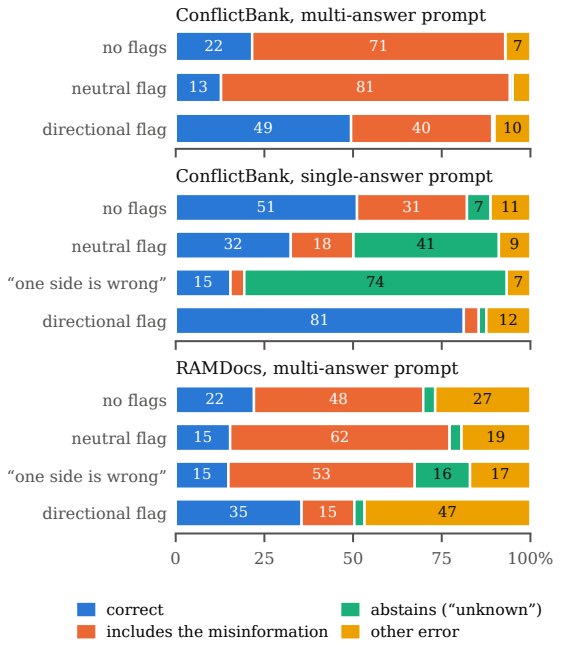


Figure 3: What a flag does to answers on conflict questions (gpt-4o-mini, oracle pairs; share of answers, %). The answer policy decides how a neutral flag fails: told to give every valid answer, a model adds the misinformation; told there is one answer, it abstains. Only the directional flag raises the share of correct answers. Categories are exclusive, correct (under the primary metric) taking precedence, so the misinformation share can be slightly lower than in Table 17, which counts every answer containing a wrong answer under the bidirectional matcher.

C Further Results

Answer mix by prompt (Table 17). Table 17 gives the answer mix behind §5.1. Its correct and selective-accuracy columns use the one-way matcher, and abstention needs no matcher. Its misinformation column counts answers containing the misinformation under the bidirectional matcher, and the bold marks in that column rest on the bidirectional value: from H21 in the ConflictBank one-answer and forced-choice panels, from the exploratory family elsewhere. R12 (Appendix F) gives the one-way version, under which one bold mark is lost (RAMDocs, Claude Haiku, oracle neutral flag). Table 30 and Figure 1 use the one-way matcher for every column.

H24 is supported only mechanically. The registered H24 tests relate a crossed cell's gain to the gap between the detector's and the generator's reliability. The generator's no-flag accuracy enters both sides of both tests (PR§13.3), so the correlation is expected whatever the detector does. We record H24's status as *supported (mechanical)*; the "supported" in the hypothesis tables should be read that way. Five of the 30 cells existed when H24 was registered. Table 20 re-analyses the cells within generators (exploratory).

Pooled summaries describe these 34 settings only (Table 21). The body reports no pooled estimate. Table 21 keeps the summaries that PR§13.4, PR§16.4 and PR§17.1.1 registered, each after the per-setting effects were seen. They describe these 34 settings, which over-represent ConflictBank prompt variants sharing one question pool. They are not estimates for any wider set of generators, benchmarks or prompts, and no claim rests on them. The multilevel mean is -1.4 [-5.1, $+2.3$] (prediction interval [-12.8, $+10.0$]; [-2.2, -0.6] conditional on these settings), and the equal-weight mean of PR§17.1.1 is -3.0 [-3.8, -2.2]. The moderators were coded after the per-setting effects were known (PR§16.4.1). With seven generators and four benchmarks the variance components are poorly identified, and the RAGuard and Qwen moderators each coincide with one level of a random effect.

C.1 Where the directional flag fails (exploratory)

After seeing the minimal-skeleton run, we split conflict questions by gold-answer count. 105 of 139 have several (same-name ambiguity *and* misinformation), and a strict answer must list all. On the 34 single-answer conflict questions, the directional flag's estimate is positive in every setting, from $+11.8$ points (Qwen) to $+46.1$ (gpt-4o-mini), both with the default skeleton. On multi-answer questions it changes little except with gpt-4.1 (gpt-4o-mini $+2.9$, and -5.7 with the minimal skeleton; gpt-4.1 $+19.0$). One untested reading: a warning about one passage makes the smaller models more conservative about legitimate alternatives too. Removal, which needs no warning to be acted on, gains in both groups.

C.2 Real detectors

CAAG flags 34.1% of clean questions. Its neutral flags do not beat random flags at the same rate (H5; -0.4 vs. $+1.0$ points overall). The 90% CI of the difference, [-3.1, $+0.1$], rules out a gain above $+0.1$ points but does not establish equivalence within ± 3 points (TOST). The addendum

	No flags				Oracle, neutral flag				Oracle, directional flag			
Generator	Correct	Misinfo.	Abstain	Sel.	Correct	Misinfo.	Abstain	Sel.	Correct	Misinfo.	Abstain	Sel.
RAMDocs, multi-answer prompt												
gpt-4o-mini	22.1	47.7	3.4	22.8	15.3	**61.9**	5.0	16.2	35.5	**14.9**	2.9	**36.5**
gpt-6-luna	14.4	72.2	1.2	14.6	11.3	76.7	0.7	11.4	53.5	**11.3**	1.4	**54.3**
Claude Haiku 5.5	25.2	55.2	2.2	25.7	26.4	**47.2**	2.2	27.0	53.5	**6.7**	2.9	**55.1**
gpt-6.1-terra	10.8	75.5	2.2	11.0	10.1	81.3	1.4	10.2	50.4	**13.7**	2.2	**51.5**
Claude Sonnet 5.5	45.3	24.5	2.2	46.3	45.3	21.6	2.2	46.3	59.7	**3.6**	1.4	**60.6**
ConflictBank, multi-answer prompt												
gpt-4o-mini	21.6	72.8	0.3	21.6	12.8	**83.2**	0.9	**12.9**	49.4	**41.2**	1.2	**50.1**
gpt-4.1	13.0	82.7	0.7	13.1	6.7	**90.7**	0.3	**6.7**	36.0	**58.3**	1.0	**36.4**
Qwen2.5-7B	23.7	67.0	6.0	25.2	15.3	**79.0**	5.0	**16.1**	41.3	**52.0**	13.0	**47.5**
ConflictBank, one-answer prompt												
gpt-4o-mini	51.1	31.0	6.7	54.8	32.4	**17.7**	**41.0**	55.0	81.2	**4.2**	**2.1**	**83.0**
gpt-6-luna	53.6	24.3	11.9	60.8	44.1	**17.4**	**28.9**	62.0	90.0	**0.0**	**0.8**	**90.7**
Claude Haiku 5.5	66.8	22.0	1.2	67.6	67.4	22.1	0.9	68.0	90.1	**0.2**	1.1	**91.1**
gpt-6.1-terra	73.0	18.0	1.7	74.2	70.7	15.7	4.7	74.1	90.0	**0.7**	0.3	**90.3**
Claude Sonnet 5.5	75.7	15.7	1.0	76.4	78.7	12.0	1.7	80.0	92.0	1.0	1.7	93.6
gpt-4.1	64.7	27.3	1.7	65.8	56.7	21.7	14.0	65.9	91.3	0.3	1.7	92.9
Qwen2.5-7B	50.3	35.3	5.0	53.0	47.0	28.7	15.0	55.3	75.7	13.0	1.3	76.7
ConflictBank, forced choice												
gpt-4o-mini	55.4	31.6	0.1	55.5	55.2	31.9	0.4	55.5	79.7	**6.7**	0.0	**79.7**
gpt-6-luna	64.6	25.8	0.0	64.6	66.1	22.0	0.0	66.1	91.0	**0.1**	0.0	**91.0**
Claude Haiku 5.5	69.3	20.9	0.0	69.3	73.2	**16.8**	0.0	**73.2**	90.4	**0.0**	0.0	**90.4**
gpt-6.1-terra	77.7	14.7	0.0	77.7	77.3	13.3	0.0	77.3	92.3	**0.3**	0.0	**92.3**

Table 17: Risk and coverage on conflict questions (%): strictly correct answers, answers that contain the misinformation, abstentions ("unknown"), and selective accuracy (correct answers over non-abstained answers, pooled over questions and seeds). **Bold**: the change from no flags is significant (H21 for ConflictBank one-answer and forced; the paper's exploratory family elsewhere). The answer policy decides what a neutral flag does: under a prompt that asks for every valid answer it adds the misinformation to the answer; under a one-answer prompt that allows "unknown" it moves answers, misinformed and correct, into abstention; under forced choice it has no abstention to trade.

registered the LLM detector's directional-versus-neutral comparison as exploratory. Under the bidirectional metric, the LLM detector's directional flag costs clean questions 6.1 points. With gpt-4.1 generating, its gain over all 300 questions (+5.4) is significant after Holm, but the 35 unlabelled questions contribute disproportionately, gaining 13.3 points. On labelled questions it is +4.4 [+0.6, +8.3]. On ConflictBank the NLI cross-encoder flags no pair under the registered window: the benchmark's passages are long, LLM-written and agree on all but the one fact, and the truncated pair rarely contains it.

The single call on clean questions. Asked in one call to name suspect passages and then answer, gpt-6-luna, gpt-6.1-terra and Claude Haiku name passages as suspect even on clean questions and often abstain. On ConflictBank's clean questions the single call changes their accuracy by −61.0 points [−69.7, −52.0] (gpt-6-luna), −51.0 [−61.0, −41.0] (gpt-6.1-terra) and −37.7 [−45.7, −30.0] (Claude Haiku). Table 7 gives the single call's effects on conflict questions; no table carries these clean-question losses. Used as two-call detectors on ConflictBank-fresh, the same three models flag no clean question, so their clean-question rows in Table 26 are zero. The loss therefore belongs to the single-call format. We ran these models in a single call only on ConflictBank, so whether they treat generated passages differently from real ones is untested.

Reasoning models on RAMDocs. Without flags, gpt-6-luna and gpt-6.1-terra are strictly correct on few RAMDocs conflict questions because the default prompt asks for every valid answer. They give 2.8 and 3.0 answers on average and recover 79% and 77% of gold answers, but 72% and 76% of their answers include the misinformation.

Detector parsing. The LLM detectors' JSON failed to parse on 1 of 2400 ConflictBank questions (a Claude Sonnet output), counted as flagging nothing.

Method	RAMDocs, gpt-4o-mini			RAMDocs, gpt-4.1		ConflictBank		CB majority		
	Lab.	Conflict	Clean	Lab.	Conflict	Conflict	Clean	Conflict	Clean	Calls
No flags (plain), accuracy	31.3	22.1	41.5	32.8†	19.4†	51.1	89.0	21.6	89.0	1
Instruction (v2) only	+2.3	+5.3	−1.1	–	+12.2	–	–	–	–	1
Oracle, flag + instr.	−1.8	−2.4	−1.1	+1.5†	+5.8†	–	–	–	–	1
Oracle, dir. flag + instr.	+6.4	+13.2	−1.1	**+14.7**†	**+30.9**†	–	–	–	–	1
MADAM-RAG	+5.7	+3.6	+7.9	–	–	–	–	–	–	9×tok.
Astute RAG	**−15.6**	−8.4	**−23.5**	–	–	−2.1	−9.7	+1.4	−9.7	2
Astute RAG (multi)	−5.9	−1.2	−11.1	+2.3†	+13.7†	–	–	–	–	2
TrustRAG	**−17.5**	−11.3	**−24.3**	–	–	−9.7	−4.0	+6.8	−4.0	3
TrustRAG (multi)	**−18.1**	−12.2	**−24.6**	−10.9†	0.0†	–	–	–	–	3
Single call (name suspects, then answer)	+6.2	+12.0	−0.3	–	–	+7.6	−1.0	+6.8	−1.0	1
LLM detector, flag + instr.	−2.0	−2.2	−1.9	–	–	–	–	–	–	1+1
LLM detector, dir. flag + instr.	−3.5	+8.4	**−16.7**	–	–	–	–	–	–	1+1
LLM detector, dir. flag	+4.9	+12.5	−3.4	+8.7†	+14.4†	+8.4	−4.7	**+5.7**	−4.7	1+1
LLM detector, remove suspect	0.0	+18.7	**−20.6**	–	**+20.6**	+7.9	−7.0	+7.0	−7.0	1+1

Table 18: Resolution instructions, published baselines and a single-call baseline. First row: accuracy (%, one-way strict) with no flags; other rows: paired difference from it in points. RAMDocs uses the multi-answer prompt (it has same-name questions with several answers); ConflictBank and its majority variant (three wrong-side passages, one correct) use the single-answer prompt and gpt-4o-mini. *Lab.*: labelled RAMDocs questions. 3 seeds, except †: gpt-4.1 seed 0 only, relative to the seed-0 no-flag accuracy in the first row (other gpt-4.1 cells are relative to its 3-seed no-flag accuracy, Table 15). *instr.*: the v2 reliability instruction. *(multi)*: the baseline's final step asks for every valid answer. *Single call*: one call that names suspect passages, then answers. Calls per question; *1+1*: one detector call and one generator call. **Bold**: significant under the paper's single scheme.

C.3 A second detector model on RAMDocs

Replacing the gpt-4o-mini detector with Qwen2.5-7B (gpt-4o-mini still generates) gives a much less selective detector, flagging 96.8% of clean RAMDocs questions, with direction precision 89.0% (as defined in Table 10). Its directional flag changes accuracy by +10.6 points on conflict questions and 0.0 on clean ones; its neutral flag, by −2.6. Removal again trades a conflict-question gain (+14.1) for a clean-question loss (−14.6). With this one detector on one benchmark, the pattern matches the gpt-4o-mini detector's; it cannot show whether the conclusions depend on the detector's model family.

C.4 CONFLICTS

On the 237 CONFLICTS questions with a reference answer (exploratory, 3 seeds), no condition moves accuracy significantly (Table 14). Only 69 of them have a conflict, most of these from outdated information and few from misinformation, and the benchmark has no passage labels or oracle pairs. It is underpowered for our question and can neither confirm nor refute our findings.

		Detector					Accuracy with				Flag minus	
Detector	Generator	Right	Wrong	Gen. acc.	Score	Gain	flag	verdict	hint	det. alone	hint	det. alone
gpt-4o-mini	gpt-6.1-terra	67.0	30.3	73.0	−4.7	**−11.7**	61.3	–	67.7	51.1	−6.3	**+10.2**
gpt-4o-mini	Claude Haiku	67.0	30.3	66.8	+1.9	−4.1	62.7	61.7	64.1	51.1	−1.4	**+11.6**
gpt-6.1-terra	gpt-6.1-terra	29.7	4.3	73.0	+2.7	+2.7	75.7	–	66.0	73.0	**+9.7**	+2.7
gpt-4o-mini	gpt-4.1	67.0	30.3	64.7	+4.0	−2.3	62.3	–	61.7	51.1	+0.7	**+11.2**
gpt-6.1-terra	Claude Haiku	29.7	4.3	66.8	+4.3	**+4.1**	70.9	68.2	59.9	73.0	**+11.0**	−2.1
gpt-6.1-terra	gpt-4.1	29.7	4.3	64.7	+4.7	**+5.0**	69.7	–	68.0	73.0	+1.7	−3.3
gpt-6-luna	gpt-6.1-terra	45.7	6.7	73.0	+5.3	+2.7	75.7	–	66.0	53.6	**+9.7**	**+22.1**
gpt-6-luna	Claude Haiku	45.7	6.7	66.8	+7.1	**+5.0**	71.8	69.7	61.4	53.6	**+10.3**	**+18.2**
gpt-6.1-terra	gpt-6-luna	29.7	4.3	53.6	+7.7	**+7.3**	60.9	60.6	66.7	73.0	−5.8	**−12.1**
gpt-6-luna	gpt-4.1	45.7	6.7	64.7	+7.7	**+6.3**	71.0	–	61.3	53.6	**+9.7**	**+17.4**
Claude Haiku	Claude Haiku	61.7	11.3	66.8	+9.8	**+5.7**	72.4	71.6	61.3	66.8	**+11.1**	**+5.7**
Claude Sonnet	Claude Haiku	65.0	9.7	66.8	+11.7	**+8.2**	75.0	73.8	60.4	75.7	**+14.6**	−0.7
gpt-6.1-terra	Qwen2.5-7B	31.0	4.0	50.3	+11.7	**+8.0**	58.3	55.0	62.3	73.0	−4.0	**−14.7**
gpt-6.1-terra	gpt-4o-mini	31.3	4.0	51.1	+12.8	**+9.6**	60.7	58.4	60.6	73.0	+0.1	**−12.3**
gpt-6-luna	gpt-6-luna	45.7	6.7	53.6	+13.8	**+11.1**	64.7	65.3	56.4	53.6	**+8.2**	**+11.1**
gpt-4o-mini	gpt-6-luna	67.0	30.3	53.6	+14.7	**+8.6**	62.1	61.9	56.1	51.1	+6.0	**+11.0**
gpt-4.1	gpt-6.1-terra	79.0	8.0	73.0	+15.7	**+7.3**	80.3	–	69.7	64.7	**+10.7**	**+15.7**
gpt-4o-mini	Qwen2.5-7B	67.0	30.3	50.3	+17.3	**+7.3**	57.7	54.7	53.3	51.1	+4.3	+6.6
gpt-4o-mini	gpt-4o-mini	67.0	30.3	51.1	+17.6	**+8.4**	59.6	57.1	52.4	51.1	+7.1	**+8.4**
gpt-6-luna	gpt-4o-mini	45.7	6.7	51.1	+19.1	**+13.7**	64.8	63.3	56.7	53.6	**+8.1**	**+11.2**
Claude Haiku	gpt-6-luna	61.7	11.3	53.6	+19.1	**+14.3**	67.9	68.7	63.0	66.8	+4.9	+1.1
gpt-6-luna	Qwen2.5-7B	48.0	6.7	50.3	+19.7	**+12.0**	62.3	59.7	58.0	53.6	+4.3	**+8.8**
gpt-4.1	Claude Haiku	79.0	8.0	66.8	+20.2	**+11.0**	77.8	76.0	63.7	64.7	**+14.1**	**+13.1**
gpt-4.1	gpt-4.1	79.0	8.0	64.7	+20.7	**+11.7**	76.3	–	63.3	64.7	**+13.0**	**+11.7**
Claude Haiku	gpt-4o-mini	61.7	11.3	51.1	+21.3	**+13.0**	64.1	62.7	57.9	66.8	+6.2	−2.7
Claude Sonnet	gpt-6-luna	65.0	9.7	53.6	+22.8	**+18.1**	71.7	72.9	67.8	75.7	+3.9	−4.0
Claude Sonnet	gpt-4o-mini	65.0	9.7	51.1	+28.1	**+15.9**	67.0	69.0	61.2	75.7	+5.8	**−8.7**
gpt-4.1	gpt-6-luna	79.0	8.0	53.6	+30.1	**+21.3**	74.9	76.8	61.2	64.7	**+13.7**	**+10.2**
gpt-4.1	gpt-4o-mini	79.0	8.0	51.1	+33.8	**+18.6**	69.7	67.6	57.9	64.7	**+11.8**	+5.0
gpt-4.1	Qwen2.5-7B	79.3	7.7	50.3	+36.7	**+16.0**	66.3	63.3	57.7	64.7	+8.7	+1.7

Table 19: Every detector–generator cell of H24 and H27 (ConflictBank, one-answer prompt, 300 conflict questions; %), sorted by the exploratory score. *Right*: the detector's hit rate, the share of questions on which it flags the true contradiction and names exactly its misinformation passage (computed on the detections that cell's generator saw; its direction precision with 'both' and 'unclear' verdicts counted as misses, unlike Table 1, i.e. the same share among questions where it flags the true pair, pooled over its cells, is gpt-4o-mini 68.8%, gpt-4.1 81.8%, gpt-6-luna 82.2%, gpt-6.1-terra 82.6%, Claude Haiku 78.1% and Claude Sonnet 77.7%); *Wrong*: share on which it marks a correct passage as suspect; *Gen. acc.*: the generator's accuracy with no flags; *Score*: the mean over the 300 questions of [right direction and the generator errs] minus [a correct passage marked and the generator is right]; *Gain*: the LLM directional flag's change in points. *Verdict*: the detector's judgement passed as an answer key ("the answer X is false" for the answer its suspect passage supports; H30). *Hint*: the generator given the detector model's own plain answer, with no passage marked (H27); *det. alone*: the detector model's own plain accuracy. A detector's statistics can differ slightly between generators because concurrent runs made independent detector calls (up to 29 of 400 questions differ for one detector). Bold: significant under the paper's scheme (Gain: exploratory family; hint and det. alone columns: H27, Holm across cells).

Cell level (30 cells): accuracy with the flag, A	
$A \sim D + G$: coefficient of D (hit rate)	0.09 [0.03, 0.16]
coefficient of G	0.46 [0.36, 0.55]
$A \sim D$ + generator FE, permutation of D	$p = 0.29$
$A \sim P + R$ + gen. FE: precision P	1.03 [0.76, 1.20]; $p = 0.047$
recall R	0.16 [0.05, 0.26]; $p = 0.040$
Leave one detector out: R^2 for the gain	
G only	0.17
gap $D - G$ (registered)	-0.21
$P + R + G$	0.12
score (exploratory)	0.62
Question level, question/detector/generator FE (points)	
detector right on q	$+34.4$ [$+30.7$, $+38.3$]
detector marks a correct passage	-31.3 [-36.7, -26.5]

	H24 cells (observational)			H31 (randomised)	
Generator	right	wrong	wrong, abst.	right	wrong
Qwen2.5-7B	61	68	6	–	–
Claude Haiku	57	59	18	63	64
gpt-4.1	69	65	13	–	–
gpt-4o-mini	66	52	24	57	51
gpt-6-luna	75	57	16	72	77
gpt-6.1-terra	68	75	4	67	95

Table 20: H24 re-analysed (PR§15.4, PR§16.4; data already seen). D: hit rate (the detector flags the gold pair and names exactly its misinformation passage); P: direction precision (of those gold-pair flags, the share naming it); R: recall (the share of questions whose gold pair is flagged); G: the generator's no-flag accuracy. CIs: 2,000 question resamples shared across cells (300 with question fixed effects; the leave-one-detector-out range of the question-level coefficients is $+32.7$ to $+37.1$ and -32.8 to -26.8). p: exact permutation of the detectors' values across the six detectors; the design is unbalanced (the Claude detectors serve three generators), so detectors are only approximately exchangeable. Lower table, one definition of following: a right flag is followed when the flagged answer is correct (questions the generator gets wrong in every plain seed); a wrong flag is followed when the flagged answer contains the misinformation (questions it gets right in every plain seed); *abst.*: the flagged answer abstains instead. With no flag, a held-out plain seed changes outcome on at most 12% of such questions.

Policy	Setting	Neutral flag	Policy	Setting	Neutral flag
forced	CB forced, gpt-6.1-terra	−0.3	multi	RAMDocs, Claude Haiku	+1.2
forced	CB forced, gpt-4o-mini	−0.2	one	CB one-answer, gpt-4o-mini	−18.7
forced	CB forced, gpt-6-luna	+1.6	one	CB one-answer, gpt-6-luna	−9.4
forced	CB forced, Claude Haiku	+3.9	one	CB one-answer, gpt-4.1	−8.0
minimal	CB minimal, gpt-4o-mini	−18.1	one	CB majority, gpt-6.1-terra	−3.7
minimal	RAMDocs, gpt-4.1 (minimal)	−11.0	one	CB one-answer, Qwen2.5-7B	−3.3
minimal	RAMDocs, Qwen2.5-7B (minimal)	−5.8	one	CB one-answer, gpt-6.1-terra	−2.3
minimal	RAMDocs, gpt-4o-mini (minimal)	−5.3	one	RAGuard, gpt-6-luna	0.0
multi	CB multi-answer, gpt-4o-mini	−8.8	one	RAGuard, Claude Haiku	+0.3
multi	RAMDocs, gpt-4.1 (multi-answer)	−8.6	one	CB one-answer, Claude Haiku	+0.7
multi	CB multi-answer, Qwen2.5-7B	−8.3	one	CB majority, gpt-4o-mini	+1.1
multi	RAMDocs, gpt-4o-mini (multi-answer)	−6.7	one	RAGuard, gpt-6.1-terra	+2.4
multi	CB multi-answer, gpt-4.1	−6.3	one	CB one-answer, Claude Sonnet	+3.0
multi	RAMDocs, Qwen2.5-7B (multi-answer)	−5.8	one	RAGuard, gpt-4o-mini	+3.2
multi	RAMDocs, gpt-6-luna	−3.1	one	CB majority, Claude Haiku	+4.3
multi	RAMDocs, gpt-6.1-terra	−0.7	one	RAGuard, Claude Sonnet	+4.4
multi	RAMDocs, Claude Sonnet	0.0	one	CB majority, gpt-6-luna	+6.6
Multilevel model (random effects for generator, benchmark and setting; REML)					
pooled, all settings		−1.4 [−5.1, +2.3]; prediction interval [−12.8, +10.0]			
pooled, multi-answer prompt (10 settings)		−3.5 [−6.4, −0.6]			
pooled, minimal prompt (4 settings)		−12.1 [−23.2, −1.0]			
pooled, one answer prompt (16 settings)		−0.4 [−5.3, +4.5]			
pooled, forced prompt (4 settings)		+1.5 [−1.7, +4.8]			
Meta-regression (same model)					
intercept (OpenAI, one answer, generated)		−2.1 [−7.3, +3.0]			
several answers allowed		−3.7 [−8.2, +0.8]			
forced (no unknown)		+6.0 [+1.7, +10.3]			
Anthropic		+4.6 [+1.5, +7.7]			
Qwen		+0.6 [−3.9, +5.1]			
natural misinformation (RAGuard)		+2.2 [−7.2, +11.6]			

Table 21: The 34 settings on misinformation benchmarks (RAMDocs, ConflictBank and its one-answer, minimal, forced-choice and majority variants, RAGuard) in which the oracle neutral flag was run; WikiContradict (genuine conflicts) and CONFLICTS are not pooled (conflict questions, change in points), grouped by answer policy. The pooled estimate was registered as exploratory (PR§13.4) and the multilevel model, the policy split and the moderators in PR§16.4; they use data already seen. The prediction interval describes this design space (our generators, benchmarks and prompts), not deployments in general. *Natural misinformation* is RAGuard only; RAMDocs' misinformation is synthetic. The moderator *several answers allowed* codes the multi-answer and minimal prompts together, against one-answer prompts; the policy rows above separate them. Settings share questions, so the model is fitted with their sampling covariance, estimated by a joint question bootstrap (PR§17.1: each replicate resamples each benchmark's questions once for every setting that uses them). Treating the settings as independent gives [−5.5, +2.2]; conditional on these settings (question sampling only) the pooled effect is [−2.2, −0.6].

	Neutral flag, utility change at misinformation cost c				Dir. flag	Neutral flag: plain answer → flagged answer (null)			
Setting	$c=0$	$c=1$	$c=2$	$c=4$	$c=2$	mis.→abst.	mis.→corr.	corr.→abst.	corr.→mis.
RAMDocs, gpt-4o-mini (multi-answer)	**−6.7**	**−20.9**	**−35.0**	**−63.3**	**+79.1**	1 (0)	2 (2)	0 (0)	29 (3)
RAMDocs, gpt-6-luna	**−3.1**	−7.7	−12.2	−21.3	**+160.9**	0 (0)	1 (3)	0 (0)	22 (16)
RAMDocs, Claude Haiku	+1.2	+9.1	**+17.0**	**+32.9**	**+125.2**	0 (0)	9 (4)	0 (0)	5 (9)
CB multi-answer, gpt-4o-mini	**−8.8**	**−19.2**	**−29.7**	**−50.6**	**+91.0**	0 (0)	4 (3)	1 (0)	55 (9)
CB one-answer, gpt-4o-mini	**−18.7**	−5.3	+8.0	**+34.7**	**+83.7**	45 (2)	7 (3)	37 (1)	5 (2)
CB one-answer, gpt-6-luna	**−9.4**	−2.6	+4.3	+18.1	**+85.1**	27 (10)	12 (13)	20 (6)	5 (6)
CB one-answer, gpt-6.1-terra	−2.3	0.0	+2.3	+7.0	**+51.7**	2 (–)	26 (–)	4 (–)	5 (–)
CB one-answer, Claude Haiku	+0.7	+0.6	+0.4	+0.2	**+66.9**	0 (0)	25 (16)	0 (0)	8 (5)
CB one-answer, Claude Sonnet	+3.0	+6.7	+10.3	+17.7	**+45.7**	2 (–)	34 (–)	0 (–)	3 (–)
CB forced, gpt-4o-mini	−0.2	−0.6	−0.9	−1.6	**+74.0**	1 (0)	19 (4)	0 (0)	11 (2)
CB majority, gpt-4o-mini	+1.1	**+24.6**	**+48.0**	**+94.9**	**+118.2**	26 (1)	11 (2)	22 (1)	8 (5)
RAGuard, gpt-4o-mini	+3.2	+6.6	+10.0	+16.8	**+34.6**	12 (3)	7 (3)	8 (6)	4 (2)
RAGuard, gpt-6-luna	0.0	−1.1	−2.3	−4.5	**+29.3**	10 (9)	6 (9)	8 (9)	7 (8)
RAGuard, Claude Haiku	+0.3	−1.8	−4.0	−8.3	**+26.5**	7 (6)	4 (4)	11 (9)	6 (3)

Table 22: Abstention-aware view of the oracle flags (conflict questions; PR§15.4 and PR§16.4, re-analyses of data already seen). Utility per answer: $+1$ correct, $-c$ if it contains the misinformation, 0 for abstaining or any other wrong answer; $c = 0$ is strict accuracy. Bold: significant in one Holm family per c over all 34 settings of Table 21 (a selection is shown). Right: of the question–seed pairs whose plain answer contained the misinformation (or was correct), the share whose flagged answer abstained, was correct or contained the misinformation; in brackets the same share between two plain runs with different seeds, the churn with no flag.

D Semantic Re-score, Label Audits and Selection

An outside-family judge leaves the directional conclusions intact (PR§17.3). To check whether the matcher's errors differ by condition, an outside-family LLM re-scored both arms of 10 contrasts on the same 120 conflict questions each (gemini-3.5-flash-lite; Table 23). PR§17.3 named Gemini 2.5 Flash, closed to new users. Given the gold and wrong answers, the judge tests the scorer, not the benchmark labels. Judge and matcher agree on 92.8% of 2400 outputs. Almost all disagreements are answers the judge accepts and the matcher rejects (aliases, paraphrases, partial forms). Their share depends on condition: 10.3% of answer-key outputs and 11.7% of oracle-directional ones, against 5.2% without a flag. The failure of most concern, a final answer such as "Y, not X" counted as asserting the wrong answer X, occurs in 0.2% of outputs. Since both arms gain, the judge moves every contrast by -1.7 to $+3.3$ points. One small contrast changes sign (RAMDocs oracle flag against answer key, -0.8 [-5.8, $+4.2$] to $+0.8$ [-8.3, $+10.0$]). Two whose matcher intervals include zero exclude it under the judge (unadjusted): Claude Haiku's neutral flag ($+6.7$ [$+0.8$, $+13.3$]) and gpt-4o-mini's LLM flag against its verdict key ($+3.3$ [-2.5, $+9.2$] to $+6.7$ [$+0.8$, $+12.5$], outside ± 3). With 120 questions per contrast the re-score's intervals are too wide to test the ± 3-point equivalence claims; re-scored in full, neither holds (Appendix E).

A within-family judge check. Before the outside-family re-score, gpt-6.1-terra judged 162 stratified conflict-question outputs and agreed with the primary metric on 92.6% ($\kappa = 0.85$). gpt-6.1-terra is itself one of the generators whose outputs it judged, so this check is within-family. The outside-family re-scores above and in Appendix E take its place.

Human audit status. No human has labelled any output yet; every sheet below is released unlabelled. The blind disagreement sheet `runs/labelling/audit11_disagreements.csv` (PR§20.3, A1) holds 300 conflict-question responses, 75 from each of the oracle directional flag, the answer key, the LLM directional flag and the verdict key: 50 on which the matcher and the outside-family judge disagree and 25 on which they agree. The rows are shuffled, and the condition, the run and both scores are hidden. The sheet can settle H23 and H30 only if the annotators side clearly with one scorer. The paired, blinded sheets `runs/labelling/audit8_*` (A2, A3) hold 150 of the PR§17.3 pairs and 50 RAGuard post pairs. The 200-output sheet of PR§15.4.7 holds no answer-key or verdict-key output and shows the condition and the matcher score, so PR§20.1 supersedes it.

The neutral-flag cost holds without cross-entity pairs (PR§18). RAMDocs' pair rule labels a pair a contradiction whenever one side is misinformation and the answers differ, without aligning entities. Assuming each misinformation passage targets one entity, at least 261 of the 709 oracle contradiction pairs on the hold-out join it to a correct passage about another entity, in the 98 questions whose correct passages give several answers. On the 41 single-entity questions no pair can cross entities. Averaged over the 10 RAMDocs settings, the oracle neutral flag changes accuracy by -4.4 [-7.3, -1.5] points there and -4.7 [-6.7, -2.7] on the rest; the oracle directional flag by $+29.2$ [$+21.0$, $+37.8$] and $+12.9$ [$+9.0$, $+17.1$] (Table 24; leaving one setting out, R15). Per question, the flag and its key disagree on 9.9% (H23; flag only 4.9%, key only 5.0%) and 9.4% (H30; 5.3% and 4.1%) of question–seed pairs. On the cells with two no-flag seeds, the seeds disagree on 6.7% and 9.5%, and flag and key on 11.7% and 8.3%.

The WikiContradict harm holds under its original rubric. The original rubric counts an answer correct when it gives both answers, acknowledges the conflict and does not prefer one without support (Hou et al., 2024, §3.4). Under it, the LLM directional flag changes the share of correct answers by -6.7 [-11.9, -1.6] points for gpt-4o-mini (16.2% to 9.5%), -31.6 [-37.9, -25.3] for gpt-6-luna and -28.1 [-34.0, -22.1] for Claude Haiku (H28; seed 0, all 253 questions). For gpt-4o-mini the flag raises the share that merely mentions both answers ($+5.1$ [0.0, $+10.3$]) while lowering the share acknowledging the conflict without an unsupported preference.

RAGuard pairs rarely contradict. RAGuard labels a post supporting or misleading by whether GPT-4 judged the claim correctly with it in context, so its "contradiction pairs" transfer behavioural labels. Of 150 sampled support-

Setting	Contrast	Matcher (points)	Judge (points)	Judge − matcher
RAMDocs, gpt-4o-mini	neutral flag vs plain	−4.2 [−9.2, 0.0]	−2.5 [−9.2, +4.2]	+1.7 [−3.3, +6.7]
ConflictBank, gpt-4o-mini	neutral flag vs plain	−20.0 [−27.5, −12.5]	−21.7 [−29.2, −14.2]	−1.7 [−4.2, 0.0]
ConflictBank, Claude Haiku	neutral flag vs plain	+3.3 [−3.3, +10.0]	+6.7 [+0.8, +13.3]	+3.3 [0.0, +7.5]
RAMDocs, gpt-4o-mini	oracle directional flag vs answer key	−0.8 [−5.8, +4.2]	+0.8 [−8.3, +10.0]	+1.7 [−6.7, +10.0]
ConflictBank, gpt-4o-mini	oracle directional flag vs answer key	+10.8 [+4.2, +17.5]	+12.5 [+6.7, +19.2]	+1.7 [−1.7, +5.0]
ConflictBank, gpt-6-luna	oracle directional flag vs answer key	−1.7 [−5.8, +2.5]	0.0 [−3.3, +3.3]	+1.7 [−1.7, +5.0]
ConflictBank, gpt-4o-mini	LLM directional flag vs verdict key	+3.3 [−2.5, +9.2]	+6.7 [+0.8, +12.5]	+3.3 [0.0, +7.5]
ConflictBank, gpt-6-luna	LLM directional flag vs verdict key	−0.8 [−3.3, +1.7]	0.0 [−2.5, +2.5]	+0.8 [−1.7, +4.2]
ConflictBank, gpt-4o-mini	randomised directional flag vs plain	+4.2 [−5.8, +15.0]	+6.7 [−3.3, +16.7]	+2.5 [0.0, +5.8]
ConflictBank, Claude Haiku	randomised directional flag vs plain	−7.5 [−18.3, +3.3]	−6.7 [−17.5, +4.2]	+0.8 [−1.7, +3.3]

Condition	Outputs	Matcher (%)	Judge (%)	Judge only (%)	Matcher only (%)
answer key	360	67.8	77.8	10.3	0.3
directional flag (LLM)	240	59.6	62.5	3.8	0.8
directional flag (oracle)	360	70.6	82.2	11.7	0.0
neutral flag (oracle)	360	36.9	43.3	6.9	0.6
no flag	600	48.8	53.3	5.2	0.7
randomised direction (H31)	240	54.6	59.6	5.0	0.0
verdict key (LLM)	240	58.3	59.2	2.1	1.2

Table 23: Semantic re-score (PR§17.3; exploratory). An outside-family judge (gemini-3.5-flash-lite, temperature 0) reads the question, the reference and known wrong answers and the response, and classifies the final answer; judge-strict means it gives the reference answer, asserts no wrong one and does not abstain. Top: each contrast on the same 120 conflict questions (seed 0) under the matcher and the judge, and their difference, with question-bootstrap 95% CIs. Bottom: per condition, the share scored correct by each and the shares where only one does. The judge is an LLM given the answer keys: it checks the matcher's semantics, not the benchmark labels.

ing × misleading pairs, the judge finds the two posts cannot both be right about the claim in 6%, the misleading post false about it in 21% (unclear in 57%) and the supporting post consistent with the verdict in 71%; all three hold for 5%. Most pairs discuss the claim from different angles without asserting opposite facts. RAGuard's oracle-flag rows therefore test what a generator does with these labels, not verified contradictions; we draw no factual-oracle conclusion from them. This audit uses one LLM judge; the released human sheet covers 50 of these pairs.

Selection in the crossed cells. Table 25 splits each crossed cell by the flag's per-question effect. The marginal break-even calculation, $p\Delta_R + (1 - p)\Delta_W$ at the cell's direction precision p and full coverage (written $pR + (1 - p)W$ in the table), has mean error 3.5 points against the observed gains (1.9 when scaled by recall); the follow-rate account, using each cell's coverage and no-flag accuracy, has 1.5 [1.4, 3.1].

Generator	Oracle flag	Single-entity	Multi-entity	Gap
Qwen2.5-7B	neutral	−12.2 [−24.4, 0.0]	−3.1 [−8.2, +2.0]	−9.1
Qwen2.5-7B	directional	+9.8 [+2.4, +19.5]	+1.0 [−4.1, +6.1]	+8.7
Qwen2.5-7B, minimal	neutral	−4.9 [−17.1, +7.3]	−6.1 [−11.2, −2.0]	+1.2
Qwen2.5-7B, minimal	directional	+17.1 [0.0, +34.1]	−3.1 [−8.2, +2.0]	+20.1
claude-haiku-5-5	neutral	+5.7 [−0.8, +13.8]	−0.7 [−5.4, +3.7]	+6.4
claude-haiku-5-5	directional	+43.9 [+29.3, +58.5]	+21.8 [+13.3, +30.6]	+22.1
claude-sonnet-5-5	neutral	−4.9 [−14.6, +4.9]	+2.0 [−3.1, +7.1]	−6.9
claude-sonnet-5-5	directional	+17.1 [+4.9, +29.3]	+13.3 [+5.1, +21.4]	+3.8
gpt-4.1	neutral	−5.7 [−13.0, 0.0]	−9.9 [−15.3, −4.4]	+4.2
gpt-4.1	directional	+13.0 [+4.1, +23.6]	+20.4 [+12.9, +28.6]	−7.4
gpt-4.1, minimal	neutral	−13.0 [−26.0, 0.0]	−10.2 [−17.7, −2.7]	−2.8
gpt-4.1, minimal	directional	+23.6 [+9.8, +37.4]	+8.2 [+0.3, +16.0]	+15.4
gpt-4o-mini	neutral	−3.3 [−9.8, +2.4]	−8.2 [−13.6, −3.7]	+4.9
gpt-4o-mini	directional	+38.2 [+24.4, +53.7]	+3.1 [−3.7, +9.9]	+35.2
gpt-4o-mini, minimal	neutral	−3.3 [−11.4, +4.9]	−6.1 [−11.2, −2.0]	+2.9
gpt-4o-mini, minimal	directional	+31.7 [+18.7, +45.5]	−6.1 [−12.6, −0.3]	+37.8
gpt-6-luna	neutral	−2.4 [−5.7, 0.0]	−3.4 [−6.1, −1.4]	+1.0
gpt-6-luna	directional	+51.2 [+36.6, +65.0]	+34.0 [+24.8, +43.2]	+17.2
gpt-6.1-terra	neutral	0.0 [−7.3, +7.3]	−1.0 [−3.1, 0.0]	+1.0
gpt-6.1-terra	directional	+46.3 [+31.7, +61.0]	+36.7 [+26.5, +45.9]	+9.6

Table 24: RAMDocs hold-out, oracle flag minus no flag on conflict questions, in points, split by whether the correct passages give one answer (41 *single-entity* questions, where no oracle pair can join passages about different entities) or several (98), with question-bootstrap 95% CIs (PR§18.2; exploratory).

				Flag names the false side			Flag marks a correct passage			Rest	Gain (points)		
Generator	Detector	Prec.	Rec.	Share	Acc.	Gain	Share	Acc.	Gain	contrib.	Marginal	Question	Observed
gpt-4o-mini	Claude Haiku	78	79	62	58	+28.8	11	42	−30.4	−1.3	+16.1	+12.3	+13.0
gpt-4o-mini	Claude Sonnet	78	84	65	51	+32.8	10	40	−20.7	−3.4	+15.8	+16.1	+15.9
gpt-4o-mini	gpt-4.1	82	97	79	55	+28.6	8	25	−22.2	−2.2	+18.3	+19.2	+18.6
gpt-4o-mini	gpt-4o-mini	69	97	67	60	+23.1	30	30	−23.1	0.0	+10.5	+10.4	+8.4
gpt-4o-mini	gpt-6-luna	82	56	46	50	+36.7	7	57	−46.7	0.0	+18.5	+11.0	+13.7
gpt-4o-mini	gpt-6.1-terra	83	38	31	51	+36.9	4	61	−36.1	−0.6	+19.2	+7.4	+9.6
gpt-6-luna	Claude Haiku	78	79	62	61	+31.5	11	45	−34.3	−1.2	+16.3	+13.5	+14.3
gpt-6-luna	Claude Sonnet	78	84	65	59	+34.5	10	41	−35.6	−0.9	+16.0	+16.2	+18.1
gpt-6-luna	gpt-4.1	82	97	79	60	+31.1	8	21	−20.8	−1.6	+19.2	+21.6	+21.3
gpt-6-luna	gpt-4o-mini	69	97	67	63	+27.0	30	33	−31.5	0.0	+9.1	+10.0	+8.6
gpt-6-luna	gpt-6-luna	82	56	46	64	+29.2	7	40	−35.0	+0.1	+19.4	+9.8	+11.1
gpt-6-luna	gpt-6.1-terra	82	36	30	66	+30.3	4	56	−38.5	0.0	+19.7	+5.4	+7.3
Claude Haiku	Claude Haiku	78	79	62	79	+14.8	11	28	−22.5	−0.9	+8.0	+6.1	+5.7
Claude Haiku	Claude Sonnet	78	84	65	74	+18.1	10	53	−37.9	+0.1	+7.8	+7.3	+8.2
Claude Haiku	gpt-4.1	82	97	79	73	+18.0	8	17	−16.7	−1.9	+10.7	+12.7	+11.0
Claude Haiku	gpt-4o-mini	69	97	67	78	+12.6	30	43	−41.4	0.0	+1.3	+1.1	−4.1
Claude Haiku	gpt-6-luna	82	56	46	78	+14.8	7	45	−15.0	−0.8	+10.9	+4.4	+5.0
Claude Haiku	gpt-6.1-terra	82	36	30	76	+18.0	4	67	−28.2	0.0	+11.2	+2.7	+4.1
gpt-6.1-terra	gpt-4.1	82	97	79	74	+16.9	8	62	−62.5	−1.0	+2.6	+9.0	+7.3
gpt-6.1-terra	gpt-4o-mini	69	97	67	79	+10.4	30	62	−61.5	0.0	−8.0	−8.4	−11.7
gpt-6.1-terra	gpt-6-luna	82	56	46	77	+13.1	7	80	−45.0	−0.3	+2.9	+2.1	+2.7
gpt-6.1-terra	gpt-6.1-terra	82	36	30	80	+14.6	4	77	−30.8	−0.3	+3.2	+0.8	+2.7
Qwen2.5-7B	gpt-4.1	82	97	79	50	+27.3	8	43	−39.1	−2.7	–	+23.4	+16.0
Qwen2.5-7B	gpt-4o-mini	69	97	67	54	+25.9	30	44	−33.0	0.0	–	+10.2	+7.3
Qwen2.5-7B	gpt-6-luna	83	58	48	50	+30.6	7	65	−25.0	−1.0	–	+12.4	+12.0
Qwen2.5-7B	gpt-6.1-terra	83	37	31	54	+25.8	4	67	−16.7	+0.7	–	+7.3	+8.0
gpt-4.1	gpt-4.1	82	97	79	72	+19.8	8	21	−20.8	−2.3	–	+13.2	+11.7
gpt-4.1	gpt-4o-mini	69	97	67	77	+13.9	30	38	−38.5	0.0	–	+1.7	−2.3
gpt-4.1	gpt-6-luna	82	56	46	75	+18.2	7	55	−35.0	+0.3	–	+4.7	+6.3
gpt-4.1	gpt-6.1-terra	82	36	30	74	+20.2	4	69	−23.1	0.0	–	+2.8	+5.0

Table 25: The crossed cells (ConflictBank, one-answer prompt, conflict questions) split by what the flag does on each question (exploratory, PR§17.1). *Prec.*, *Rec.*: direction precision and recall of the gold pair (%). *Share*: share of the cell's questions; *Acc.*: the generator's no-flag accuracy on them (%); *Gain*: the flag's mean effect on them (points). *Rest*: the remaining questions' contribution to the gain. *Marginal*: the break-even calculation, $pR + (1-p)W$ with H31's R and W at the cell's precision p, which assumes every question is flagged on its true pair and that flagged questions are typical; *Question*: the follow-rate account from each question's own flag and no-flag answer; both are retrospective. Generators without an H31 run have no marginal prediction; their question-level prediction uses mean follow rates.

		Conflict questions					Clean	
Generator	Detector	Right	Wrong	Acc.	Pred.	Obs.	Pred.	Obs.
Claude Haiku	Qwen2.5-7B	59	32	59.6	+3.2	+6.2	−16.4	−8.3
Claude Haiku	Claude Haiku	69	7	59.6	+12.7	+14.6	0.0	0.0
Claude Haiku	gpt-4o-mini	72	27	59.6	+7.9	+7.8	−5.8	−2.3
Claude Haiku	gpt-6-luna	43	9	59.6	+4.2	+7.4	0.0	0.0
Claude Haiku	gpt-6.1-terra	33	2	59.6	+5.8	+8.1	0.0	0.0
gpt-4o-mini	Qwen2.5-7B	59	32	47.4	+9.7	+9.2	−12.2	−5.3
gpt-4o-mini	Claude Haiku	69	7	47.4	+17.7	+16.0	0.0	0.0
gpt-4o-mini	gpt-4o-mini	72	27	47.4	+14.2	+11.4	−4.1	−3.7
gpt-4o-mini	gpt-6-luna	43	9	47.4	+8.0	+9.1	0.0	0.0
gpt-4o-mini	gpt-6.1-terra	33	2	47.4	+9.2	+10.2	0.0	0.0
gpt-6-luna	Qwen2.5-7B	59	32	52.7	+8.1	+6.9	−19.6	−2.7
gpt-6-luna	Claude Haiku	69	7	52.7	+18.4	+17.3	0.0	0.0
gpt-6-luna	gpt-4o-mini	72	27	52.7	+13.7	+11.6	−6.4	−3.0
gpt-6-luna	gpt-6-luna	43	9	52.7	+7.4	+6.1	0.0	0.0
gpt-6-luna	gpt-6.1-terra	33	2	52.7	+8.2	+8.8	0.0	0.0
gpt-6.1-terra	Qwen2.5-7B	59	32	71.7	−10.1	−8.3	−25.6	0.0
gpt-6.1-terra	Claude Haiku	69	7	71.7	+7.4	+7.0	0.0	0.0
gpt-6.1-terra	gpt-4o-mini	72	27	71.7	−3.1	−4.0	−8.5	−2.0
gpt-6.1-terra	gpt-6-luna	43	9	71.7	−1.7	−0.7	0.0	0.0
gpt-6.1-terra	gpt-6.1-terra	33	2	71.7	+3.0	+2.3	0.0	0.0

Table 26: Prospective test on ConflictBank-fresh (PR§19.1; 300 conflict and 100 clean questions no earlier run used). *Right*/*Wrong*: % of questions on which the detector names exactly the false side of a gold pair / marks a correct passage. *Acc.*: no-flag accuracy. *Pred.*: the frozen prediction, committed before any flagged answer existed; *Obs.*: observed directional-flag effect, in points.

			Effect against no flag (points)			
Generator	No flag	Closed-book	Oracle neutral	Oracle directional	LLM directional	Pred.
gpt-4o-mini	60.5	76.1	−2.4 [−6.0, +1.1]	+5.5 [+1.1, +10.1]	−2.7 [−6.8, +1.3]	−8.2
gpt-6-luna	68.0	91.0	−0.1 [−2.2, +2.0]	+11.7 [+7.7, +15.9]	−8.6 [−14.9, −2.8]	−19.4
Claude Haiku	69.6	91.2	−0.4 [−2.7, +2.0]	+13.8 [+9.7, +18.5]	+1.1 [−3.0, +5.2]	−16.6

Table 27: HealthContradict (PR§19.3): yes/no health questions with one real web page per side, accuracy in % and effects in points with topic-cluster bootstrap 95% CIs. *Pred.*: the frozen follow-rate prediction for the LLM directional flag (gpt-4o-mini detector), committed before any flagged answer existed.

E Prospective Test, Full Re-score and HealthContradict (PR§19)

Frozen, the account predicts unseen cells (H32–H36). ConflictBank-fresh has 400 questions built like our ConflictBank sample, with no shared subject. After detections and no-flag answers, we committed the follow-rate account's predictions, from H31's follow rates fixed in PR§19.1, (`splits/predictions19.json`) before running the flagged condition; the analysis reproduces them exactly (maximum drift 0.00 points). All three H32 criteria hold: the slope's CI [0.77, 1.00] excludes 0 and includes 1 (intercept $+1.0$); the mean error is $1.4 \leq 3$, and the account beats the generator-mean (4.1) and unit-follow-rate (5.3) baselines (Holm $p < 0.001$). A cell-level marginal estimate scores 1.8; predicting zero, 8.7. The sign is right in 20 of 20 cells. Of the 18 cells predicted at 3 points or more in either direction, 2 are predicted harms, both with gpt-6.1-terra generating; only 1 of them is significant, so H33 fails. The within-generator correlation is 0.97 (H34, $p < 0.001$). On clean questions (H35) the account predicts -4.9 points on average against -1.4 observed (mean error 3.6): a flag there changes fewer correct answers than H31's wrong-flag follow rate implies. Unflagged questions keep their no-flag answer, so in every cell whose detector flags no clean question the prediction and the observation are both zero; over the 8 of 20 cells whose detector flags one, the mean error is 8.9 (R6, Appendix F). A fresh randomised-direction run gives gpt-4o-mini follow rates of 50% [40, 61] and 50% [38, 60], inside the frozen intervals (H36). For every generator, the Claude Haiku detector has the highest predicted and observed effect (effects over all 400 questions, against the gpt-4o-mini detector: gpt-4o-mini generator $+12.0$ vs $+7.7$ points; gpt-6.1-terra $+5.3$ vs -3.5).

Under the judge, neither pooled equivalence holds (H38). The PR§17.3 judge scored every response in the pooled H30 and H23 tests (23,204 rows; agreement with the matcher 94.3%). On the same cells and pooling, the LLM flag minus the verdict key is $+1.2$ [$+0.1$, $+2.2$] (90% CI) under the matcher and $+2.4$ [$+1.3$, $+3.5$] under the judge; the oracle flag minus the answer key is -0.1 [-1.3, $+1.0$] and $+2.5$ [$+1.1$, $+3.9$]. Neither is equivalent within ± 3 under the judge; the smallest passing margins are 3.5 and 3.9. The matcher misses more correct answers under the flags than under the keys (PR§19.2): the judge accepts 8.7% of oracle-directional responses that the matcher rejects, against 6.7% of answer-key ones, and 2.9% of LLM-flag responses against 2.4% of verdict-key ones (the reverse disagreement: 0.9%, 1.4%, 0.5% and 0.8%; agreement 90.4%, 92.0%, 96.5% and 96.9%). So the matcher understates the flag's advantage by 1.2 and 2.6 points (matcher minus judge -1.2 [-1.9, -0.6] and -2.6 [-3.9, -1.4]). Under either scorer, both point estimates stay below 3 points. The pooled TOSTs are unregistered: we added them after seeing that the per-comparison H23 TOSTs showed equivalence in only 2 of 9 comparisons. Seed by seed, the pooled matcher estimates range from $+0.5$ to $+1.4$ (H30a) and from -0.4 to $+1.1$ (H23; R11).

HealthContradict is a label-transfer stress test (H37). Each item is a yes/no question with one page per side, so the oracle directional flag, which heads the page typed misinformation, tells the generator which answer the released labels favour. H37a therefore measures deference to those labels; the flag works close to an answer key. It beats no flag (H37a) and the neutral flag (H37c) for all three generators. The neutral flag changes accuracy by at most 2.4 points (H37b; within ± 3 for gpt-6-luna and Claude Haiku). For every generator, closed-book accuracy exceeds accuracy with the pages (gpt-4o-mini 76.1% against 60.5%). The gpt-4o-mini detector names the false page as suspect on 36% of items and the correct page on 48%, so the frozen account predicts harm; it overstates it for all three generators (Table 27): -8.2 predicted against -2.7 observed for gpt-4o-mini, -19.4 against -8.6 for gpt-6-luna, and -16.6 against $+1.1$ for Claude Haiku. The labels are as released. Auditing them (247 items), an LLM judge finds the correct page supporting the gold answer for 68%, the misinformation page the wrong answer for 55%, and both for 37% (R13 gives the full breakdown). Label noise could explain the overshoot, since a reversed label turns a wrong flag into a right one; so could the generators' priors, or follow rates that do not transfer. We cannot separate these until humans label the pages (PR§20.3, A4).

F Re-analyses after the External Review (PR§20)

After an external review we registered re-analyses R1–R19 in PR§20 (Table 34). All are exploratory: they re-analyse outputs already seen, make no API call and sit outside the FDR count. R3 and R13 were first tabulated on 2026-11-02, before PR§20 was written. R16 counts settings whose 95% CI excludes zero, unadjusted; R18 classifies settings by the signs of the point estimates of K and C. Conventions follow PR§20: one-way strict matching, seed means within questions and question-bootstrap 95% CIs. Results the body states in full are summarised here only briefly.

R1: ConflictBank's false side is readable from style. We trained a passage-only classifier, an L2 logistic regression on TF-IDF word uni- and bi-grams, on 124,769 ConflictBank rows (249,538 passages: each row's correct and misinformation passage). No row's subject appears in our eval set or in ConflictBank-fresh (1,400 subjects excluded), no training passage equals an evaluated one, and the closest has TF-IDF cosine 0.78 to an eval passage. On every conflict question the classifier gives the misinformation passage the higher probability of being false: pairwise accuracy 1.000 [1.000, 1.000] on the 300 eval and 1.000 [1.000, 1.000] on the 300 fresh questions. Masking each passage's own answer string leaves this unchanged (1.000 and 1.000). Classified one at a time, eval passages get the right label at a rate of 0.992 (mean probability of false 0.94 for misinformation passages, 0.07 for correct ones). Its heaviest n-grams toward false are "studies", "study", "exclusive interview", with "journal of", "dr" and "our publication" close behind; toward correct they are "evidence", "breaking news", "title the", with "confirms", "sources" and "website" close behind. These are genre words: the misinformation passages read like reports of studies and interviews, the correct ones like news and records. So on ConflictBank a detector's direction, or a generator's choice without a flag, can come from style rather than from knowledge or from reasoning about the contradiction. Perfect accuracy here is a lower bound on how legible the side is from surface text. It does not undo H31: there a coin flip assigns the flag's direction, so the follow rates measure deference to the flag. Generators also follow wrong flags that point at the passage whose style marks it as correct.

R2, R3: direction precision and the detector's prior. Every detector names the false side more often on facts its own model knows closed-book (Table 28). Counting flags that name one side, the known-minus-unknown difference runs from $+3.2$ (gpt-6.1-terra) to $+18.7$ (gpt-4o-mini), and its CI lies above zero for 4 of 5 detectors; under the PR§11.3 coding it does so for 5. Known questions are few (28 to 60 of 300 per detector). On RAMDocs, where several passages can agree, the PR§11.3 split for the gpt-4o-mini detector (registered as exploratory) shows no visible gap, on only 7 known questions (87.5%, against 88.3% on 132 unknown ones; 'both' counted as a miss). Counting only one-side verdicts (R3), direction precision on ConflictBank runs from 68.8% (gpt-4o-mini) to 97.0% (gpt-6.1-terra), and from 87.7% to 97.0% for the five detectors other than gpt-4o-mini; the earlier definition, which counts "both" and "unclear" verdicts as misses, gives 77.7–82.6% for those five.

R4, R5: follow rates and the generator's prior. Known questions are 9–20% of each H31 generator's conflict questions, so the known-question rates rest on few questions. Right flags are followed more often on known questions by every generator (differences $+29$ to $+38$ points), but over at most 5 known questions per generator. Wrong-flag follow rates show no consistent direction: gpt-6.1-terra $+7$ [$+1$, $+13$], Claude Haiku -31 [-60, $+2$]. Pooled over generators, the odds ratio for following a wrong flag on a known question is 0.84 [0.38, 1.85] (307 question–generator observations, 48 of them known); the right-flag odds ratio has no finite upper confidence limit (complete separation in 1,793 of 5,000 bootstrap resamples, with only 12 known observations), so we do not report it. Conditioning each question's follow rates on whether the generator knows the fact moves the crossed-cell account's mean error from 1.82 to 1.79 points over 30 cells (R5), and no cell's prediction by more than 0.7 points.

R6: the prospective account against simpler baselines. On the 20 fresh cells, the mean error of the cell-level marginal minus that of the account is $+0.4$ [-0.3, $+1.2$] points. A common-rate baseline, with one f_R (64.5%) and one f_W (71.8%) for every generator, scores 1.6 [1.5, 2.7] ($+0.2$ [-0.2, $+0.7$] against the account), with the observed sign in 20 of 20 cells. Both CIs of the differences include zero. On clean questions the mean error

	R2: by the detector's closed-book knowledge (single-side)					R3: all gold-pair flags	
Detector	Known q.	Flags	Known	Unknown	Difference	Single-side	All verdicts
gpt-4o-mini	28	28/264	85.7 [71.4, 96.8]	67.0 [61.2, 72.6]	+18.7 [+3.3, +31.9]	68.8 [63.3, 74.0]	68.8 [63.3, 74.0]
gpt-4.1	60	56/205	98.2 [93.9, 100.0]	88.8 [84.3, 93.1]	+9.4 [+3.6, +15.0]	90.9 [87.3, 94.3]	81.8 [77.2, 86.2]
gpt-6-luna	47	28/116	100.0 [100.0, 100.0]	94.0 [89.2, 98.1]	+6.0 [+1.9, +10.8]	94.9 [91.1, 98.0]	82.2 [76.4, 87.7]
gpt-6.1-terra	60	33/63	100.0 [100.0, 100.0]	96.8 [91.8, 100.0]	+3.2 [0.0, +8.2]	97.0 [93.1, 100.0]	82.6 [75.5, 89.4]
Claude Haiku	32	26/185	100.0 [100.0, 100.0]	85.9 [80.6, 90.9]	+14.1 [+9.1, +19.4]	87.7 [83.0, 92.1]	78.1 [72.5, 83.3]
Claude Sonnet	–	–	–	–	–	93.8 [90.2, 96.9]	77.7 [72.6, 82.7]

Table 28: Direction precision and prior knowledge on ConflictBank (PR§20.2 R2, R3, exploratory; %, 95% bootstrap CIs below). Direction precision: among a detector's flags on the gold pair, the share naming the misinformation passage. *R2*, in the detector's run with gpt-4o-mini as generator: a question is *known* to the detector model when its closed-book answer is correct on at least two of three seeds (or its one seed); *Known q.*: such conflict questions (of 300); *Flags*: flags naming exactly one side, on known/unknown questions; *Known*, *Unknown*: direction precision over those flags; *Difference*: known minus unknown (points); question-bootstrap CIs. Claude Sonnet has no closed-book run. *R3*, pooled over the detector's crossed generator cells: direction precision counting only flags that name exactly one side (*Single-side*) or every flag, with "both" and "unclear" as misses (*All verdicts*, the earlier definition); question-cluster bootstrap CIs.

is 3.6 [2.5, 4.8] over all cells and 8.9 [6.3, 11.9] over the 8 cells whose detector flags a clean question. The two predicted harms are gpt-6.1-terra with the Qwen2.5-7B detector (predicted −10.1 [−16.2, −4.2], observed −8.3 [−15.0, −1.7]) and gpt-6.1-terra with the gpt-4o-mini detector (−3.1 [−8.1, +1.6] against −4.0 [−9.7, +1.3]). In the selection analysis of PR§19.1, the all-question prediction has mean error 1.7 points (2.6 where the detector flags clean questions, 1.0 elsewhere); its mean signed error is −0.1 from conflict questions and −0.9 from clean ones. The largest error, −7.7 for gpt-6.1-terra with the Qwen2.5-7B detector, is −6.4 from clean questions.

R7: gpt-6.1-terra's two no-flag draws. Because the response cache keys on the requested temperature, gpt-6.1-terra's no-flag condition on ConflictBank was sampled twice (PR§20.1); 80 of 400 responses are identical, and 30 questions change score. With the other draw, its p^* moves from 0.79 [0.72, 0.85] to 0.81 [0.75, 0.87], f_R from 67% to 68%, and f_W from 95% to 93%. Swapping the RAMDocs draw changes the status of 0 of the H19 tests.

R8: response length under the flag. API completion tokens, which include hidden reasoning tokens, fall with a flag in the H31 runs of all three generators measured: gpt-6.1-terra 105 to 51 (−54 [−61, −46]), gpt-6-luna 255 to 134, Claude Haiku 555 to 508. Claude Haiku's responses to wrong flags are not shorter (+36 [−7, +81]). gpt-6.1-terra uses fewer tokens whether the flag is right (−55) or wrong (−51). Responses that follow a wrong flag are shorter than those that resist it for gpt-6-luna (−94 [−182, −22]) and Claude Haiku (−121 [−189, −54]); for gpt-6.1-terra, which resists in only 5 responses, the difference is −47 [−96, +1]. Token counts cannot show why a generator defers.

R9, R10: missing final answers. Of 788,506 responses in 222 runs, 1,168 (0.15%) have no final-answer line (no `Final answer(s):` line and no MADAM answer list); as registered (PR§5) they are scored as they are: the matcher then reads the whole response, so a missing line does not by itself make a response wrong. Of 979 run–condition cells, 51 exceed 1% (32 with gpt-4.1, 16 with Qwen2.5-7B, 2 with gpt-6-luna; Table 29). The worst cell is gpt-4.1 with the oracle neutral flag + instr. (v2) condition (20.3%; 25.3% of that cell's responses stop at the 256-token cap). Among missing responses from methods with one generator call, 65.5% stopped at the cap. Dropping the questions on which any seed of either arm lacks a final answer (R10) changes H13's two gpt-4.1 tests: the neutral flag with the instruction against no flag goes from +5.8 [−2.9, +14.4] (Holm p 0.743)

Run	Condition	Missing	At cap
holdout_gpt41_round2	oracle neutral flag + instr. (v2)	20.3	25.3
cb_qwen	oracle remove suspect	18.5	0.0
cb_qwen	LLM remove suspect	17.5	0.0
holdout_gpt41	instruction (v2)	12.8	15.6
cross_mj_four_four	no flag	11.2	13.5
cross_mj_four_luna	no flag	11.2	13.5
cross_mj_four_terra	no flag	11.2	13.5
holdout_qwen	LLM remove suspect	11.0	0.0
cross_mj_four_terra	LLM dir. flag	8.5	10.2
holdout_gpt41_round2	oracle dir. flag + instr. (v2)	8.0	10.0
cb_qwen	no flag	7.2	0.0
cb_qwen	oracle dir. flag	6.5	0.0
cross_mj_four_luna	LLM dir. flag	6.5	7.5
cb_qwen	oracle neutral flag	6.2	0.2
cb_qwen	LLM dir. flag	5.5	0.0
cb_qwen	LLM neutral flag	5.2	0.2
holdout_qwen	LLM dir. flag	5.0	0.7
holdout_gpt41_round2	LLM dir. flag	4.0	6.0
holdout_qwen	oracle remove suspect	4.0	0.3
holdout_gpt41	LLM dir. flag	3.9	5.7
cross_mj_four_four	LLM dir. flag	3.8	5.5
cb_single_gpt41	oracle neutral flag	3.5	4.2
holdout_qwen	oracle neutral flag	3.0	1.0
holdout_qwen	LLM neutral flag	3.0	0.0
holdout_gpt41	oracle neutral flag	2.9	3.9

Table 29: Responses with no final-answer line (PR§20.3 R9, exploratory): the 25 worst of the 51 run $\times$ condition cells above 1% (of 979 cells in 222 runs; 0.15% of all 788,506 responses); by generator, the cells above 1% are 32 gpt-4.1, 16 Qwen2.5-7B, 2 gpt-6-luna, 1 gpt-4o-mini. A cell holds one response per question and seed (300–900 here). *Missing*: % of the cell's responses. *At cap*: % that stopped at the completion-token cap, which is 256 tokens for Qwen2.5-7B, gpt-4.1, gpt-4o-mini and 4,096 tokens for Claude Haiku, Claude Sonnet, gpt-6-luna, gpt-6.1-terra (not defined for multi-call methods). The 26 further cells above 1% (not shown) miss 1.1–2.7%.

to +17.2 [+6.9, +27.6] (Holm p 0.007; 52 questions dropped), and the directional against the neutral flag, both with the instruction, from +25.2 to +17.4 [+9.3, +26.7]. This is a sensitivity analysis: H13 keeps its registered status.

R11: seeds. Using one seed at a time, the pooled H30a estimate ranges from +0.5 to +1.4 points (all seeds: +1.2) and the pooled H23 estimate from −0.4 to +1.1 (all seeds: −0.1). Of the pooled cells, 4 of 22 (H30a) and 3 of 9 (H23) have one seed only. In Table 17, 9 of 19 settings have one seed; over the others, the neutral flag's effect varies across seeds by at most 6.3 points in accuracy (CB one-answer, gpt-6-luna), 5.0 in the misinformation share (RAMDocs, Claude Haiku) and 2.3 in abstention.

R12: one matcher for the misinformation share. Recomputing the misinformation share with the one-way matcher changes 6 of 210 exploratory decisions (5 lose significance, 1 gains it) and 0 of 16 H21(a) tests. In Table 17 one bold mark is lost, RAMDocs, Claude Haiku, oracle neutral flag: its change is −7.9 (Holm p 0.024) under the bidirectional matcher and −7.4 (Holm p 0.061) one-way.

R13: HealthContradict page stance. An LLM judge's verdicts from the PR§19.3 audit, first tabulated on 2026-11-02, cover 247 items. The page typed correct supports the gold answer for 68% of items, does not for 7% and is unclear for 25%. The page typed misinformation supports the wrong answer for 55%, does not for 17% and is unclear for 28%. Both pages support their labelled answers for 92 items (37%), so 63% lack a confirmed pair. Of 494 pages, 60 (12%) read as not supporting their labelled side, and 357 (72%) reach the 4,000-character cut the generators also saw. A judge cannot stand in for human labels, so this does not test label noise.

R14: RAGuard's answer key against the flag. On RAGuard's 294 conflict claims the answer key raises accuracy more than the oracle directional flag for 3 of the 4 generators with a key arm: key minus flag is +33.2 [+28.7, +37.8] for gpt-6-luna, +12.2 [+7.8, +16.7] for gpt-6.1-terra and +7.1 [+3.6, +10.8] for Claude Haiku. For gpt-4o-mini the flag carries more (−8.5 [−13.0, −4.2]). Against no flag, the key and the flag change accuracy by +42.1 and +8.8 points (gpt-6-luna), +19.0 and +6.8 (gpt-6.1-terra), +19.8 and +12.7 (Claude Haiku), and +1.4 and +9.9 (gpt-4o-mini). Claude Sonnet has no key arm. On RAGuard the oracle flag is therefore far from an answer key.

R15: single-entity RAMDocs questions. Leaving out one of the 10 RAMDocs settings at a time, the oracle neutral flag's mean effect on single-entity questions ranges from −5.5 (without RAMDocs, Claude Haiku) to −3.4 (without RAMDocs, gpt-4.1 (minimal)), and every leave-one-out CI excludes zero; the oracle directional flag's ranges from +26.7 to +31.3. Averaged with equal weight over all settings, the neutral flag's effect is −4.4 [−7.3, −1.5] (Appendix D); its CI uses one question resample shared by every setting (PR§18.2), so the settings are not independent.

R16: risk–coverage operating points. Table 30 gives the operating points behind Figure 1, every column under the one-way matcher. Under the multi-answer prompt the neutral flag lowers selective accuracy in 7 of 10 settings, raises the misinformation share in 7 and changes abstention in 0

of them. Under the minimal skeleton it raises abstention in 4 of 4. Under the one-answer prompt it raises abstention in 7 of 16 settings (lowers it in 2) and raises selective accuracy in 5, lowering it in 0. Under forced choice there is no abstention to trade.

R17: RAMDocs answer mix and clean questions by type. Under the minimal skeleton, which states no answer policy, the oracle neutral flag moves gpt-4o-mini towards abstention (+16.1 [+11.0, +21.6]; misinformation −0.7 [−7.2, +5.5]) and Qwen2.5-7B likewise (+10.1 [+4.3, +16.5]; misinformation +4.3 [−2.9, +11.5]), while gpt-4.1 adds the misinformation (+12.2 [+5.5, +19.2]; abstention +4.1 [+0.7, +7.9]). On RAMDocs' clean questions, the gpt-4o-mini detector flags 81.7% [74.0, 88.5] of the 104 with several gold answers and 50.0% [31.8, 68.2] of the 22 with one. Its directional flag changes accuracy on them by +1.1 [−1.0, +3.2] and +1.0 [−1.5, +4.0] points (mean over 6 settings).

R18: prevalence. Table 31 gives, for the gpt-4o-mini LLM detector's directional flag, the conflict share above which it helps. Of 31 settings, 17 have such a break-even share (median 22%, from 2% for RAMDocs, gpt-6.1-terra to 74% for RAMDocs, gpt-4o-mini (minimal)); the flag helps at every share in 7 and hurts at every share in 7. Grouped by how a benchmark's clean questions were built, the break-even runs from 5% to 45% over 9 ConflictBank settings, whose clean questions add only off-subject passages, and from 2% to 74% over 7 RAMDocs settings, whose clean questions mostly have several valid answers. RAGuard has 1 such setting (30%). The RAMDocs range contains the ConflictBank range, so the clean-question build does not explain the spread.

R19: H4b, second form. With every suspect document treated as misinformation, the H4b correlation over 18 cells is $\rho = -0.08$. A permutation test over 10,000 permutations gives two-sided $p = 0.75$, and a bias-corrected (BCa) interval is [−0.32, +0.01]. The percentile interval in Table 12, [−0.56, −0.02], lies mostly below the point estimate because the bootstrap distribution is skewed (70% of replicates fall below it), so it cannot be read as support. H4b is supported under the main labels but label-sensitive (Appendix I).

	Coverage		Selective accuracy		Misinformation		Strict accuracy	
Setting	No flag	Flag	No flag	Flag	No flag	Flag	No flag	Flag
Answer policy: multi-answer								
ConflictBank, gpt-4o-mini	99.7	99.1	21.4	12.8†	69.0	78.1†	21.6	12.8†
ConflictBank, gpt-4.1	99.3	99.7	13.1	6.7†	79.7	88.3†	13.0	6.7†
ConflictBank, Qwen2.5-7B	94.0	95.0	24.1	14.0†	66.3	77.7†	23.7	15.3†
RAMDocs, gpt-4o-mini	96.6	95.0	22.8	16.2†	43.9	57.6†	22.1	15.3†
RAMDocs, gpt-4.1	98.3	98.3	22.2	13.4†	55.4	69.8†	21.8	13.2†
RAMDocs, Qwen2.5-7B	84.2	82.0	21.4	14.0†	45.3	65.5†	19.4	13.7†
RAMDocs, gpt-6-luna	98.8	99.3	14.6	11.4†	68.6	73.4†	14.4	11.3†
RAMDocs, gpt-6.1-terra	97.8	98.6	11.0	10.2	73.4	77.0	10.8	10.1
RAMDocs, Claude Haiku	97.8	97.8	25.7	27.0	53.5	46.0†	25.2	26.4
RAMDocs, Claude Sonnet	97.8	97.8	46.3	46.3	23.7	21.6	45.3	45.3
Answer policy: minimal skeleton (no stated policy)								
ConflictBank, gpt-4o-mini	96.8	67.6†	36.2	25.0†	49.6	43.4†	35.0	16.9†
RAMDocs, gpt-4o-mini	92.6	76.5†	22.0	19.7	29.3	28.5	20.4	15.1†
RAMDocs, gpt-4.1	97.6	93.5†	34.4	24.1†	29.0	41.2†	33.6	22.5†
RAMDocs, Qwen2.5-7B	89.9	79.9†	18.4	14.4	38.1	42.4	17.3	11.5†
Answer policy: one answer, "unknown" allowed								
ConflictBank, gpt-4o-mini	93.3	59.0†	54.8	55.0	30.9	17.2†	51.1	32.4†
ConflictBank, gpt-4.1	98.3	86.0†	65.8	65.9	27.3	21.3†	64.7	56.7†
ConflictBank, Qwen2.5-7B	95.0	85.0†	53.0	55.3	35.0	28.7†	50.3	47.0
ConflictBank, gpt-6-luna	88.1	71.1†	60.8	62.0	23.7	17.0†	53.6	44.1†
ConflictBank, gpt-6.1-terra	98.3	95.3†	74.2	74.1	17.7	15.3	73.0	70.7
ConflictBank, Claude Haiku	98.8	99.1	67.6	68.0	21.8	21.6	66.8	67.4
ConflictBank, Claude Sonnet	99.0	98.3	76.4	80.0†	15.7	11.7†	75.7	78.7
CB majority, gpt-4o-mini	96.8	73.2†	22.3	31.0†	63.1	39.6†	21.6	22.7
CB majority, gpt-6-luna	95.8	88.3†	31.1	41.1†	58.3	43.1†	29.8	36.3†
CB majority, gpt-6.1-terra	99.0	99.3	26.3	22.5	65.0	70.7†	26.0	22.3
CB majority, Claude Haiku	98.9	99.1	31.3	35.7†	57.8	54.1†	31.0	35.3†
RAGuard, gpt-4o-mini	67.9	67.7	54.1	59.0†	31.2	27.8†	36.7	39.9†
RAGuard, gpt-6-luna	43.2	44.3	53.8	52.4	20.0	21.1	23.2	23.2
RAGuard, gpt-6.1-terra	39.1	43.9†	63.5	62.0	14.3	16.7	24.8	27.2
RAGuard, Claude Haiku	42.4	44.9	57.8	55.3	17.9	20.1†	24.5	24.8
RAGuard, Claude Sonnet	39.8	46.3†	68.4	68.4	12.6	14.6	27.2	31.6†
Answer policy: one answer, forced								
ConflictBank, gpt-4o-mini	99.9	99.6	55.5	55.5	31.6	31.2	55.4	55.2
ConflictBank, gpt-6-luna	100.0	100.0	64.6	66.1	25.3	21.4†	64.6	66.1
ConflictBank, gpt-6.1-terra	100.0	100.0	77.7	77.3	14.0	12.7	77.7	77.3
ConflictBank, Claude Haiku	100.0	100.0	69.3	73.2†	20.6	16.4†	69.3	73.2†

Table 30: Risk–coverage operating points on conflict questions (PR§20.4 R16, exploratory; 34 settings): no flag against the oracle neutral flag, in %. *Coverage*: 1 − abstention; *Selective accuracy*: correct responses over answered (non-abstaining) responses, summed over questions and seeds; *Misinformation*: share of responses containing a wrong answer as a whole-word span; *Strict accuracy*: one-way strict. All four use the one-way matcher (PR§20.3 R12). †: the paired question-bootstrap 95% CI of the change excludes zero (unadjusted). The minimal skeleton states no answer policy. Figure 1 plots the coverage and selective-accuracy columns.

Setting	K	C	Break-even [95% CI]
ConflictBank: correct passage plus three off-subject distractors			
multi-answer, gpt-4o-mini	+20.0	−1.0	5 [0, 18]
multi-answer, gpt-4.1	+17.0	0.0	↑
multi-answer, Qwen2.5-7B	+6.7	−1.0	13 [0, 50]
one-answer, gpt-4o-mini	+8.4	−4.7	36 [10, 63]
one-answer, gpt-4.1	−2.3	−3.0	↓
one-answer, Qwen2.5-7B	+7.3	−1.0	12 [0, 53]
one-answer, gpt-6-luna	+8.6	−1.3	13 [0, 39]
one-answer, gpt-6.1-terra	−11.7	0.0	↓
one-answer, Claude Haiku	−4.1	−5.0	↓
one-answer, Claude Sonnet	−2.7	−1.0	↓
majority, gpt-4o-mini	+5.7	−4.7	45 [14, 77]
majority, gpt-6-luna	+7.1	−1.3	16 [0, 44]
majority, gpt-6.1-terra	+9.0	0.0	↑
majority, Claude Haiku	+6.8	−5.0	42 [16, 72]
forced, gpt-4o-mini	+4.0	−2.7	40 [6, 100]†
forced, gpt-6-luna	−2.4	−1.3	↓
forced, gpt-6.1-terra	−14.3	0.0	↓
forced, Claude Haiku	−4.7	−1.3	↓
RAMDocs: questions without misinformation, mostly several valid answers			
multi-answer, gpt-4o-mini	+12.5	−3.4	22 [0, 46]
minimal, gpt-4o-mini	+3.6	−10.3	74 [46, 100]†
multi-answer, gpt-4.1	+8.4	0.0	↑
multi-answer, Qwen2.5-7B	+2.9	−7.1	71 [36, 100]†
multi-answer, gpt-6-luna	+29.7	−11.6	28 [18, 39]
multi-answer, gpt-6.1-terra	+38.8	−0.8	2 [0, 9]
multi-answer, Claude Haiku	+22.5	−4.5	17 [2, 30]
multi-answer, Claude Sonnet	+12.2	−3.2	21 [0, 44]
RAGuard: claims with supporting posts and no misleading post			
one-answer, gpt-4o-mini	+0.6	+1.7	↑
one-answer, gpt-6-luna	+3.1	−1.3	30 [0, 100]†
one-answer, gpt-6.1-terra	+2.0	+1.0	↑
one-answer, Claude Haiku	+5.0	+1.3	↑
one-answer, Claude Sonnet	+7.1	+10.0	↑

Table 31: Prevalence (PR§20.4 R18, exploratory): the gpt-4o-mini LLM detector's directional flag. K, C: its effect (points) on conflict and on clean questions; at a conflict share π the effect is $\pi K + (1-\pi)C$. *Break-even*: $-C/(K-C)$, the share (%) above which the flag helps, where $K > 0 > C$; CI from a joint question bootstrap. †: at least 2.5% of replicates help at no share, so the upper limit is 100. ↑: helps at every share ($K, C \geq 0$); ↓: hurts at every share ($K, C \leq 0$). Rows name the prompt (*majority*: the one-answer prompt on the majority variant); settings are grouped by how their clean questions were built.

Detector	Direction accuracy	Answer only	Single passage	Object swap before	Object swap after	Outcome
gpt-4o-mini	69	80 [76, 83]	73 [69, 76]	68	52 [46, 59]	chance
Claude Haiku	84	84 [81, 88]	84 [79, 88]	88	61 [54, 69]	survives
gpt-6-luna	87	88 [84, 90]	87 [84, 91]	96	78 [70, 85]	survives
gpt-6.1-terra	89	90 [86, 93]	90 [86, 93]	97	78 [68, 87]	survives

Table 32: Artefact probes on ConflictBank's 300 conflict questions (C1, PR§21.1; exploratory, one seed; percentages, 95% item-bootstrap CIs below). *Direction accuracy*: the detector's direction accuracy on the eval set (PR§11.3). *Answer only*: no passage; the share of the two candidate answers picked correctly, averaged over both orders. *Single passage*: one passage alone scored for the probability that it is false; the share of questions whose misinformation passage scores higher (ties count half). *Object swap*: on 249 usable items the two answers are exchanged between the passages, so each passage keeps its style and argues the other side; the share of single-side verdicts on the true pair that name the passage stating the wrong answer, before and after the swap. *Outcome* (registered): survives if the CI lies above 50, inverts if below, chance if it includes 50.

G Runs after the External Review (PR§21)

After the review we ran the API package of PR§21, registered in full and pushed before any of its runs (Table 34). H40 and H41 are registered tests (tier ii) and join the FDR count; everything else here is descriptive, with reading rules fixed in PR§21, and so exploratory. Conventions follow PR§21: one-answer prompt, one-way strict matching, seed means within questions and question-bootstrap 95% CIs. From PR§21 on, each result row records the model name the API reports serving.

C1: what the detectors read. Four detector models took three probes on the 300 ConflictBank conflict questions, one seed each (Table 32). *Answer only:* shown the question and the two candidate answers, with no passage, they pick the true one 80–90% of the time, averaged over both orders. For 4 of the 4 this is within five points of their direction accuracy as detectors or above it, the registered reading for "the direction signal needs no passage text", although closed-book they answer only 4.3–20.0% of these questions (Table 3): recognising the true answer among two is far easier than recalling it. *Single passage:* scoring one passage at a time, they give the misinformation passage the higher probability of being false on 73–90% of questions; the reasoning models and Claude Haiku also give about half of the correct passages a probability of at least 50 (gpt-6-luna 51%, Claude Haiku 54%). *Object swap:* on 249 items whose two answers can be exchanged cleanly between the passages, each passage keeps its style but argues the other side. The share of single-side verdicts naming the passage that states the wrong answer falls from 68–97% to 52–78% (72 to 246 verdicts per detector; the reasoning models flag the true pair less often). It stays above chance for 3 detectors (the registered "survives") and falls to chance for gpt-4o-mini (52% [46, 59]). So the stronger detectors mostly follow which answer is plausible, and passage style accounts for part of their direction; gpt-4o-mini's direction rests as much on style as on the answer. In both cases the direction can come without reasoning about the contradiction. These probes do not tell recognition of the fact from inference from the question's wording (a bishop's name suggesting his nationality); either way, ConflictBank's false side is legible without the passages.

C4: randomised direction on new questions (H40). The H31 design was repeated on the 300 conflict questions of ConflictBank-fresh for gpt-6-luna, Claude Haiku and gpt-6.1-terra, with each generator's prediction frozen from its no-flag answers and the round-10 follow rates before any flagged run (Table 33). H40a: the fresh f_R and f_W lie inside the frozen eval intervals for gpt-6-luna and Claude Haiku; gpt-6.1-terra's f_R does too, but it follows 86% [79, 92] of wrong flags, below its eval interval, so H40a holds for 2 of 3. H40b: for all three, a right flag helps, a wrong flag hurts and their sum is negative (-14.9 to -50.8 points; all significant after Holm). H40c: the frozen predictions miss the observed effects over all 300 questions by 2.2 points on average (predicting zero: 14.2; follow rates of 1: 3.7), within the registered 3 points. About two thirds of the flagged answers were not new samples (see *Shared responses* below), so H40c is not prospective in the sense of H32: the predictions read none of them and fit nothing to them, but the answers existed when the predictions were frozen.

C5: a stated confidence (H41). The same randomised flags were re-run with each suspect labelled "detector confidence: 55%" or "95%". A

Generator	f_R	f_W	Right	Wrong	p^*
ConflictBank-fresh, randomised direction (H40)					
gpt-6-luna	65	75	+33.1	−49.1	0.60
	[53, 75]	[65, 84]			[0.54, 0.66]
Claude Haiku	69	63	+33.1	−48.0	0.59
	[57, 80]	[54, 72]			[0.53, 0.66]
gpt-6.1-terra	67	86	+18.9	−69.7	0.79
	[54, 80]	[79, 92]			[0.72, 0.86]
gpt-4o-mini†	50	50	+28.6	−29.4	0.51
	[40, 61]	[39, 60]			[0.41, 0.60]
Eval set, one seed (C6, C8)					
Claude Sonnet	65	23	+16.4	−20.6	0.56
	[50, 80]	[16, 31]			[0.44, 0.68]
Qwen2.5-7B	56	52	+25.8	−22.0	0.46
	[44, 66]	[41, 64]			[0.33, 0.58]
Eval set, stated confidence 55% / 95% (H41)					
gpt-4o-mini, 55%	57	50	+28.9	−31.2	0.52
	[46, 67]	[39, 61]			[0.43, 0.60]
gpt-4o-mini, 95%	58	60	+30.0	−32.4	0.52
	[47, 68]	[49, 70]			[0.43, 0.60]
Claude Haiku, 55%	56	44	+22.4	−33.8	0.60
	[43, 69]	[35, 53]			[0.52, 0.68]
Claude Haiku, 95%	64	61	+23.9	−46.1	0.66
	[50, 77]	[51, 69]			[0.59, 0.73]
gpt-6-luna, 55%	70	56	+32.3	−32.4	0.50
	[59, 81]	[44, 67]			[0.43, 0.58]
gpt-6-luna, 95%	69	65	+32.1	−42.3	0.57
	[57, 80]	[56, 75]			[0.51, 0.63]
gpt-6.1-terra, 55%	64	84	+17.6	−57.4	0.77
	[51, 78]	[77, 91]			[0.70, 0.83]
gpt-6.1-terra, 95%	69	92	+19.5	−63.1	0.76
	[56, 82]	[85, 97]			[0.70, 0.82]

Table 33: Deference to randomised directional flags in the PR§21 runs (one-answer prompt, one-way matcher; columns as Table 3). Fresh: 300 new conflict questions, predictions frozen before the flagged runs. †: gpt-4o-mini's fresh run predates PR§21 (H36). Stated confidence: the same flags, each suspect labelled with a detector confidence of 55% or 95%.

wrong flag stated at 55% costs less than one stated at 95% for 3 of the 4 generators (H41a; differences +1.2 to +12.3 points): gpt-6-luna follows 56% of wrong flags at 55% and 65% at 95%, against 77% with no confidence stated, and gpt-6.1-terra 84% and 92%, against 95%. A right flag stated at 95% is not detectably more helpful than one at 55% (H41b: 0 of 4). A stated confidence thus lowers deference to wrong flags in some generators without raising the gain from right ones; these confidences are synthetic, and whether real detectors' confidences are calibrated enough to use is untested.

C6, C8: Claude Sonnet and Qwen2.5-7B (one seed). On the eval questions Claude Sonnet, the most accurate generator in our runs (75.7% without a flag on these questions, against 69.7% for gpt-6.1-terra), follows 65% [50, 80] of right flags and only 23% [16, 31] of wrong ones; its p^* is 0.56 [0.44, 0.68]. On this one draw, deference does not rise with accuracy beyond the four H31 generators, and neither does p^*. Qwen2.5-7B follows right and wrong flags at similar rates (56% and 52%), like gpt-4o-mini, with p^* 0.46.

C9: prior knowledge on new questions and for Sonnet. Closed-book accuracy on ConflictBank-fresh's conflict questions is 7.0% to 26.3% (Claude Sonnet on the eval conflict questions: 16.3%). The R2 gap recurs on fresh: every detector names the false side at least as often on facts its model knows (+1 to +16 points; CI above zero for 2 of 5), and Sonnet's is 98% against 93%. Conditioned on the generator's prior (R4 on fresh), wrong flags are followed less on known facts by gpt-4o-mini (33% against 53%, over 13 known questions) and Claude Haiku (38% against 69%, 15), hardly by gpt-6.1-terra (85% against 86%); Claude Sonnet follows 4% on the 23 facts it knows. Known questions remain few.

C3: HealthContradict answer key. An explicit key ("the answer X is false") beats the oracle directional flag by +16.3 points [+11.8, +21.3] for gpt-6-luna and +4.9 [+1.4, +8.6] for Claude Haiku, and does not differ detectably for gpt-4o-mini (+2.0 [−2.1, +6.2]; topic-cluster CIs). Even with one page per side, the flag conveys less than the answer it implies to these two generators. Labels are as released.

C7: the Claude Haiku detector on RAMDocs. On the RAMDocs hold-out with gpt-4o-mini generating, the Claude Haiku detector flags 2% of clean questions, against 76% for the gpt-4o-mini detector (most clean RAMDocs questions have several valid answers). Its directional flag gains +12.7 points [+6.7, +18.7] on conflict questions and changes clean ones by −0.3, against +12.5 and −3.4 for the gpt-4o-mini detector; with removal, +23.5 and 0.0 against +18.7 and −20.6. The detector the rule of §6 picked on ConflictBank-fresh thus also has almost no clean-question cost here (break-even conflict share 0.02, against 0.22).

C11: sentence-pair NLI. Scoring the three question-relevant sentences of each passage against those of every other passage, the NLI cross-encoder flags every conflict question and every clean question (100% and 100%): sentences

about different subjects read as contradictions. Its neutral flag changes conflict accuracy by -3.3 points [-6.3, -0.4]; removal changes conflict accuracy by $+0.4$ and costs -67.0 points [-75.7, -57.7] on clean questions. Off-the-shelf NLI is no usable detector for these passages at any window we tried.

Shared responses (PR§21.11). The response cache keys on the model, the full prompt, the requested temperature, the seed and the token budget, so two conditions that render the same prompt share one sampled answer. A randomised flag that names the misinformation passage renders exactly the oracle directional prompt, and one that names the correct passage can match a real detector's wrong flag. About two thirds of the flagged conflict answers in the randomised runs of gpt-4o-mini and Claude Haiku (PR§16), Qwen2.5-7B and Claude Sonnet (PR§21) and in every ConflictBank-fresh randomised run came from earlier conditions this way; gpt-6-luna's and gpt-6.1-terra's eval runs requested another temperature and share none. Each shared answer is a real sample for its prompt, so no estimate is biased, but Δ_R shares answers with the oracle directional rows and is not independent evidence. The stated confidence prompts (C5) are new.

Registration	Commit	Push	First run file	Before first run? Commit	Push
Base (H1–H9)	10-27 02:48	10-27 05:14	10-27 02:48	yes	no
PR§9 (H10–H11)	10-27 15:12	10-27 17:18	10-27 15:13	yes	no
PR§9.4 (Qwen seed)	10-27 15:18	10-27 17:18	10-27 15:13	no	no
PR§9.5 (inverted oracle)	10-27 15:24	10-27 17:18	10-27 15:24	yes	no
PR§9.6 (Qwen aborts)	10-27 16:27	10-27 17:18	10-27 15:13	no	no
PR§10 (H12–H14)	10-27 18:34	10-27 21:48	10-27 18:34	yes	no
PR§11 (H15–H18)	10-27 22:09	10-27 22:09	10-27 22:10	yes	yes
PR§12 (H19)	10-27 23:29	10-27 23:29	10-27 23:37	yes	yes
PR§13 (H20–H24)	10-28 01:48	10-28 01:48	10-28 01:49	yes	yes
PR§14 (H25–H26)	10-28 01:54	10-28 01:55	10-28 01:55	yes	yes
PR§15 (H27–H29)	10-28 14:43	10-28 14:43	10-28 14:49	yes	yes
PR§16 (H30–H31)	10-28 18:12	10-28 18:12	10-28 18:13	yes	yes
PR§17 (re-analyses, baselines, judge)	10-29 13:38	10-29 14:52	10-29 13:40	yes	no
PR§17.6 (deviations)	10-29 14:04	10-29 14:52	10-29 13:45	no	no
PR§18 (re-analyses)	10-29 21:09	10-29 21:35	–	–	–
PR§19 (H32–H38)	10-29 22:57	10-29 22:57	10-29 22:57	yes	yes
PR§20 (re-analyses)	11-02 18:00	11-02 18:00	–	–	–
PR§21 (H40–H41)	11-03 03:59	11-03 03:59	11-03 04:00	yes	yes

Table 34: Registration timeline (UTC, 2026). *Commit*: local git time; *Push*: when the remote first held the commit (GitHub's activity log, a third-party record); *First run file*: the earliest file in the runs that section registered (local file metadata). Sections without runs (PR§18, PR§20) register analyses only; PR§21's C4 predictions were committed after its no-flag files and before its flagged runs. PR§9.4 and PR§9.6 changed the Qwen runs already under way, and PR§17.6 followed the judge outputs (PR§20.1).

H Reproducibility

Runs. One script produces every number from the committed per-call responses, token counts, detector outputs and configurations. Table 34 gives, for every registration, its local commit time, the time the public remote first held it (GitHub's activity log, a third-party record) and the earliest file of the runs it registered. The base registration (H1–H9) was committed before any confirmatory run but reached the remote only after they had started; PR§9 and PR§10 likewise; PR§11–PR§16 and PR§19 were on the remote before their first run. Three items changed runs already under way (PR§20.1): PR§9.4 set the Qwen runs to one seed after they had started (none of their output had been read), PR§9.6 decided how to score Qwen's aborted generations during the run, and PR§17.6 replaced the named judge after its outputs had been written. Parts of three tests reuse runs that predate their registration: 5 of the 30 H24 cells (PR§13.3), every H27 comparison of a cell with the detector model alone, and H28's gpt-4o-mini RAMDocs rows (PR§15.3). Each addendum after the base followed a review of the results so far, so these registrations fix how an analysis is run, not which analyses are chosen. We used no third-party registry; the remote's push times are the only outside record (van Miltenburg et al., 2021).

Deviations from the pre-registration.

1. *Primary metric.* The registration first named the bidirectional metric primary. Before the confirmatory run we switched to one-way matching, because it is closer to MADAM-RAG's published score on the pilot split (§4); bidirectional became the robustness metric. Table 11 repeats the main results under it; Figure 1 and Tables 30 and 3 use the one-way matcher only.

2. *Second generator.* The plan named an open-weights model. None was available before the confirmatory run, so we used gpt-4.1 (H9). We later added Qwen2.5-7B-Instruct (PR§9). At about 3 s per local call, its runs use one seed (PR§9.4, committed after the run had started, before any Qwen output was read).

3. *CONFLICTS judge.* The plan called for an outside-family judge validated on about 100 human labels. Neither was available, so CONFLICTS is scored, as exploratory, only on items with a reference answer (three seeds).

4. *H4.* The pre-registration counted 54 false-flag cells. On clean questions, cells that differ only in direction precision render identical prompts (their scores differ by exactly 0), so they are one comparison: 18 in all, 9 per format. We report H4a per format. The cell-permutation p-value for the dose-response correlation assumed exchangeable cells, but they are nested, so we use a question-level bootstrap CI. H4c had no threshold, so we report a count.

5. *H5 and the equivalence margin.* A non-significant difference is not equivalence. We add a TOST with a ± 3-point margin, chosen in PR§9.3 after the confirmatory run (half the observed H2 effect), so it is post hoc; it was fixed before every later run that uses it.

6. *H6.* We report the pre-registered test (actual effect inside a bootstrap prediction interval) and, as an addition, a paired test of the two spaces' errors.

7. *Placement modes.* For H6, detectors without direction ran under both `remove_suspect` and `remove`; as the labels render identical prompts, the response cache served every duplicate.

8. *Significance.* One scheme applies throughout (§4): a registered test takes its family's Holm adjustment, every other comparison joins one exploratory Holm family per metric.

9. *Single- and multi-answer split.* The split of conflict questions by number of gold answers (Appendix C.1) was chosen after seeing the minimal-skeleton run and is exploratory.

10. *PR§10.* H12–H14 (ConflictBank, the neutral flag with a resolution instruction, and the published baselines), the Qwen2.5-7B detector and the ConflictBank sampling rule. Its gpt-4.1 runs use seed 0 only. One ConflictBank gpt-4o-mini run was resumed from the response cache after a transient API error.

11. *PR§11 and PR§12.* H15–H18 and H19. PR§12 dropped two PR§11 runs before they started: gpt-4.1 under majority misinformation (replaced by gpt-6.1-terra in H17) and gpt-4.1 seeds 1–2, so H14 with gpt-4.1 stays on seed 0 as first registered, and every gpt-4.1 baseline comparison is seed-0 matched.

12. *Reasoning models and Claude.* gpt-6-luna and gpt-6.1-terra accept only their default temperature and require `max_completion_tokens`; we used their defaults with a 4,096-token limit. They returned 25 empty responses (almost all closed-book) because reasoning used every allowed token; we score them wrong. On our endpoint Claude takes no temperature and no JSON mode: it ran at its default (run configurations record the requested 0.1), and as a detector its JSON goes to the regex fallback. For gpt-6-luna and gpt-6.1-terra the no-flag condition was sampled twice in some run families, because the response cache keys on the requested temperature; Appendix F (R7) reports how much this moves gpt-6.1-terra's estimates.

13. *PR§13 and PR§14.* H20–H24 and H25–H26 (Claude and RAGuard). Gemini, planned as a generator, was not run: no API key was available then. The answer parser unwraps paired tags (`<answer>X</answer>`), a format no earlier result contained.

14. *Permutation resolution.* With 10,000 permutations the smallest p-value is 10^{-4}, which cannot survive Holm over 636 exploratory comparisons. After the PR§14 runs, before analysing them, we extended to 200,000 permutations whenever none of the first 10,000 reached the observed statistic. Only p-values at their floor change, and only downward.

15. *H24 additions.* The cross-cell break-even point, the count of positive cells and the mechanism score (credit for naming the false side where the generator errs, charge for marking a correct passage where it is right) were defined after seeing the H24 cells and are exploratory. H24 itself is supported only mechanically (Appendix C).

16. *PR§15.* H27–H29 and re-analyses. Its runs did not follow the registered priority order; no run was dropped. The question-level model adds statsmodels 0.14.5. H21(c) is reported as a ratio, correct over answered; the registered per-question mean averaged the two conditions over different questions (those not abstained in every seed). Both measures are in the released statistics.

17. *PR§16.* H30–H31 and re-analyses. The pooled estimate uses a multilevel model, not the DerSimonian–Laird bootstrap. The moderators were recoded (answer policy; natural misinformation is RAGuard only) with the per-setting effects already known. "Following" a flag has one definition for right and wrong flags (PR§16.3).

18. *Pooled equivalence tests.* The pooled TOSTs for H23 and H30 are unregistered; we added them after the per-comparison TOSTs were seen.

19. *PR§17.* Re-analyses, corrected baselines and the semantic re-score. Its deviations (PR§17.6) were committed after the judge outputs existed: the judge is gemini-3.5-flash-lite, because the named Gemini 2.5 Flash was closed to new users, and the registered 1,500-character NLI window flags no ConflictBank pair, so we add a 300-character window, chosen after seeing detection output only, and report the registered window's results unchanged.

20. *PR§18 and PR§19.* The PR§18 pair-rule counts were first tabulated on 2026-10-29, before PR§18 was written (PR§18.5); its other analyses were not. PR§19 (H32–H38) records no deviation. H37 followed a smoke test on HealthContradict; PR§19.3 records its numbers, and we cite them nowhere as results.

21. *PR§20.* Re-analyses registered after an external review; every number it adds is exploratory (Appendix F). R3 and R13 were first tabulated on 2026-11-02, before PR§20 was written. The human-audit sheet of PR§15.4.7 is superseded by a blind disagreement sheet (PR§20.3; Appendix D).

22. *PR§21.* The funded runs after the review (Appendix G). Four registered runs wait for human labels and did not run: the HealthContradict inverted oracle and frozen cells (C2, C10) and the judge re-scores (C12, C13). Runs went in parallel lanes by provider, not in the listed order, which governed only the budget gate; the gate never bound (PR§21.10). About two thirds of the C4 flagged answers were shared through the response cache with earlier cells (PR§21.11); the note disclosing this was written after preliminary results had been read.

23. *Shelved pilots.* Unregistered pilots on PoisonedRAG (Zou et al., 2025) and on HoH were stopped before any analysis; their runs are released.

24. *Run interruptions.* Several runs were stopped and resumed (a transient API error; a reordered run queue), keeping completed rows and re-requesting only missing ones; no row was resampled.

RAMDocs document types are imperfect. In the release, 163 documents typed *correct* contain a wrong answer, some as injected misinformation, others incidentally. Pairs touching them are excluded from pair-level metrics. Questions whose only sign of conflict is such a document (55 of 500; 35 of the confirmatory split) count only in "All". Treating them as misinformation is a robustness check. The sheet listing these documents, `ramdocs_suspect_docs.csv` in `runs/labelling`, is released unlabelled.

Power. With 139 RAMDocs conflict questions, the minimum detectable effect at 80% power is 4–12 points (median 5.8; Card et al., 2020).

Metric choice. With the one-way metric, our port of MADAM-RAG scores 28.5% on the pilot split and 32.8% on all 500 questions (published: 28.00, on an undisclosed subset); the bidirectional metric gives 38.4% on all 500. With the subset and scorer unknown, we do not claim to reproduce the published number. Normalisation lower-cases and removes punctuation and articles.

WikiContradict and CONFLICTS. A WikiContradict answer is correct if both dataset answer strings (one per passage) match a final answer under the bidirectional rule. From CONFLICTS we use the 237 questions with a reference answer: no-conflict, complementary-information, outdated-information, misinformation and conflicting-opinion items, of which the last three count as conflict questions. An answer is correct if it contains the reference answer.

Baselines. We port MADAM-RAG unchanged to our client: three debate rounds, temperature 0, 128 tokens per call. Astute RAG has no official code; we use its appendix prompts verbatim with its defaults (t=1 consolidation round, m=1 generated passage, temperature 0, 1,024 output tokens). We drop a generated passage that says "I don't know"; the answer is the text in its `<ANSWER>` tag. We port TrustRAG from its repository (commit `11dcea0`, MIT licence): SimCSE CLS embeddings (`sup-simcse-bert-base-uncased`), a ROUGE-L gate of 0.25, 2-means clustering and a cosine threshold of 0.88, then its three prompts verbatim at temperature 0.1. The registered runs omit its system message (our client then sent only user messages); the PR§17 rerun restores it (Appendix B). The *multi* variants change one sentence of the final step to ask for every valid answer (Astute RAG) or append our final-answer line (TrustRAG); we also report the unmodified variants.

ConflictBank. We use the QA file of the Hugging Face release (revision `92866a0`). We sample questions round-robin over relations with seed 0; distractors are other subjects' correct passages. The sampling script, item ids and seed are released; we do not redistribute the data, because the release states a licence in its README but ships no licence file.

RAGuard. We use the Hugging Face release `UCSC-IRKM/RAGuard` (revision `dae344a2`, MIT licence on the card). A post is usable if, without the "[Link Post]" placeholder, it has at least 15 words and is not a duplicate within its claim; posts are cut to 300 words and shuffled by claim id. Each supporting $\times$ misleading pair is a contradiction, with the misleading post as the misinformation side. The build script and item ids are released.

Model configuration. Models were called by these names between 2026-10-26 and 2026-10-29, and on 2026-11-02–03 for PR§21; before PR§21 the served snapshot identifiers were not recorded, and PR§21 rows record the model name the API reports serving (gpt-4o-mini: `gpt-4o-mini-2024-07-18`). gpt-4o-mini and

gpt-4.1 ran at temperature 0.1 with at most 256 output tokens. gpt-6-luna and gpt-6.1-terra ran at default temperature and reasoning effort, with 4,096 completion tokens including reasoning. Claude Haiku 5.5 and Sonnet 5.5 ran through Anthropic's OpenAI-compatible endpoint at their default temperature with 4,096 output tokens. Qwen2.5-7B-Instruct ran through Ollama 0.40.11 (`qwen2.5:7b-instruct`, image `845dbda0ea48`, Q4_K_M quantisation, 8,192-token context, temperature 0.1) on an Apple M4 Pro with 48 GB. Missing final-answer lines (R9) are in Table 29.

Cost and compute. All API calls, pilots included, cost $218.39. Prices per million input, cached-input and output tokens, checked on 2026-10-27: gpt-4o-mini $0.15, $0.075 and $0.60; gpt-4.1 $2.00, $0.50 and $8.00; gpt-6-luna $0.10, $0.01 and $0.50; gpt-6.1-terra $1.00, $0.05 and $5.00; Claude Haiku 5.5 $0.10, $0.01 and $0.50; Claude Sonnet 5.5 $2.00, $0.10 and $10.00. The PR§17.3 judge, gemini-3.5-flash-lite, cost $0.30 input and $2.50 output (checked on 2026-10-29). CAAG and NLI ran on a laptop CPU (0.2 s and 1.2 s per question).

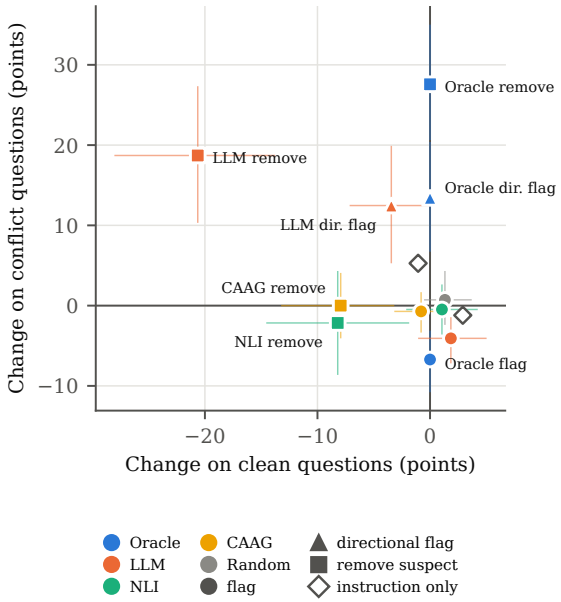


Figure 4: Change in one-way strict accuracy on RAMDocs conflict questions (y) against clean questions (x), relative to no flags (gpt-4o-mini, default skeleton, confirmatory split, 95% CIs). Colour: detector; shape: format. The oracle sits on the y-axis because it never flags a clean question. Without direction, removal deletes the second passage of each pair.

I Synthetic Detectors and the Grid

Construction. We build each synthetic detector (§3) from the gold labels. Each gold contradiction pair is flagged with probability r. False flags are drawn from non-contradicting pairs at a single per-pair rate, set over the whole split so that expected pair precision is p; clean questions therefore receive false flags too. A true flag names the gold misinformation side with probability d, its direction precision, and otherwise the other side. A false flag has no misinformation side, so its suspect is picked uniformly. Each pair gets one uniform draw for flagging and one for direction, and every cell reuses them. Cells are therefore nested: raising r or d, or lowering p, only adds flags or corrections, which keeps comparisons across cells paired. H31 applies this detector to ConflictBank with every true pair flagged, no false flags and $d = 0.5$.

Gains rise with direction precision; pair precision matters mainly for removal. Figure 5 replaces the oracle with synthetic detectors of known quality. Gains rise with direction precision in both formats. Only the cells with perfect direction are significant on all questions (directional flag +6, removal +13 points at perfect pair precision). Neighbouring cells at direction precision 0.75 have similar point estimates on conflict questions, so the pattern is a dose-response, not a threshold. With 36 cells under Holm, the grid has little power to separate adjacent levels.

Pair precision matters mainly for removal. On clean questions a cell's prompt does not depend on direction precision, so the false-flag cells reduce to 9 distinct comparisons per format. Removal is significantly harmful in 6 of them, the directional flag in 0 (H4a). Under removal, false flags delete correct passages, which costs clean questions 15 points at pair precision 0.5 and 26 at 0.25, while synthetic false directional flags cost at most two points. The harm tracks the share of clean questions that lose a correct document (Spearman $\rho = -0.39$ over 18 distinct cells, 95% CI $[-0.74, -0.12]$ from a question-level bootstrap; H4b). This support is label-sensitive. Treating every suspect document as misinformation, ρ is -0.08 (Table 12): a permutation test gives two-sided $p = 0.75$, and a bias-corrected (BCa) interval, $[-0.32, +0.01]$, includes zero. The percentile interval printed there lies mostly below the point estimate, because the bootstrap distribu-

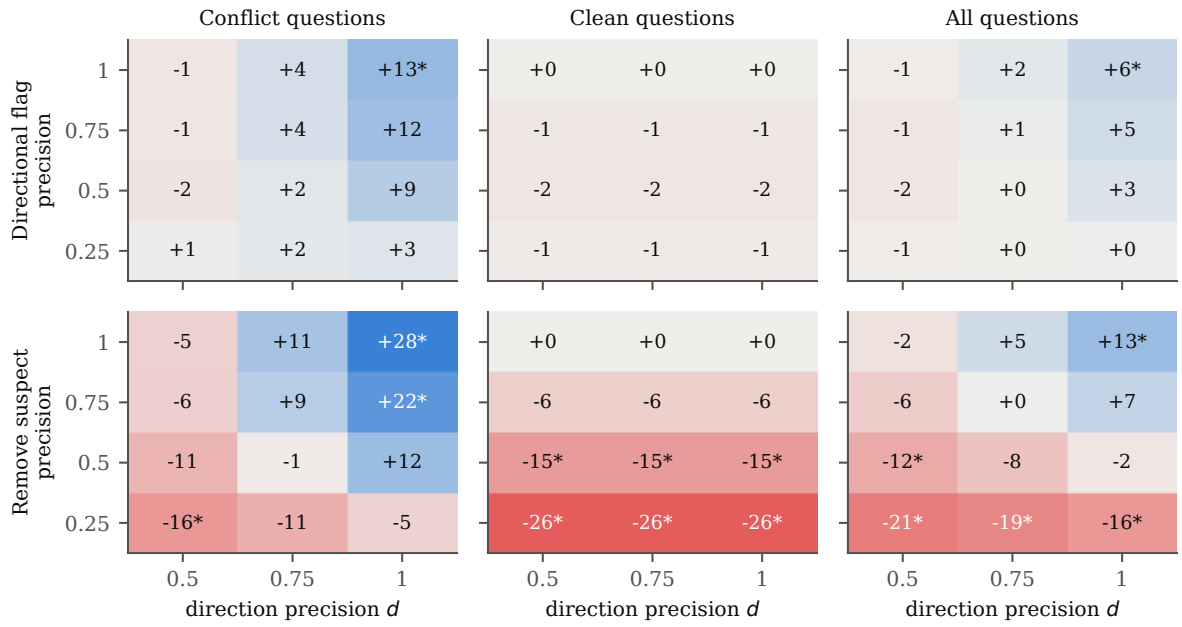


Figure 5: Synthetic detectors at recall 1: change in one-way strict accuracy (points) against no flags, by pair precision (rows) and direction precision d (columns, labelled direction accuracy in the plot), for the directional flag (top) and removal (bottom). *: significant after Holm correction over the 36 cells of that format and subset. Clean questions only see false flags, so their rows do not vary with direction precision.

tion is skewed, so it cannot be read as support (R19, Appendix F). We therefore record H4b as *supported (label-sensitive)*. Removal hurts clean questions significantly more than the directional flag in 5 of 9 matched cells (H4c; we set no threshold, so we report the count). Removal is both more powerful and more fragile: its best cell gains twice the directional flag's on all questions, but below pair precision 0.75 it is the only format that does net harm.

Placing real detectors on the grid. We compare two ways of summarising a real detector. *Pair-level*: its pair precision, recall and direction precision. *Document-level*: what it does to each subset's documents in a given format. For conflict questions, that is the share whose misinformation documents are all marked and the share that have a correct document marked; for clean questions, only the latter. Here "marked" means removed (remove) or headed as misinformation (directional flag). The prediction for a subset is the inverse-distance-weighted mean effect of the four nearest grid cells in that space.

Both summaries place most real effects inside the prediction interval. Figure 2 (Appendix B) tests whether the grid predicts real detectors' effects (H6). As pre-registered, we bootstrap the whole prediction (cell effects, every detector's location and the interpolation) over questions, and ask whether the actual effect falls inside the 95% interval. Document-level locations put 7 of 8 effects inside, and pair-level locations 6. Document-level errors are smaller on average (4.1 vs. 5.3 points), but the paired difference, -1.2 $[-3.6, +0.4]$ (exact sign-flip $p = 0.41$), is not significant over 8 cases. Both spaces miss the same case, the LLM detector's removal gain on conflict questions: one correctly directed pair suffices to remove a misinformation document, so the detector removes the misinformation on 61.2% of conflict questions despite its low pair recall. We chose the document-level axes after seeing the LLM detector on the pilot split. The grid is a useful first estimate of a detector's effect, but with three detectors we cannot claim that one summary predicts better than the other.

J Registered Tests

The tables below list every registered test, grouped by the registration that added it (times in Table 34), except H32–H38, which are reported with their data in Appendix E. Their status column is the registered outcome. Two statuses need a qualifier: H4b is *supported (label-sensitive)* (Appendix I),

and H24 is *supported (mechanical)*, because the generator's accuracy enters both sides of its tests (Appendix C).

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H1: detection quality					
H1	F1 llm_dir > nli	+2.6 [−2.4, +7.5]	0.311 (0.411)	not supported	–
H1	F1 nli > caag	+12.2 [+7.5, +16.9]	<0.001 (<0.001)	supported	–
H1	direction accuracy llm_dir > 50% (cluster CI[r])	88.3% [83.3, 92.9]	<0.001 (<0.001)	supported	–
H2: a correct neutral flag					
H2	oracle+flag < plain (conflict)	−6.7 [−10.8, −3.1]	<0.001 (0.002)	supported	yes
H3: what direction adds					
H3	oracle+flag_directional > plain (conflict)	+13.4 [+6.7, +20.4]	<0.001 (<0.001)	supported	yes
H3	oracle+flag_directional > oracle+flag (conflict)	+20.1 [+13.9, +26.9]	<0.001 (<0.001)	supported	yes
H3	oracle+remove_suspect > plain (conflict)	+27.6 [+20.6, +35.0]	<0.001 (<0.001)	supported	yes
H4: false flags					
H4a	false-flag cells below plain on clean q., directional flag[r]	0 of 9 cells	Holm	not supported	yes
H4a	false-flag cells below plain on clean q., removal[r]	6 of 9 cells	Holm	supported	yes
H4b	harm grows with clean-q. exposure, Spearman over 18 cells[r]	$\rho = -0.39$ [−0.74, −0.12]	–	supported[L]	yes
H4c	removal harms clean q. more than directional flag[r]	5 of 9 matched cells	Holm	count only	–
H5: the heuristic detector vs random					
H5	caag+flag vs random+flag (all)	−1.4 [−3.4, +0.3]	0.178 (0.256)	n.s.	yes
H5	equivalence within ±3 points (TOST)[r]	90% CI [−3.1, +0.1]	–	not equivalent	yes
H6: does the grid predict real detectors					
H6	actual effect inside the 95% interval of the document-level prediction[r]	7 of 8 (pair level: 6 of 8)	–	descriptive	–
H6	document-level error < pair-level error (paired, exact sign-flip)[r]	−1.2 [−3.6, +0.4]	0.414	not supported	yes
H7: prompt-only instructions					
H7	instruct vs plain (conflict)	−1.2 [−4.3, +1.7]	1.000 (0.634)	n.s.	yes
H7	instruct vs plain (clean)	+2.9 [0.0, +6.1]	0.280 (0.145)	n.s.	yes
H7	instruct_v2 vs plain (conflict)	+5.3 [+1.4, +9.6]	0.059 (0.027)	n.s.	yes
H7	instruct_v2 vs plain (clean)	−1.1 [−5.0, +2.6]	1.000 (0.784)	n.s.	yes
H9: gpt-4.1					
H9	gpt-4.1: oracle+flag < plain (conflict)	−8.6 [−13.2, −4.3]	<0.001 (<0.001)	supported	yes
H9	gpt-4.1: oracle+flag_directional > plain (conflict)	+18.2 [+12.0, +24.7]	<0.001 (<0.001)	supported	yes
H9	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	+26.9 [+20.1, +33.6]	<0.001 (<0.001)	supported	yes
H9	gpt-4.1: oracle+remove_suspect > plain (conflict)	+30.2 [+22.5, +37.9]	<0.001 (<0.001)	supported	yes
H8: genuine conflicts (WikiContradict)					
H8	WikiContradict: instruct vs plain	+8.7 [+5.4, +12.3]	<0.001 (<0.001)	significant	–
H8	WikiContradict: instruct_v2 vs plain	−6.7 [−10.0, −3.7]	<0.001 (<0.001)	significant	–
H8	WikiContradict: oracle+flag vs plain	+6.5 [+3.0, +10.0]	<0.001 (<0.001)	supported	–
H8	WikiContradict: caag+flag vs plain	+0.7 [0.0, +1.6]	0.250 (0.343)	n.s.	–
H8	WikiContradict: nli+flag vs plain	+3.2 [+0.3, +6.1]	0.077 (0.066)	n.s.	–
H8	WikiContradict: llm_dir+flag vs plain	+6.2 [+2.8, +9.6]	<0.001 (<0.001)	significant	–

Table 35: Pre-registered tests registered before the confirmatory run (RAMDocs hold-out unless stated; one-way strict accuracy; differences in points). Rows marked [r] are re-analyses registered later, before they were computed. Directional predictions are *supported* or *not supported*; tests with no predicted sign are *significant* or *n.s.* *Robust*: the same outcome under the bidirectional metric (robustness to the RAMDocs label rule is in Table 12, where H4b's correlation weakens). [L]: supported but label-sensitive (Table 12; PR§20, R19). [M]: supported only mechanically, because generator accuracy enters both sides of the H24 tests (Appendix J). p-values come from 10,000 sign-flip permutations, extended to 200,000 when none reaches the observed statistic, Holm-adjusted within the family; this is the significance used everywhere in the paper. In brackets: the Benjamini–Hochberg q over all registered tests of the programme (primary metric).

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H10: without the answer policy (minimal skeleton)					
H10	gpt-4o-mini: oracle+flag vs plain (conflict)	-5.3 [-9.4, -1.4]	0.042 (0.020)	significant	yes
H10	gpt-4o-mini: oracle+flag_directional > plain (conflict)	$+5.0$ [-1.7, $+11.8$]	0.314 (0.230)	not supported	yes
H10	gpt-4o-mini: oracle+flag_directional > oracle+flag (conflict)	$+10.3$ [$+4.8$, $+16.1$]	0.005 (0.001)	supported	yes
H10	gpt-4o-mini: oracle+remove_suspect > plain (conflict)	$+18.9$ [$+12.5$, $+25.9$]	<0.001 (<0.001)	supported	yes
H10	gpt-4.1: oracle+flag vs plain (conflict)	-11.0 [-17.5, -4.6]	0.010 (0.003)	significant	yes
H10	gpt-4.1: oracle+flag_directional > plain (conflict)	$+12.7$ [$+5.8$, $+19.7$]	0.005 (0.001)	supported	yes
H10	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	$+23.7$ [$+17.0$, $+30.7$]	<0.001 (<0.001)	supported	yes
H10	gpt-4.1: oracle+remove_suspect > plain (conflict)	$+14.9$ [$+8.6$, $+21.6$]	<0.001 (<0.001)	supported	yes
H10	Qwen2.5-7B: oracle+flag vs plain (conflict)	-5.8 [-10.8, -1.4]	0.117 (0.066)	n.s.	yes
H10	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	$+2.9$ [-3.6, $+9.4$]	0.509 (0.611)	not supported	yes
H10	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	$+8.6$ [$+2.9$, $+14.4$]	0.041 (0.016)	supported	yes
H10	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	$+10.8$ [$+4.3$, $+17.3$]	0.017 (0.006)	supported	yes
H11: an open-weights generator					
H11	Qwen2.5-7B: oracle+flag < plain (conflict)	-5.8 [-10.8, -0.7]	0.114 (0.091)	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	$+3.6$ [-0.7, $+8.6$]	0.224 (0.316)	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	$+9.4$ [$+5.0$, $+14.4$]	0.001 (0.001)	supported	yes
H11	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	$+18.0$ [$+10.8$, $+25.2$]	<0.001 (<0.001)	supported	yes

Table 36: Pre-registered tests added in PR§9 (H10–H11), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H12: ConflictBank					
H12	gpt-4o-mini: neutral flag vs plain (conflict)	−8.8 [−12.8, −4.9]	<0.001 (<0.001)	significant	yes
H12	gpt-4o-mini: dir. flag > plain (conflict)	+27.9 [+22.8, +32.9]	<0.001 (<0.001)	supported	yes
H12	gpt-4o-mini: dir. flag > neutral flag (conflict)	+36.7 [+31.6, +41.9]	<0.001 (<0.001)	supported	yes
H12	gpt-4o-mini: removal > plain (conflict)	+66.4 [+61.1, +71.7]	<0.001 (<0.001)	supported	yes
H12	gpt-4.1: neutral flag vs plain (conflict)	−6.3 [−9.7, −3.3]	<0.001 (<0.001)	significant	yes
H12	gpt-4.1: dir. flag > plain (conflict)	+23.0 [+18.0, +28.3]	<0.001 (<0.001)	supported	yes
H12	gpt-4.1: dir. flag > neutral flag (conflict)	+29.3 [+24.0, +34.7]	<0.001 (<0.001)	supported	yes
H12	gpt-4.1: removal > plain (conflict)	+78.3 [+73.3, +83.0]	<0.001 (<0.001)	supported	yes
H12	Qwen2.5-7B: neutral flag vs plain (conflict)	−8.3 [−12.3, −4.3]	<0.001 (<0.001)	significant	yes
H12	Qwen2.5-7B: dir. flag > plain (conflict)	+17.7 [+12.0, +23.3]	<0.001 (<0.001)	supported	yes
H12	Qwen2.5-7B: dir. flag > neutral flag (conflict)	+26.0 [+20.7, +31.3]	<0.001 (<0.001)	supported	yes
H12	Qwen2.5-7B: removal > plain (conflict)	+63.7 [+58.3, +69.0]	<0.001 (<0.001)	supported	yes
H13: neutral flag + resolution instruction					
H13	RAMDocs, gpt-4o-mini: neutral flag + instruction vs plain (conflict)	−2.4 [−6.2, +1.2]	0.743 (0.350)	n.s.	yes
H13	RAMDocs, gpt-4o-mini: dir. flag + instruction > neutral flag + instruction (conflict)	+15.6 [+9.6, +21.8]	<0.001 (<0.001)	supported	yes
H13	RAMDocs, gpt-4.1: neutral flag + instruction vs plain (conflict)	+5.8 [−2.9, +14.4]	0.743 (0.342)	n.s.	yes
H13	RAMDocs, gpt-4.1: dir. flag + instruction > neutral flag + instruction (conflict)	+25.2 [+18.0, +32.4]	<0.001 (<0.001)	supported	yes
H13	ConflictBank, gpt-4o-mini: neutral flag + instruction vs plain (conflict)	+1.6 [−2.8, +6.0]	0.743 (0.619)	n.s.	yes
H13	ConflictBank, gpt-4o-mini: dir. flag + instruction > neutral flag + instruction (conflict)	+62.2 [+57.1, +67.2]	<0.001 (<0.001)	supported	yes
H14: published baselines					
H14	RAMDocs, gpt-4o-mini: LLM dir. flag vs astute multi (labelled)	+10.8 [+5.9, +16.0]	<0.001 (<0.001)	significant	yes
H14	RAMDocs, gpt-4o-mini: LLM dir. flag vs trustrag multi (labelled)	+23.0 [+18.1, +28.3]	<0.001 (<0.001)	significant	yes
H14	RAMDocs, gpt-4.1: LLM dir. flag vs astute multi (labelled)	+6.4 [+0.4, +12.5]	0.151 (0.082)	n.s.	no
H14	RAMDocs, gpt-4.1: LLM dir. flag vs trustrag multi (labelled)	+19.6 [+13.6, +25.7]	<0.001 (<0.001)	significant	yes
H14	ConflictBank, gpt-4o-mini: LLM dir. flag vs astute (all questions)	−3.4 [−8.1, +1.3]	0.324 (0.236)	n.s.	yes
H14	ConflictBank, gpt-4o-mini: LLM dir. flag vs trustrag (all questions)	+0.8 [−3.9, +5.6]	0.761 (0.841)	n.s.	yes

Table 37: Pre-registered tests added in PR§10 (H12–H14), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H15: ConflictBank, single-answer prompt					
H15	ConflictBank, gpt-4o-mini: neutral flag vs plain (conflict)	−18.7 [−23.4, −14.0]	<0.001 (<0.001)	significant	yes
H15	ConflictBank, gpt-4o-mini: dir. flag > neutral flag (conflict)	+48.8 [+43.6, +54.1]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4o-mini: dir. flag > plain (conflict)	+30.1 [+25.1, +35.2]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4o-mini: LLM dir. flag > plain (all q.)	+5.2 [+1.0, +9.3]	0.065 (0.029)	not supported	no
H15	ConflictBank, gpt-4.1: neutral flag vs plain (conflict)	−8.0 [−12.7, −3.7]	0.006 (0.003)	significant	yes
H15	ConflictBank, gpt-4.1: dir. flag > neutral flag (conflict)	+34.7 [+29.0, +40.3]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4.1: dir. flag > plain (conflict)	+26.7 [+21.7, +31.7]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4.1: LLM dir. flag > plain (all q.)	−2.5 [−6.5, +1.5]	0.522 (0.355)	not supported	yes
H15	ConflictBank, Qwen2.5-7B: neutral flag vs plain (conflict)	−3.3 [−9.0, +2.0]	0.522 (0.389)	n.s.	yes
H15	ConflictBank, Qwen2.5-7B: dir. flag > neutral flag (conflict)	+28.7 [+23.3, +34.0]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, Qwen2.5-7B: dir. flag > plain (conflict)	+25.3 [+19.7, +31.0]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, Qwen2.5-7B: LLM dir. flag > plain (all q.)	+5.2 [+0.5, +10.0]	0.130 (0.072)	not supported	yes
H16: information vs direction (one-wrong flag)					
H16	ConflictBank, gpt-4o-mini: dir. flag > one-wrong flag (conflict)	+65.8 [+60.6, +70.9]	<0.001 (<0.001)	supported	yes
H16	ConflictBank, gpt-4o-mini: one-wrong flag vs plain (conflict)	−35.7 [−41.1, −30.3]	<0.001 (<0.001)	significant	yes
H16	ConflictBank, gpt-4.1: dir. flag > one-wrong flag (conflict)	+62.0 [+56.0, +67.7]	<0.001 (<0.001)	supported	yes
H16	ConflictBank, gpt-4.1: one-wrong flag vs plain (conflict)	−35.3 [−41.3, −29.3]	<0.001 (<0.001)	significant	yes
H16	RAMDocs, gpt-4o-mini: dir. flag > one-wrong flag (conflict)	+20.6 [+14.4, +27.3]	<0.001 (<0.001)	supported	yes
H16	RAMDocs, gpt-4o-mini: one-wrong flag vs plain (conflict)	−7.2 [−12.2, −2.6]	0.006 (0.011)	significant	yes
H17: majority misinformation					
H17	CB majority, gpt-4o-mini: neutral flag vs plain (conflict)	+1.1 [−2.7, +4.9]	0.603 (0.700)	n.s.	yes
H17	CB majority, gpt-4o-mini: dir. flag > plain (conflict)	+32.7 [+27.3, +37.9]	<0.001 (<0.001)	supported	yes
H17	CB majority, gpt-4o-mini: LLM dir. flag vs plain (conflict)	+5.7 [+1.4, +9.9]	0.017 (0.016)	significant	yes
H18: a single-call baseline					
H18	RAMDocs, gpt-4o-mini: LLM dir. flag vs self-identify (labelled)	−1.3 [−4.8, +2.1]	1.000 (0.624)	n.s.	yes
H18	ConflictBank, gpt-4o-mini: LLM dir. flag vs self-identify (all questions)	−0.2 [−4.2, +3.7]	1.000 (0.994)	n.s.	yes
H18	CB majority, gpt-4o-mini: LLM dir. flag vs self-identify (all questions)	−1.8 [−4.6, +1.1]	0.730 (0.337)	n.s.	yes
H19: current reasoning models					
H19	ConflictBank, gpt-6-luna: neutral flag vs plain (conflict)	−9.4 [−12.9, −6.0]	<0.001 (<0.001)	significant	yes
H19	ConflictBank, gpt-6-luna: dir. flag > neutral flag (conflict)	+45.9 [+40.8, +50.9]	<0.001 (<0.001)	supported	yes

Table 38: Pre-registered tests added in PR§11–PR§12 (H15–H19) (part 1 of 2), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H19: current reasoning models					
H19	ConflictBank, gpt-6-luna: dir. flag > one-wrong flag (conflict)	+65.6 [+60.7, +70.1]	<0.001 (<0.001)	supported	yes
H19	ConflictBank, gpt-6-luna: LLM dir. flag > plain (all q.)	+6.1 [+2.1, +10.0]	0.019 (0.008)	supported	yes
H19	ConflictBank, gpt-6.1-terra: neutral flag vs plain (conflict)	−2.3 [−6.7, +2.0]	0.717 (0.466)	n.s.	yes
H19	ConflictBank, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+19.3 [+14.7, +24.0]	<0.001 (<0.001)	supported	yes
H19	ConflictBank, gpt-6.1-terra: dir. flag > one-wrong flag (conflict)	+22.3 [+17.7, +27.3]	<0.001 (<0.001)	supported	yes
H19	ConflictBank, gpt-6.1-terra: LLM dir. flag > plain (all q.)	−8.8 [−13.2, −4.2]	0.002 (<0.001)	not supported	yes
H19	RAMDocs, gpt-6-luna: neutral flag vs plain (conflict)	−3.1 [−5.3, −1.4]	0.007 (0.003)	significant	yes
H19	RAMDocs, gpt-6-luna: dir. flag > neutral flag (conflict)	+42.2 [+34.5, +49.9]	<0.001 (<0.001)	supported	yes
H19	RAMDocs, gpt-6.1-terra: neutral flag vs plain (conflict)	−0.7 [−2.9, +1.4]	1.000 (1.000)	n.s.	yes
H19	RAMDocs, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+40.3 [+32.4, +48.2]	<0.001 (<0.001)	supported	yes
H19	CB majority, gpt-6-luna: dir. flag > plain (conflict)	+60.4 [+55.3, +65.4]	<0.001 (<0.001)	supported	yes
H19	CB majority, gpt-6-luna: LLM dir. flag vs plain (conflict)	+7.1 [+2.4, +11.8]	0.019 (0.008)	significant	yes
H19	CB majority, gpt-6.1-terra: dir. flag > plain (conflict)	+66.3 [+61.0, +71.7]	<0.001 (<0.001)	supported	yes
H19	CB majority, gpt-6.1-terra: LLM dir. flag vs plain (conflict)	+9.0 [+3.3, +14.7]	0.019 (0.008)	significant	yes

Table 39: Pre-registered tests added in PR§11–PR§12 (H15–H19) (part 2 of 2), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H20: forced choice (no "unknown")					
H20	ConflictBank forced, gpt-4o-mini: neutral flag vs plain (conflict)	−0.2 [−4.1, +3.7]	1.000 (1.000)	n.s.	yes
H20	ConflictBank forced, gpt-4o-mini: one-wrong flag vs plain (conflict)	+7.6 [+2.7, +12.3]	0.024 (0.007)	significant	yes
H20	ConflictBank forced, gpt-4o-mini: dir. flag > plain (conflict)	+24.2 [+19.7, +29.0]	<0.001 (<0.001)	supported	yes
H20	ConflictBank forced, gpt-6-luna: neutral flag vs plain (conflict)	+1.6 [−1.6, +4.8]	1.000 (0.475)	n.s.	yes
H20	ConflictBank forced, gpt-6-luna: one-wrong flag vs plain (conflict)	+10.3 [+5.6, +15.1]	<0.001 (<0.001)	significant	yes
H20	ConflictBank forced, gpt-6-luna: dir. flag > plain (conflict)	+26.4 [+21.9, +31.0]	<0.001 (<0.001)	supported	yes
H20	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain (conflict)	−0.3 [−4.0, +3.3]	1.000 (1.000)	n.s.	yes
H20	ConflictBank forced, gpt-6.1-terra: one-wrong flag vs plain (conflict)	+6.7 [+2.3, +11.3]	0.034 (0.013)	significant	yes
H20	ConflictBank forced, gpt-6.1-terra: dir. flag > plain (conflict)	+14.7 [+10.3, +19.0]	<0.001 (<0.001)	supported	yes
H20	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain (conflict)	+3.9 [+1.2, +6.7]	0.034 (0.011)	significant	yes
H20	ConflictBank forced, claude-haiku-5-5: one-wrong flag vs plain (conflict)	+5.1 [+1.4, +8.8]	0.034 (0.016)	significant	yes
H20	ConflictBank forced, claude-haiku-5-5: dir. flag > plain (conflict)	+21.1 [+16.9, +25.4]	<0.001 (<0.001)	supported	yes
H22: pair topology					
H22	RAMDocs, gpt-4o-mini: one-wrong without pairs vs with pairs (conflict)	+6.5 [+2.4, +10.8]	0.015 (0.005)	significant	yes
H22	RAMDocs, gpt-6-luna: one-wrong without pairs vs with pairs (conflict)	+10.1 [+5.8, +14.9]	<0.001 (<0.001)	significant	yes
H22	RAMDocs, gpt-6.1-terra: one-wrong without pairs vs with pairs (conflict)	−0.7 [−6.5, +5.8]	1.000 (1.000)	n.s.	yes
H22	CB majority, gpt-4o-mini: neutral without pairs vs with pairs (conflict)	−3.1 [−6.4, +0.2]	0.390 (0.123)	n.s.	yes
H22	CB majority, gpt-4o-mini: LLM dir. flag without pairs vs with pairs (conflict)	+0.3 [−2.1, +2.8]	1.000 (0.932)	n.s.	yes
H22	CB majority, gpt-6-luna: neutral without pairs vs with pairs (conflict)	−6.1 [−9.1, −3.1]	0.002 (<0.001)	significant	yes
H22	CB majority, gpt-6-luna: LLM dir. flag without pairs vs with pairs (conflict)	+0.8 [−1.7, +3.1]	1.000 (0.681)	n.s.	yes
H22	CB majority, gpt-6.1-terra: neutral without pairs vs with pairs (conflict)	+10.0 [+5.7, +14.3]	0.002 (<0.001)	significant	yes
H22	CB majority, gpt-6.1-terra: LLM dir. flag without pairs vs with pairs (conflict)	+1.7 [−1.3, +4.7]	1.000 (0.470)	n.s.	yes
H23: answer key					
H23	RAMDocs, gpt-4o-mini: dir. flag vs answer key (conflict)	−1.7 [−6.2, +3.1]	1.000 (0.654)	n.s.	yes
H23	RAMDocs, gpt-6-luna: dir. flag vs answer key (conflict)	−4.1 [−9.1, +0.7]	0.808 (0.201)	n.s.	yes
H23	RAMDocs, gpt-6.1-terra: dir. flag vs answer key (conflict)	−5.0 [−10.8, +0.7]	0.827 (0.240)	n.s.	yes
H23	ConflictBank, gpt-4o-mini: dir. flag vs answer key (conflict)	+5.3 [+1.6, +9.1]	0.059 (0.013)	n.s.	no
H23	ConflictBank, gpt-6-luna: dir. flag vs answer key (conflict)	+0.1 [−1.6, +1.8]	1.000 (1.000)	n.s.	yes

Table 40: Pre-registered tests added in PR§13–PR§14 (H20–H26) (part 1 of 5), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H23: answer key					
H23	ConflictBank, gpt-6.1-terra: dir. flag vs answer key (conflict)	-1.7 [-3.3, -0.3]	0.515 (0.103)	n.s.	yes
H23	CB majority, gpt-4o-mini: dir. flag vs answer key (conflict)	$+3.2$ [-1.8, $+8.2$]	0.905 (0.316)	n.s.	yes
H23	CB majority, gpt-6-luna: dir. flag vs answer key (conflict)	$+0.4$ [-1.2, $+2.2$]	1.000 (0.795)	n.s.	yes
H23	CB majority, gpt-6.1-terra: dir. flag vs answer key (conflict)	$+2.3$ [$+0.3$, $+4.7$]	0.515 (0.114)	n.s.	yes
H25: Claude					
H25	ConflictBank, claude-haiku-5-5: neutral flag vs plain (conflict)	$+0.7$ [-2.2, $+3.4$]	1.000 (0.788)	n.s.	yes
H25	ConflictBank, claude-haiku-5-5: dir. flag > neutral flag (conflict)	$+22.7$ [$+18.6$, $+26.9$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-haiku-5-5: dir. flag > one-wrong flag (conflict)	$+16.8$ [$+13.0$, $+20.8$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-haiku-5-5: LLM dir. flag > plain (all q.)	-4.3 [-8.1, -0.7]	0.145 (0.042)	not supported	yes
H25	ConflictBank, claude-sonnet-5-5: neutral flag vs plain (conflict)	$+3.0$ [-0.3, $+6.3$]	0.634 (0.191)	n.s.	yes
H25	ConflictBank, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	$+13.3$ [$+9.7$, $+17.3$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-sonnet-5-5: dir. flag > one-wrong flag (conflict)	$+12.3$ [$+8.7$, $+16.3$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-sonnet-5-5: LLM dir. flag > plain (all q.)	-2.2 [-5.8, $+1.0$]	0.993 (0.342)	not supported	yes
H25	RAMDocs, claude-haiku-5-5: neutral flag vs plain (conflict)	$+1.2$ [-2.6, $+5.3$]	1.000 (0.729)	n.s.	yes
H25	RAMDocs, claude-haiku-5-5: dir. flag > neutral flag (conflict)	$+27.1$ [$+20.1$, $+34.1$]	<0.001 (<0.001)	supported	yes
H25	RAMDocs, claude-sonnet-5-5: neutral flag vs plain (conflict)	0.0 [-4.3, $+4.3$]	1.000 (1.000)	n.s.	yes
H25	RAMDocs, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	$+14.4$ [$+7.9$, $+21.6$]	0.002 (<0.001)	supported	yes
H25	CB majority, claude-haiku-5-5: dir. flag > plain (conflict)	$+57.3$ [$+52.0$, $+62.7$]	<0.001 (<0.001)	supported	yes
H25	CB majority, claude-haiku-5-5: LLM dir. flag vs plain (conflict)	$+6.8$ [$+2.1$, $+11.4$]	0.043 (0.012)	significant	yes
H26: RAGuard (real misleading posts)					
H26	RAGuard, gpt-4o-mini: neutral flag vs plain (conflict)	$+3.2$ [$+0.1$, $+6.3$]	0.426 (0.086)	n.s.	yes
H26	RAGuard, gpt-4o-mini: dir. flag > plain (conflict)	$+9.9$ [$+5.6$, $+14.2$]	<0.001 (<0.001)	supported	yes
H26	RAGuard, gpt-4o-mini: dir. flag > neutral flag (conflict)	$+6.7$ [$+2.9$, $+10.5$]	0.008 (0.001)	supported	yes
H26	RAGuard, gpt-4o-mini: one-wrong flag vs plain (conflict)	-5.7 [-10.2, -1.2]	0.161 (0.029)	n.s.	yes
H26	RAGuard, gpt-4o-mini: LLM dir. flag vs plain (conflict)	$+0.6$ [-2.8, $+4.0$]	1.000 (0.877)	n.s.	yes
H26	RAGuard, gpt-6-luna: neutral flag vs plain (conflict)	0.0 [-1.7, $+1.7$]	1.000 (1.000)	n.s.	yes
H26	RAGuard, gpt-6-luna: dir. flag > plain (conflict)	$+8.8$ [$+5.7$, $+12.1$]	<0.001 (<0.001)	supported	yes
H26	RAGuard, gpt-6-luna: dir. flag > neutral flag (conflict)	$+8.8$ [$+5.7$, $+12.2$]	<0.001 (<0.001)	supported	yes

Table 41: Pre-registered tests added in PR§13–PR§14 (H20–H26) (part 2 of 5), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H26: RAGuard (real misleading posts)					
H26	RAGuard, gpt-6-luna: one-wrong flag vs plain (conflict)	-7.0 [-10.3, -3.9]	<0.001 (<0.001)	significant	yes
H26	RAGuard, gpt-6-luna: LLM dir. flag vs plain (conflict)	$+3.1$ [-0.1, $+6.2$]	0.511 (0.116)	n.s.	yes
H26	RAGuard, gpt-6.1-terra: neutral flag vs plain (conflict)	$+2.4$ [0.0, $+5.1$]	0.731 (0.187)	n.s.	yes
H26	RAGuard, gpt-6.1-terra: dir. flag > plain (conflict)	$+6.8$ [$+3.4$, $+10.5$]	0.005 (<0.001)	supported	yes
H26	RAGuard, gpt-6.1-terra: dir. flag > neutral flag (conflict)	$+4.4$ [$+1.4$, $+7.8$]	0.126 (0.021)	not supported	yes
H26	RAGuard, gpt-6.1-terra: one-wrong flag vs plain (conflict)	$+0.7$ [-3.4, $+4.8$]	1.000 (0.932)	n.s.	yes
H26	RAGuard, gpt-6.1-terra: LLM dir. flag vs plain (conflict)	$+2.0$ [-1.0, $+5.1$]	1.000 (0.389)	n.s.	yes
H26	RAGuard, claude-haiku-5-5: neutral flag vs plain (conflict)	$+0.3$ [-2.0, $+2.7$]	1.000 (0.930)	n.s.	yes
H26	RAGuard, claude-haiku-5-5: dir. flag > plain (conflict)	$+12.7$ [$+9.2$, $+16.3$]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-haiku-5-5: dir. flag > neutral flag (conflict)	$+12.4$ [$+9.0$, $+15.9$]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-haiku-5-5: one-wrong flag vs plain (conflict)	$+5.9$ [$+2.9$, $+9.0$]	0.005 (<0.001)	significant	yes
H26	RAGuard, claude-haiku-5-5: LLM dir. flag vs plain (conflict)	$+5.0$ [$+1.8$, $+8.2$]	0.025 (0.004)	significant	yes
H26	RAGuard, claude-sonnet-5-5: neutral flag vs plain (conflict)	$+4.4$ [$+0.7$, $+8.2$]	0.321 (0.061)	n.s.	yes
H26	RAGuard, claude-sonnet-5-5: dir. flag > plain (conflict)	$+20.4$ [$+15.6$, $+25.2$]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	$+16.0$ [$+11.6$, $+20.4$]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-sonnet-5-5: LLM dir. flag vs plain (conflict)	$+7.1$ [$+3.1$, $+11.2$]	0.022 (0.004)	significant	yes
H21: risk and coverage					
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, misinformation rate (conflict)	-13.3 [-17.6, -9.1]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, misinformation rate (conflict)	-26.8 [-31.6, -22.0]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, misinformation rate (conflict)	-6.9 [-9.9, -4.0]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, misinformation rate (conflict)	-24.3 [-28.6, -20.2]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, misinformation rate (conflict)	-2.3 [-6.0, $+1.3$]	1.000 (0.394)	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, misinformation rate (conflict)	-17.3 [-21.7, -13.0]	<0.001 (<0.001)	significant	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, misinformation rate (conflict)	$+0.1$ [-2.6, $+2.8$]	1.000 (1.000)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, misinformation rate (conflict)	-21.8 [-26.1, -17.6]	<0.001 (<0.001)	significant	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, misinformation rate (conflict)	$+0.3$ [-3.4, $+4.2$]	1.000 (0.973)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, misinformation rate (conflict)	-24.9 [-29.7, -20.1]	<0.001 (<0.001)	significant	–

Table 42: Pre-registered tests added in PR§13–PR§14 (H20–H26) (part 3 of 5), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H21: risk and coverage					
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, misinformation rate (conflict)	−3.8 [−6.8, −0.9]	0.079 (0.029)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, misinformation rate (conflict)	−25.7 [−30.0, −21.3]	<0.001 (<0.001)	significant	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, misinformation rate (conflict)	−1.3 [−5.0, +2.3]	1.000 (0.693)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, misinformation rate (conflict)	−14.3 [−18.3, −10.3]	<0.001 (<0.001)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, misinformation rate (conflict)	−4.1 [−6.9, −1.3]	0.025 (0.008)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, misinformation rate (conflict)	−20.9 [−25.1, −16.8]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, abstention rate (conflict)	+34.3 [+29.3, +39.2]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, abstention rate (conflict)	−4.6 [−7.2, −2.0]	0.013 (0.002)	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, abstention rate (conflict)	+17.0 [+13.6, +20.6]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, abstention rate (conflict)	−11.1 [−14.1, −8.2]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, abstention rate (conflict)	+3.0 [+0.7, +5.3]	0.281 (0.041)	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, abstention rate (conflict)	−1.3 [−2.7, −0.3]	1.000 (0.187)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, abstention rate (conflict)	−0.3 [−0.9, +0.1]	1.000 (0.475)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, abstention rate (conflict)	−0.1 [−0.9, +0.8]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, abstention rate (conflict)	+0.3 [−0.1, +0.9]	1.000 (0.475)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, abstention rate (conflict)	−0.1 [−0.3, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, selective accuracy (ratio) (conflict)	+0.2 [−4.8, +5.2]	1.000 (0.994)	n.s.	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, selective accuracy (ratio) (conflict)	+28.2 [+22.8, +33.5]	0.002 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, selective accuracy (ratio) (conflict)	+1.2 [−2.2, +4.8]	1.000 (0.596)	n.s.	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, selective accuracy (ratio) (conflict)	+29.9 [+25.0, +34.7]	0.002 (<0.001)	significant	–

Table 43: Pre-registered tests added in PR§13–PR§14 (H20–H26) (part 4 of 5), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H21: risk and coverage					
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, selective accuracy (ratio) (conflict)	-0.1 [-4.0, $+3.7$]	1.000 (0.998)	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+16.1$ [$+11.5$, $+20.6$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, selective accuracy (ratio) (conflict)	$+0.4$ [-2.3, $+3.3$]	1.000 (0.858)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+23.5$ [$+19.2$, $+27.9$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, selective accuracy (ratio) (conflict)	0.0 [-3.9, $+3.9$]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+24.2$ [$+19.4$, $+28.9$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, selective accuracy (ratio) (conflict)	$+1.6$ [-1.7, $+4.7$]	1.000 (0.475)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+26.4$ [$+21.9$, $+31.1$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, selective accuracy (ratio) (conflict)	-0.3 [-4.0, $+3.3$]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+14.7$ [$+10.3$, $+19.0$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, selective accuracy (ratio) (conflict)	$+3.9$ [$+1.2$, $+6.7$]	0.046 (0.011)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+21.1$ [$+17.0$, $+25.4$]	0.002 ($<$0.001)	significant	–
H24: detector versus generator reliability (crossed)					
H24	gain rises with detector-minus-generator reliability, Spearman over 30 cells	$\rho = +0.64$	$<$0.001	supported^{M}	–
H24	per-question slope of gain on reliability gap, within cells	$+47.6$ [$+43.2$, $+51.9$]	$<$0.001	supported^{M}	–

Table 44: Pre-registered tests added in PR§13–PR§14 (H20–H26) (part 5 of 5), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	Qwen2.5-7B, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+8.7 [+2.7, +14.3]	0.093 (0.011)	n.s.	no
H27	Qwen2.5-7B, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+4.3 [−1.7, +10.3]	1.000 (0.285)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+4.3 [−1.3, +10.0]	1.000 (0.250)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	−4.0 [−9.7, +1.7]	1.000 (0.285)	n.s.	yes
H27	claude-haiku-5-5, detector claude-haiku-5-5: LLM dir. flag vs hint (conflict)	+11.1 [+7.2, +15.1]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector claude-sonnet-5-5: LLM dir. flag vs hint (conflict)	+14.6 [+9.6, +19.7]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+14.1 [+9.8, +18.6]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	−1.4 [−6.0, +3.2]	1.000 (0.680)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+10.3 [+6.6, +14.1]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+11.0 [+7.1, +15.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+13.0 [+8.0, +18.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+0.7 [−4.7, +6.0]	1.000 (0.964)	n.s.	yes
H27	gpt-4.1, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+9.7 [+4.7, +14.7]	0.006 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+1.7 [−3.0, +6.3]	1.000 (0.686)	n.s.	yes
H27	gpt-4o-mini, detector claude-haiku-5-5: LLM dir. flag vs hint (conflict)	+6.2 [+1.1, +11.4]	0.279 (0.037)	n.s.	yes
H27	gpt-4o-mini, detector claude-sonnet-5-5: LLM dir. flag vs hint (conflict)	+5.8 [+0.2, +11.3]	0.485 (0.075)	n.s.	yes
H27	gpt-4o-mini, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+11.8 [+6.6, +17.0]	0.005 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+7.1 [+1.8, +12.6]	0.180 (0.022)	n.s.	yes
H27	gpt-4o-mini, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+8.1 [+2.9, +13.2]	0.037 (0.005)	significant	yes
H27	gpt-4o-mini, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+0.1 [−4.9, +5.2]	1.000 (1.000)	n.s.	yes
H27	gpt-6-luna, detector claude-haiku-5-5: LLM dir. flag vs hint (conflict)	+4.9 [−0.4, +10.3]	0.785 (0.137)	n.s.	yes
H27	gpt-6-luna, detector claude-sonnet-5-5: LLM dir. flag vs hint (conflict)	+3.9 [−0.8, +8.6]	0.992 (0.188)	n.s.	yes
H27	gpt-6-luna, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+13.7 [+8.4, +18.9]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+6.0 [+0.6, +11.7]	0.485 (0.069)	n.s.	yes
H27	gpt-6-luna, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+8.2 [+3.7, +12.8]	0.005 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	−5.8 [−10.6, −1.0]	0.279 (0.036)	n.s.	yes

Table 45: Pre-registered tests added in PR§15 (H27–H29) (part 1 of 5), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	gpt-6.1-terra, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+10.7 [+5.0, +16.0]	0.008 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	−6.3 [−12.0, −0.7]	0.485 (0.070)	n.s.	yes
H27	gpt-6.1-terra, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+9.7 [+4.7, +14.7]	0.005 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+9.7 [+5.0, +14.3]	0.008 (<0.001)	significant	yes
H27	Qwen2.5-7B, detector gpt-4.1: hint vs plain (conflict)	+7.3 [+2.3, +12.3]	0.071 (0.011)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-4o-mini: hint vs plain (conflict)	+3.0 [−2.0, +8.0]	1.000 (0.397)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6-luna: hint vs plain (conflict)	+7.7 [+3.0, +12.7]	0.038 (0.006)	significant	no
H27	Qwen2.5-7B, detector gpt-6.1-terra: hint vs plain (conflict)	+12.0 [+7.0, +17.0]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector claude-haiku-5-5: hint vs plain (conflict)	−5.4 [−8.8, −2.1]	0.032 (0.004)	significant	yes
H27	claude-haiku-5-5, detector claude-sonnet-5-5: hint vs plain (conflict)	−6.3 [−9.4, −3.2]	0.004 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4.1: hint vs plain (conflict)	−3.1 [−5.8, −0.4]	0.341 (0.054)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-4o-mini: hint vs plain (conflict)	−2.7 [−5.6, +0.1]	0.681 (0.120)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-6-luna: hint vs plain (conflict)	−5.3 [−8.2, −2.6]	0.011 (0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6.1-terra: hint vs plain (conflict)	−6.9 [−9.9, −4.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4.1: hint vs plain (conflict)	−1.3 [−3.3, +0.3]	1.000 (0.389)	n.s.	yes
H27	gpt-4.1, detector gpt-4o-mini: hint vs plain (conflict)	−3.0 [−6.7, +0.3]	1.000 (0.205)	n.s.	yes
H27	gpt-4.1, detector gpt-6-luna: hint vs plain (conflict)	−3.3 [−7.3, +0.3]	1.000 (0.198)	n.s.	yes
H27	gpt-4.1, detector gpt-6.1-terra: hint vs plain (conflict)	+3.3 [−0.7, +7.0]	1.000 (0.196)	n.s.	yes
H27	gpt-4o-mini, detector claude-haiku-5-5: hint vs plain (conflict)	+6.8 [+2.8, +10.9]	0.027 (0.003)	significant	yes
H27	gpt-4o-mini, detector claude-sonnet-5-5: hint vs plain (conflict)	+10.1 [+6.4, +13.8]	<0.001 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-4.1: hint vs plain (conflict)	+6.8 [+3.7, +10.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-4o-mini: hint vs plain (conflict)	+1.3 [−0.4, +3.2]	1.000 (0.291)	n.s.	yes
H27	gpt-4o-mini, detector gpt-6-luna: hint vs plain (conflict)	+5.6 [+2.2, +9.0]	0.036 (0.005)	significant	no
H27	gpt-4o-mini, detector gpt-6.1-terra: hint vs plain (conflict)	+9.4 [+5.9, +13.1]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector claude-haiku-5-5: hint vs plain (conflict)	+9.4 [+5.7, +13.2]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector claude-sonnet-5-5: hint vs plain (conflict)	+14.2 [+10.4, +18.2]	<0.001 (<0.001)	significant	yes

Table 46: Pre-registered tests added in PR§15 (H27–H29) (part 2 of 5), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	gpt-6-luna, detector gpt-4.1: hint vs plain (conflict)	+7.7 [+4.2, +11.2]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-4o-mini: hint vs plain (conflict)	+2.6 [−1.3, +6.4]	1.000 (0.323)	n.s.	yes
H27	gpt-6-luna, detector gpt-6-luna: hint vs plain (conflict)	+2.9 [+0.2, +5.6]	0.462 (0.076)	n.s.	yes
H27	gpt-6-luna, detector gpt-6.1-terra: hint vs plain (conflict)	+13.1 [+9.4, +16.9]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4.1: hint vs plain (conflict)	−3.3 [−7.7, +1.0]	1.000 (0.241)	n.s.	yes
H27	gpt-6.1-terra, detector gpt-4o-mini: hint vs plain (conflict)	−5.3 [−9.3, −1.3]	0.194 (0.029)	n.s.	yes
H27	gpt-6.1-terra, detector gpt-6-luna: hint vs plain (conflict)	−7.0 [−11.0, −3.3]	0.006 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-6.1-terra: hint vs plain (conflict)	−7.0 [−11.0, −3.0]	0.027 (0.003)	significant	yes
H27	Qwen2.5-7B, detector gpt-4.1: cell vs detector model alone (conflict)	+1.7 [−4.0, +7.3]	1.000 (0.737)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-4o-mini: cell vs detector model alone (conflict)	+6.6 [+0.8, +12.3]	0.312 (0.054)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6-luna: cell vs detector model alone (conflict)	+8.8 [+3.6, +14.0]	0.021 (0.004)	significant	yes
H27	Qwen2.5-7B, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−14.7 [−20.3, −9.0]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector claude-haiku-5-5: cell vs detector model alone (conflict)	+5.7 [+2.1, +9.3]	0.027 (0.005)	significant	no
H27	claude-haiku-5-5, detector claude-sonnet-5-5: cell vs detector model alone (conflict)	−0.7 [−4.9, +3.6]	1.000 (0.877)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-4.1: cell vs detector model alone (conflict)	+13.1 [+8.1, +18.0]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4o-mini: cell vs detector model alone (conflict)	+11.6 [+5.4, +17.4]	0.009 (0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6-luna: cell vs detector model alone (conflict)	+18.2 [+13.3, +23.1]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−2.1 [−7.6, +3.3]	1.000 (0.582)	n.s.	yes
H27	gpt-4.1, detector gpt-4.1: cell vs detector model alone (conflict)	+11.7 [+6.7, +16.7]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4o-mini: cell vs detector model alone (conflict)	+11.2 [+5.2, +17.1]	0.006 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-6-luna: cell vs detector model alone (conflict)	+17.4 [+12.4, +22.3]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−3.3 [−8.7, +2.0]	1.000 (0.375)	n.s.	yes
H27	gpt-4o-mini, detector claude-haiku-5-5: cell vs detector model alone (conflict)	−2.7 [−7.7, +2.3]	1.000 (0.429)	n.s.	yes
H27	gpt-4o-mini, detector claude-sonnet-5-5: cell vs detector model alone (conflict)	−8.7 [−13.7, −3.7]	0.018 (0.003)	significant	no
H27	gpt-4o-mini, detector gpt-4.1: cell vs detector model alone (conflict)	+5.0 [−0.2, +10.1]	0.671 (0.118)	n.s.	yes
H27	gpt-4o-mini, detector gpt-4o-mini: cell vs detector model alone (conflict)	+8.4 [+3.2, +13.8]	0.021 (0.003)	significant	yes

Table 47: Pre-registered tests added in PR§15 (H27–H29) (part 3 of 5), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	gpt-4o-mini, detector gpt-6-luna: cell vs detector model alone (conflict)	+11.2 [+5.8, +16.7]	0.004 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−12.3 [−18.0, −6.7]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector claude-haiku-5-5: cell vs detector model alone (conflict)	+1.1 [−3.8, +5.8]	1.000 (0.782)	n.s.	yes
H27	gpt-6-luna, detector claude-sonnet-5-5: cell vs detector model alone (conflict)	−4.0 [−8.3, +0.3]	0.671 (0.129)	n.s.	yes
H27	gpt-6-luna, detector gpt-4.1: cell vs detector model alone (conflict)	+10.2 [+4.8, +15.6]	0.009 (0.001)	significant	yes
H27	gpt-6-luna, detector gpt-4o-mini: cell vs detector model alone (conflict)	+11.0 [+4.9, +17.0]	0.011 (0.002)	significant	yes
H27	gpt-6-luna, detector gpt-6-luna: cell vs detector model alone (conflict)	+11.1 [+7.3, +15.0]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−12.1 [−17.2, −6.9]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4.1: cell vs detector model alone (conflict)	+15.7 [+10.7, +20.7]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4o-mini: cell vs detector model alone (conflict)	+10.2 [+4.2, +16.2]	0.018 (0.003)	significant	yes
H27	gpt-6.1-terra, detector gpt-6-luna: cell vs detector model alone (conflict)	+22.1 [+17.2, +27.0]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-6.1-terra: cell vs detector model alone (conflict)	+2.7 [−0.3, +5.7]	0.925 (0.198)	n.s.	yes
H28: LLM flags on genuine conflicts					
H28	WikiContradict, gpt-4o-mini: LLM dir. flag vs plain	−6.5 [−10.3, −2.8]	0.001 (0.002)	significant	–
H28	WikiContradict, gpt-4o-mini: LLM removal vs plain	−9.1 [−13.3, −5.1]	<0.001 (<0.001)	significant	–
H28	WikiContradict, gpt-6-luna: LLM dir. flag vs plain	−29.0 [−34.5, −23.6]	<0.001 (<0.001)	significant	–
H28	WikiContradict, gpt-6-luna: LLM removal vs plain	−30.0 [−35.6, −24.5]	<0.001 (<0.001)	significant	–
H28	WikiContradict, claude-haiku-5-5: LLM dir. flag vs plain	−27.7 [−33.2, −22.3]	<0.001 (<0.001)	significant	–
H28	WikiContradict, claude-haiku-5-5: LLM removal vs plain	−28.2 [−33.6, −22.9]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM dir. flag vs plain	−4.0 [−8.4, 0.0]	0.100 (0.156)	n.s.	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM removal vs plain	−21.2 [−29.6, −13.1]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM dir. flag vs plain, gold recall	−3.6 [−6.8, −0.7]	0.024 (0.042)	significant	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM removal vs plain, gold recall	−17.4 [−22.4, −12.4]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM dir. flag vs plain	−14.8 [−21.5, −8.8]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM removal vs plain	−29.3 [−38.0, −20.9]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM dir. flag vs plain, gold recall	−8.9 [−12.4, −5.7]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM removal vs plain, gold recall	−20.3 [−25.0, −15.7]	<0.001 (<0.001)	significant	–

Table 48: Pre-registered tests added in PR§15 (H27–H29) (part 4 of 5), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H28: LLM flags on genuine conflicts					
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM dir. flag vs plain	−5.1 [−10.1, −0.3]	0.068 (0.109)	n.s.	–
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM removal vs plain	−25.6 [−34.0, −17.8]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM dir. flag vs plain, gold recall	−2.7 [−5.8, +0.2]	0.089 (0.139)	n.s.	–
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM removal vs plain, gold recall	−16.6 [−21.2, −12.0]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM dir. flag vs plain	−1.0 [−6.1, +3.0]	1.000 (1.000)	n.s.	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM removal vs plain	−27.3 [−36.4, −18.2]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM dir. flag vs plain, gold recall	0.0 [−2.4, +2.2]	1.000 (1.000)	n.s.	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM removal vs plain, gold recall	−18.5 [−23.6, −13.5]	<0.001 (<0.001)	significant	–
H29: artefact probe of detector direction					
H29	gpt-4o-mini detector, reversed order: direction accuracy vs original	−1.3 [−7.3, +4.7]	1.000 (0.809)	n.s.	–
H29	gpt-4o-mini detector, paraphrased: direction accuracy vs original	+1.0 [−2.7, +4.7]	1.000 (0.803)	n.s.	–
H29	gpt-6-luna detector, reversed order: direction accuracy vs original	−1.7 [−6.0, +2.7]	1.000 (0.641)	n.s.	–
H29	gpt-6-luna detector, paraphrased: direction accuracy vs original	−1.3 [−5.7, +3.0]	1.000 (0.744)	n.s.	–
H29	gpt-4o-mini detector, reversed order: gpt-4o-mini's gain vs original	−1.9 [−8.4, +4.7]	1.000 (0.697)	n.s.	–
H29	gpt-4o-mini detector, paraphrased: gpt-4o-mini's gain vs original	+1.7 [−3.1, +6.3]	1.000 (0.625)	n.s.	–
H29	gpt-6-luna detector, reversed order: gpt-4o-mini's gain vs original	−4.6 [−8.9, −0.2]	0.196 (0.080)	n.s.	–
H29	gpt-6-luna detector, paraphrased: gpt-4o-mini's gain vs original	−1.2 [−5.1, +2.7]	1.000 (0.677)	n.s.	–

Table 49: Pre-registered tests added in PR§15 (H27–H29) (part 5 of 5), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H30: the detector's verdict as an answer key					
H30	Qwen2.5-7B, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	+3.0 [−2.0, +8.0]	1.000 (0.394)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+3.0 [−1.7, +7.7]	1.000 (0.351)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	+2.7 [−1.3, +7.0]	1.000 (0.375)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+3.3 [+0.3, +6.3]	1.000 (0.082)	n.s.	yes
H30	claude-haiku-5-5, detector claude-haiku-5-5: LLM dir. flag vs verdict key (conflict)	+0.9 [−1.3, +3.2]	1.000 (0.602)	n.s.	yes
H30	claude-haiku-5-5, detector claude-sonnet-5-5: LLM dir. flag vs verdict key (conflict)	+1.2 [−1.4, +3.9]	1.000 (0.501)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	+1.8 [−1.0, +4.6]	1.000 (0.329)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+1.0 [−1.3, +3.4]	1.000 (0.588)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	+2.1 [−0.4, +4.8]	1.000 (0.207)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+2.7 [+0.6, +4.9]	0.425 (0.035)	n.s.	yes
H30	gpt-4o-mini, detector claude-haiku-5-5: LLM dir. flag vs verdict key (conflict)	+1.4 [−2.0, +4.9]	1.000 (0.545)	n.s.	yes
H30	gpt-4o-mini, detector claude-sonnet-5-5: LLM dir. flag vs verdict key (conflict)	−2.0 [−6.3, +2.2]	1.000 (0.483)	n.s.	yes
H30	gpt-4o-mini, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	+2.1 [−1.7, +5.8]	1.000 (0.394)	n.s.	yes
H30	gpt-4o-mini, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+2.4 [−0.8, +5.8]	1.000 (0.240)	n.s.	yes
H30	gpt-4o-mini, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	+1.4 [−1.9, +4.8]	1.000 (0.537)	n.s.	yes
H30	gpt-4o-mini, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+2.2 [−0.9, +5.3]	1.000 (0.289)	n.s.	yes
H30	gpt-6-luna, detector claude-haiku-5-5: LLM dir. flag vs verdict key (conflict)	−0.8 [−3.0, +1.4]	1.000 (0.652)	n.s.	yes
H30	gpt-6-luna, detector claude-sonnet-5-5: LLM dir. flag vs verdict key (conflict)	−1.2 [−3.1, +0.7]	1.000 (0.350)	n.s.	yes
H30	gpt-6-luna, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	−1.9 [−3.8, −0.2]	1.000 (0.088)	n.s.	yes
H30	gpt-6-luna, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+0.2 [−1.0, +1.7]	1.000 (0.935)	n.s.	yes
H30	gpt-6-luna, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	−0.7 [−2.7, +1.4]	1.000 (0.697)	n.s.	yes
H30	gpt-6-luna, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+0.3 [−2.0, +2.8]	1.000 (0.931)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-4.1: verdict key vs hint (conflict)	+5.7 [+0.3, +10.7]	0.393 (0.073)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-4o-mini: verdict key vs hint (conflict)	+1.3 [−4.3, +6.7]	1.000 (0.809)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6-luna: verdict key vs hint (conflict)	+1.7 [−3.7, +6.7]	1.000 (0.707)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6.1-terra: verdict key vs hint (conflict)	−7.3 [−12.7, −2.3]	0.095 (0.015)	n.s.	yes

Table 50: Pre-registered tests added in PR§16 (H30–H31) (part 1 of 3), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H30: the detector's verdict as an answer key					
H30	claude-haiku-5-5, detector claude-haiku-5-5: verdict key vs hint (conflict)	+10.2 [+6.3, +14.2]	<0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector claude-sonnet-5-5: verdict key vs hint (conflict)	+13.3 [+8.8, +17.9]	<0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector gpt-4.1: verdict key vs hint (conflict)	+12.3 [+8.6, +16.2]	<0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector gpt-4o-mini: verdict key vs hint (conflict)	−2.4 [−6.8, +1.9]	1.000 (0.397)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-6-luna: verdict key vs hint (conflict)	+8.2 [+4.3, +12.1]	0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector gpt-6.1-terra: verdict key vs hint (conflict)	+8.3 [+4.6, +12.2]	<0.001 (<0.001)	significant	yes
H30	gpt-4o-mini, detector claude-haiku-5-5: verdict key vs hint (conflict)	+4.8 [+0.1, +9.3]	0.393 (0.082)	n.s.	yes
H30	gpt-4o-mini, detector claude-sonnet-5-5: verdict key vs hint (conflict)	+7.8 [+3.2, +12.3]	0.025 (0.004)	significant	yes
H30	gpt-4o-mini, detector gpt-4.1: verdict key vs hint (conflict)	+9.7 [+5.1, +14.2]	0.005 (<0.001)	significant	yes
H30	gpt-4o-mini, detector gpt-4o-mini: verdict key vs hint (conflict)	+4.7 [−0.7, +9.9]	0.440 (0.138)	n.s.	yes
H30	gpt-4o-mini, detector gpt-6-luna: verdict key vs hint (conflict)	+6.7 [+2.0, +11.3]	0.075 (0.011)	n.s.	no
H30	gpt-4o-mini, detector gpt-6.1-terra: verdict key vs hint (conflict)	−2.1 [−6.7, +2.4]	1.000 (0.498)	n.s.	yes
H30	gpt-6-luna, detector claude-haiku-5-5: verdict key vs hint (conflict)	+5.7 [+0.4, +10.9]	0.388 (0.066)	n.s.	yes
H30	gpt-6-luna, detector claude-sonnet-5-5: verdict key vs hint (conflict)	+5.1 [+0.2, +10.0]	0.393 (0.080)	n.s.	yes
H30	gpt-6-luna, detector gpt-4.1: verdict key vs hint (conflict)	+15.6 [+10.8, +20.3]	<0.001 (<0.001)	significant	yes
H30	gpt-6-luna, detector gpt-4o-mini: verdict key vs hint (conflict)	+5.8 [+0.2, +11.4]	0.393 (0.083)	n.s.	yes
H30	gpt-6-luna, detector gpt-6-luna: verdict key vs hint (conflict)	+8.9 [+4.1, +13.7]	0.006 (<0.001)	significant	yes
H30	gpt-6-luna, detector gpt-6.1-terra: verdict key vs hint (conflict)	−6.1 [−11.0, −1.2]	0.209 (0.034)	n.s.	yes
H31: randomised direction					
H31	gpt-4o-mini: right-flag effect (conflict)	+29.4 [+22.6, +36.3]	<0.001 (<0.001)	supported	–
H31	gpt-6-luna: right-flag effect (conflict)	+33.5 [+26.8, +40.3]	<0.001 (<0.001)	supported	–
H31	claude-haiku-5-5: right-flag effect (conflict)	+24.1 [+18.0, +30.2]	<0.001 (<0.001)	supported	–
H31	gpt-6.1-terra: right-flag effect (conflict)	+17.6 [+11.3, +23.9]	<0.001 (<0.001)	supported	–
H31	gpt-4o-mini: wrong-flag effect (conflict)	−31.2 [−39.0, −23.4]	<0.001 (<0.001)	supported	–
H31	gpt-6-luna: wrong-flag effect (conflict)	−44.9 [−52.0, −37.8]	<0.001 (<0.001)	supported	–
H31	claude-haiku-5-5: wrong-flag effect (conflict)	−49.2 [−56.5, −42.1]	<0.001 (<0.001)	supported	–
H31	gpt-6.1-terra: wrong-flag effect (conflict)	−64.5 [−72.3, −56.0]	<0.001 (<0.001)	supported	–

Table 51: Pre-registered tests added in PR§16 (H30–H31) (part 2 of 3), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H31: randomised direction					
H31	gpt-4o-mini: right-flag effect + wrong-flag effect (conflict)	-1.9 [-12.2, $+8.6$]	0.733 (0.812)	n.s.	–
H31	gpt-6-luna: right-flag effect + wrong-flag effect (conflict)	-11.4 [-21.0, -1.8]	0.049 (0.042)	significant	–
H31	claude-haiku-5-5: right-flag effect + wrong-flag effect (conflict)	-25.1 [-34.3, -15.8]	$<$0.001 ($<$0.001)	significant	–
H31	gpt-6.1-terra: right-flag effect + wrong-flag effect (conflict)	-46.9 [-57.1, -36.4]	$<$0.001 ($<$0.001)	significant	–

Table 52: Pre-registered tests added in PR§16 (H30–H31) (part 3 of 3), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H40: randomised direction on new questions					
H40a	fresh f_R, f_W inside the frozen eval CIs, gpt-6-luna	65, 75	–	holds	–
H40a	fresh f_R, f_W inside the frozen eval CIs, Claude Haiku	69, 63	–	holds	–
H40a	fresh f_R, f_W inside the frozen eval CIs, gpt-6.1-terra	67, 86	–	fails (f_W)	–
H40b	fresh, gpt-6-luna: right-flag effect > 0	$+33.1$ [$+26.1$, $+40.1$]	$<$0.001 ($<$0.001)	supported	–
H40b	fresh, Claude Haiku: right-flag effect > 0	$+33.1$ [$+25.9$, $+40.3$]	$<$0.001 ($<$0.001)	supported	–
H40b	fresh, gpt-6.1-terra: right-flag effect > 0	$+18.9$ [$+12.2$, $+26.4$]	$<$0.001 ($<$0.001)	supported	–
H40b	fresh, gpt-6-luna: wrong-flag effect < 0	-49.1 [-56.1, -42.1]	$<$0.001 ($<$0.001)	supported	–
H40b	fresh, Claude Haiku: wrong-flag effect < 0	-48.0 [-55.0, -41.0]	$<$0.001 ($<$0.001)	supported	–
H40b	fresh, gpt-6.1-terra: wrong-flag effect < 0	-69.7 [-77.0, -61.8]	$<$0.001 ($<$0.001)	supported	–
H40b	fresh, gpt-6-luna: right + wrong effect (two-sided)	-16.0 [-25.8, -6.0]	0.004 (0.005)	significant	–
H40b	fresh, Claude Haiku: right + wrong effect (two-sided)	-14.9 [-24.9, -4.8]	0.004 (0.007)	significant	–
H40b	fresh, gpt-6.1-terra: right + wrong effect (two-sided)	-50.8 [-60.9, -40.2]	$<$0.001 ($<$0.001)	significant	–
H40c	MAE of the frozen predictions ≤ 3 points (3 generators)	2.2	–	holds	–
H41: a stated confidence					
H41a	gpt-4o-mini: wrong flag at 55% minus at 95% > 0	$+1.2$ [-1.9, $+4.5$]	0.294 (0.393)	not supported	–
H41a	Claude Haiku: wrong flag at 55% minus at 95% > 0	$+12.3$ [$+8.0$, $+16.8$]	$<$0.001 ($<$0.001)	supported	–
H41a	gpt-6-luna: wrong flag at 55% minus at 95% > 0	$+9.9$ [$+6.1$, $+14.2$]	$<$0.001 ($<$0.001)	supported	–
H41a	gpt-6.1-terra: wrong flag at 55% minus at 95% > 0	$+5.7$ [$+1.4$, $+9.9$]	0.023 (0.021)	supported	–
H41b	gpt-4o-mini: right flag at 95% minus at 55% > 0	$+1.0$ [-1.0, $+2.9$]	0.548 (0.299)	not supported	–
H41b	Claude Haiku: right flag at 95% minus at 55% > 0	$+1.5$ [-0.4, $+3.6$]	0.442 (0.170)	not supported	–
H41b	gpt-6-luna: right flag at 95% minus at 55% > 0	-0.2 [-2.9, $+2.5$]	0.620 (0.716)	not supported	–
H41b	gpt-6.1-terra: right flag at 95% minus at 55% > 0	$+1.9$ [-0.6, $+5.0$]	0.548 (0.262)	not supported	–

Table 53: Pre-registered tests added in PR§21 (H40–H41), registered before the runs they cover (times and exceptions in Table 34). Same conventions as Table 35. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.